

---

This content is from the eCFR and is authoritative but unofficial.

---

## Title 49 —Transportation

### Subtitle B —Other Regulations Relating to Transportation

#### Chapter II —Federal Railroad Administration, Department of Transportation

#### Part 213 Track Safety Standards

##### Subpart A General

- § 213.1 Scope of part.
- § 213.3 Application.
- § 213.4 Excepted track.
- § 213.5 Responsibility for compliance.
- § 213.7 Designation of qualified persons to supervise certain renewals and inspect track.
- § 213.9 Classes of track: operating speed limits.
- § 213.11 Restoration or renewal of track under traffic conditions.
- § 213.13 Measuring track not under load.
- § 213.14 Application of requirements to curved track.
- § 213.15 Penalties.
- § 213.17 Waivers.

##### Subpart B Roadbed

- § 213.31 Scope.
- § 213.33 Drainage.
- § 213.37 Vegetation.

##### Subpart C Track Geometry

- § 213.51 Scope.
- § 213.53 Gage.
- § 213.55 Track alinement.
- § 213.57 Curves; elevation and speed limitations.
- § 213.59 Elevation of curved track; runoff.
- § 213.63 Track surface.
- § 213.65 Combined track alinement and surface deviations.

##### Subpart D Track Structure

- § 213.101 Scope.
- § 213.103 Ballast; general.
- § 213.109 Crossties.
- § 213.110 Gage restraint measurement systems.
- § 213.113 Defective rails.
- § 213.115 Rail end mismatch.
- § 213.118 Continuous welded rail (CWR); plan review and approval.

§ 213.119 Continuous welded rail (CWR); plan contents.

§ 213.121 Rail joints.

§ 213.122 Torch cut rail.

§ 213.123 Tie plates.

§ 213.127 Rail fastening systems.

§ 213.133 Turnouts and track crossings generally.

§ 213.135 Switches.

§ 213.137 Frogs.

§ 213.139 Spring rail frogs.

§ 213.141 Self-guarded frogs.

§ 213.143 Frog guard rails and guard faces; gage.

#### Subpart E Track Appliances and Track-Related Devices

§ 213.201 Scope.

§ 213.205 Derails.

#### Subpart F Inspection

§ 213.231 Scope.

§ 213.233 Visual track inspections.

§ 213.234 Automated inspection of track constructed with concrete crossties.

§ 213.235 Inspection of switches, track crossings, and lift rail assemblies or other transition devices on moveable bridges.

§ 213.237 Inspection of rail.

§ 213.238 Qualified operator.

§ 213.239 Special inspections.

§ 213.240 Continuous rail testing.

§ 213.241 Inspection records.

#### Subpart G Train Operations at Track Classes 6 and Higher

§ 213.301 Scope of subpart.

§ 213.303 Responsibility for compliance.

§ 213.305 Designation of qualified individuals; general qualifications.

§ 213.307 Classes of track: operating speed limits.

§ 213.309 Restoration or renewal of track under traffic conditions.

§ 213.311 Measuring track not under load.

§ 213.313 Application of requirements to curved track.

§ 213.317 Waivers.

§ 213.319 Drainage.

§ 213.321 Vegetation.

§ 213.323 Track gage.

§ 213.327 Track alinement.

§ 213.329 Curves; elevation and speed limitations.

- § 213.331 Track surface.
- § 213.332 Combined track alignment and surface deviations.
- § 213.333 Automated vehicle-based inspection systems.
- § 213.334 Ballast; general.
- § 213.335 Crossties.
- § 213.337 Defective rails.
- § 213.339 Inspection of rail in service.
- § 213.341 Initial inspection of new rail and welds.
- § 213.343 Continuous welded rail (CWR).
- § 213.345 Vehicle/track system qualification.
- § 213.347 Automotive or railroad crossings at grade.
- § 213.349 Rail end mismatch.
- § 213.351 Rail joints.
- § 213.352 Torch cut rail.
- § 213.353 Turnouts, crossovers, and lift rail assemblies or other transition devices on moveable bridges.
- § 213.355 Frog guard rails and guard faces; gage.
- § 213.357 Derails.
- § 213.359 Track stiffness.
- § 213.361 Right of way.
- § 213.365 Visual track inspections.
- § 213.367 Special inspections.
- § 213.369 Inspection records.

#### **Appendix A to Part 213**

Maximum Allowable Curving Speeds

*Appendixes B-C to Part 213 [Reserved]*

#### **Appendix D to Part 213**

Minimally Compliant Analytical Track (MCAT) Simulations Used  
for Qualifying Vehicles To Operate at High Speeds and at High  
Cant Deficiencies

## **PART 213—TRACK SAFETY STANDARDS**

**Authority:** 49 U.S.C. 20102-20114 and 20142; 28 U.S.C. 2461 note; and 49 CFR 1.89.

**Source:** 63 FR 34029, June 22, 1998, unless otherwise noted.

### **Subpart A—General**

### § 213.1 Scope of part.

- (a) This part prescribes minimum safety requirements for railroad track that is part of the general railroad system of transportation. In general, the requirements prescribed in this part apply to specific track conditions existing in isolation. Therefore, a combination of track conditions, none of which individually amounts to a deviation from the requirements in this part, may require remedial action to provide for safe operations over that track. This part does not restrict a railroad from adopting and enforcing additional or more stringent requirements not inconsistent with this part.
- (b) Subparts A through F apply to track Classes 1 through 5. Subpart G and 213.2, 213.3, 213.15, and 213.240 apply to track over which trains are operated at speeds in excess of those permitted over Class 5 track.

[63 FR 34029, June 22, 1998, as amended at 78 FR 16100, Mar. 13, 2013; 85 FR 63387, Oct. 7, 2020]

### § 213.3 Application.

- (a) Except as provided in paragraph (b) of this section, this part applies to all standard gage track in the general railroad system of transportation.
- (b) This part does not apply to track:
  - (1) Located inside an installation that is not part of the general railroad system of transportation (i.e., a plant railroad). As used in this part, a plant railroad means a plant or installation that owns or leases a locomotive, uses that locomotive to switch cars throughout the plant or installation, and is moving goods solely for use in the facility's own industrial processes. The plant or installation could include track immediately adjacent to the plant or installation if the plant railroad leases the track from the general system railroad and the lease provides for (and actual practice entails) the exclusive use of that track by the plant railroad and the general system railroad for purposes of moving only cars shipped to or from the plant. A plant or installation that operates a locomotive to switch or move cars for other entities, even if solely within the confines of the plant or installation, rather than for its own purposes or industrial processes, will not be considered a plant railroad because the performance of such activity makes the operation part of the general railroad system of transportation. Similarly, this exclusion does not apply to track over which a general system railroad operates, even if that track is located within a plant railroad;
  - (2) Used exclusively for tourist, scenic, historic, or excursion operations that are not part of the general railroad system of transportation. As used in this part, tourist, scenic, historic, or excursion operations that are not part of the general railroad system of transportation means a tourist, scenic, historic, or excursion operation conducted only on track used exclusively for that purpose (i.e., there is no freight, intercity passenger, or commuter passenger railroad operation on the track); or
  - (3) Used exclusively for rapid transit operations in an urban area that are not connected to the general railroad system of transportation.

[63 FR 34029, June 22, 1998, as amended at 79 FR 4256, Jan. 24, 2014]

### § 213.4 Excepted track.

A track owner may designate a segment of track as excepted track provided that—

- (a) The segment is identified in the timetable, special instructions, general order, or other appropriate records which are available for inspection during regular business hours;
- (b) The identified segment is not located within 30 feet of an adjacent track which can be subjected to simultaneous use at speeds in excess of 10 miles per hour;
- (c) The identified segment is inspected in accordance with 213.233(c) and 213.235 at the frequency specified for Class 1 track;
- (d) The identified segment of track is not located on a bridge including the track approaching the bridge for 100 feet on either side, or located on a public street or highway, if railroad cars containing commodities required to be placarded by the Hazardous Materials Regulations (49 CFR part 172), are moved over the track; and
- (e) The railroad conducts operations on the identified segment under the following conditions:
  - (1) No train shall be operated at speeds in excess of 10 miles per hour;
  - (2) No occupied passenger train shall be operated;
  - (3) No freight train shall be operated that contains more than five cars required to be placarded by the Hazardous Materials Regulations (49 CFR part 172); and
  - (4) The gage on excepted track shall not be more than 4 feet 10<sup>1</sup>/<sub>4</sub> inches. This paragraph (e)(4) is applicable September 21, 1999.
- (f) A track owner shall advise the Track and Structures Division of FRA's Office of Railroad Safety (Track and Structures Division) at least 10 days prior to removal of a segment of track from excepted status.

[63 FR 34029, June 22, 1998, as amended at 90 FR 28136, July 1, 2025]

## § 213.5 Responsibility for compliance.

- (a) Except as provided in paragraph (b) of this section, any owner of track to which this part applies who knows or has notice that the track does not comply with the requirements of this part, shall—
  - (1) Bring the track into compliance;
  - (2) Halt operations over that track; or
  - (3) Operate under authority of a person designated under § 213.7(a), subject to conditions set forth in this part. If the operation is on continuous welded rail (CWR) track, the person under whose authority operations are conducted must also be designated under § 213.7(c).
- (b) If an owner of track to which this part applies designates a segment of track as “excepted track” under the provisions of § 213.4, operations may continue over that track without complying with the provisions of subparts B, C, D, and E of this part, unless otherwise expressly stated.
- (c) If an owner of track to which this part applies assigns responsibility for the track to another person (by lease or otherwise), written notification of the assignment shall be provided to FRA's Track and Structures Division at least 30 days in advance of the assignment. The notification may be made by any party to that assignment, but shall be in writing and include the following—
  - (1) The name and address of the track owner;
  - (2) The name and address of the person to whom responsibility is assigned (assignee);

- (3) A statement of the exact relationship between the track owner and the assignee;
  - (4) A precise identification of the track;
  - (5) A statement as to the competence and ability of the assignee to carry out the duties of the track owner under this part; and
  - (6) A statement signed by the assignee acknowledging the assignment to him of responsibility for purposes of compliance with this part.
- (d) The Administrator may hold the track owner or the assignee or both responsible for compliance with this part and subject to penalties under § 213.15.
- (e) A common carrier by railroad which is directed by the Surface Transportation Board to provide service over the track of another railroad under 49 U.S.C. 11123 is considered the owner of that track for the purposes of the application of this part during the period the directed service order remains in effect.
- (f) When any person, including a contractor for a railroad or track owner, performs any function required by this part, that person is required to perform that function in accordance with this part.

[63 FR 34029, June 22, 1998, as amended at 85 FR 63387, Oct. 7, 2020; 90 FR 28136, July 1, 2025]

### **§ 213.7 Designation of qualified persons to supervise certain renewals and inspect track.**

- (a) Each track owner to which this part applies shall designate qualified persons to supervise restorations and renewals of track under traffic conditions. Each person designated shall have—
- (1) At least—
    - (i) 1 year of experience in railroad track maintenance under traffic conditions; or
    - (ii) A combination of experience in track maintenance and training from a course in track maintenance or from a college level educational program related to track maintenance.
  - (2) Demonstrated to the owner that he or she—
    - (i) Knows and understands the requirements of this part that apply to the restoration and renewal of the track for which he or she is responsible;
    - (ii) Can detect deviations from those requirements; and
    - (iii) Can prescribe appropriate remedial action to correct or safely compensate for those deviations; and
  - (3) Authorization from the track owner to prescribe remedial actions to correct or safely compensate for deviations from the requirements of this part.
- (b) Each track owner to which this part applies shall designate qualified persons to inspect track for defects. Each person designated shall have—
- (1) At least—
    - (i) 1 year of experience in railroad track inspection; or
    - (ii) A combination of experience in track inspection and training from a course in track inspection or from a college level educational program related to track inspection;

- (2) Demonstrated to the owner that he or she—
  - (i) Knows and understands the requirements of this part that apply to the inspection of the track for which he or she is responsible;
  - (ii) Can detect deviations from those requirements; and
  - (iii) Can prescribe appropriate remedial action to correct or safely compensate for those deviations; and
- (3) Authorization from the track owner to prescribe remedial actions to correct or safely compensate for deviations from the requirements of this part, pending review by a qualified person designated under paragraph (a) of this section.
- (c) Individuals designated under paragraphs (a) or (b) of this section that inspect continuous welded rail (CWR) track or supervise the installation, adjustment, and maintenance of CWR track in accordance with the written procedures of the track owner shall have:
  - (1) Current qualifications under either paragraph (a) or (b) of this section;
  - (2) Successfully completed a comprehensive training course specifically developed for the application of written CWR procedures issued by the track owner;
  - (3) Demonstrated to the track owner that the individual:
    - (i) Knows and understands the requirements of those written CWR procedures;
    - (ii) Can detect deviations from those requirements; and
    - (iii) Can prescribe appropriate remedial action to correct or safely compensate for those deviations; and
  - (4) Authorization from the track owner to prescribe remedial actions to correct or safely compensate from deviation from the requirements in these procedures and successfully completed a recorded examination on those procedures as part of the qualification process.
- (d) Persons not fully qualified to supervise certain renewals and inspect track as required in paragraphs (a) through (c) of this section, but with at least one year of maintenance-of-way or signal experience, may pass trains over broken rails and pull aparts provided that—
  - (1) The track owner determines the person to be qualified and, as part of doing so, trains, examines, and re-examines the person periodically within two years after each prior examination on the following topics as they relate to the safe passage of trains over broken rails or pull aparts: rail defect identification, crosstie condition, track surface and alignment, gage restraint, rail end mismatch, joint bars, and maximum distance between rail ends over which trains may be allowed to pass. The sole purpose of the examination is to ascertain the person's ability to effectively apply these requirements and the examination may not be used to disqualify the person from other duties. A minimum of four hours training is required for initial training;
  - (2) The person deems it safe and train speeds are limited to a maximum of 10 m.p.h. over the broken rail or pull apart;
  - (3) The person shall watch all movements over the broken rail or pull apart and be prepared to stop the train if necessary; and

- (4) Person(s) fully qualified under § 213.7 are notified and dispatched to the location promptly for the purpose of authorizing movements and effecting temporary or permanent repairs.
- (e) With respect to designations under paragraph (a) through (d) of this section, each track owner shall maintain records of—
  - (1) Each designation in effect;
  - (2) The date each designation was made; and
  - (3) The basis for each designation, including the method used to determine that the designated person is qualified.
- (f) Each track owner shall keep designation records required under paragraph (e) of this section readily available for inspection or copying by the Federal Railroad Administration during regular business hours, following reasonable notice.

[63 FR 34029, June 22, 1998, as amended at 74 FR 43002, Aug. 25, 2009; 78 FR 16100, Mar. 13, 2013; 85 FR 63387, Oct. 7, 2020]

### § 213.9 Classes of track: operating speed limits.

- (a) Except as provided in paragraph (b) of this section and §§ 213.57(b), 213.59(a), 213.113(a), and 213.137(b) and (c), the following maximum allowable operating speeds apply—

#### [IN MILES PER HOUR]

| Over track that meets all of the requirements prescribed in this part for— | The maximum allowable operating speed for freight trains is— | The maximum allowable operating speed for passenger trains is— |
|----------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------|
| Excepted track                                                             | 10                                                           | N/A                                                            |
| Class 1 track                                                              | 10                                                           | 15                                                             |
| Class 2 track                                                              | 25                                                           | 30                                                             |
| Class 3 track                                                              | 40                                                           | 60                                                             |
| Class 4 track                                                              | 60                                                           | 80                                                             |
| Class 5 track                                                              | 80                                                           | 90                                                             |

- (b) If a segment of track does not meet all of the requirements of its intended class, it is reclassified to the next lowest class of track for which it does meet all of the requirements of this part. However, if the segment of track does not at least meet the requirements of Class 1 track, operations may continue at Class 1 speeds for a period of not more than 30 days without bringing the track into compliance, under the authority of a person designated under § 213.7(a), after that person determines that operations may safely continue and subject to any limiting conditions specified by such person.

[63 FR 34029, June 22, 1998, as amended at 85 FR 63388, Oct. 7, 2020]



### **§ 213.11 Restoration or renewal of track under traffic conditions.**

If during a period of restoration or renewal, track is under traffic conditions and does not meet all of the requirements prescribed in this part, the work on the track shall be under the continuous supervision of a person designated under § 213.7(a) and, as applicable, § 213.7(c). The work on the track shall also be subject to any limiting conditions specified by such person. The operating speed cannot be more than the maximum allowable speed under § 213.9 for the class of track concerned. The term “continuous supervision” as used in this section means the physical presence of that person at the job site. However, since the work may be performed over a large area, it is not necessary that each phase of the work be done under the visual supervision of that person.

[85 FR 63388, Oct. 7, 2020]

### **§ 213.13 Measuring track not under load.**

When unloaded track is measured to determine compliance with requirements of this part, the amount of rail movement, if any, that occurs while the track is loaded must be added to the measurements of the unloaded track.

### **§ 213.14 Application of requirements to curved track.**

Unless otherwise provided in this part, requirements specified for curved track apply only to track having a curvature greater than 0.25 degree.

[78 FR 16100, Mar. 13, 2013]

### **§ 213.15 Penalties.**

- (a) Any person that violates any requirement of this part or causes the violation of any such requirement is subject to a civil penalty of at least the minimum civil monetary penalty and not more than the ordinary maximum civil monetary penalty per violation. However, penalties may be assessed against individuals only for willful violations, and a penalty not to exceed the aggravated maximum civil monetary penalty per violation may be assessed, where:
  - (1) A grossly negligent violation, or a pattern of repeated violations, has created an imminent hazard of death or injury to persons, or
  - (2) A death or injury has occurred. See 49 CFR part 209, appendix A. “Person” means an entity of any type, covered under 49 U.S.C. 21301, including the following: a railroad; a manager, supervisor, official, or other employee or agent of a railroad; any owner, manufacturer, lessor, or lessee of railroad equipment, track, or facilities; any independent contractor providing goods or services to a railroad; any employee of such owner, manufacturer, lessor, lessee, or independent contractor; and anyone held by the Federal Railroad Administrator to be responsible under § 213.5(d) or § 213.303(c). Each day a violation continues shall constitute a separate offense. See FRA's website at <https://railroads.dot.gov/> for a statement of agency civil penalty policy.
- (b) Any person who knowingly and willfully falsifies a record or report required by this part may be subject to criminal penalties under 49 U.S.C. 21311.

[63 FR 34029, June 22, 1998, as amended at 69 FR 30593, May 28, 2004; 72 FR 51196, Sept. 6, 2007; 73 FR 79701, Dec. 30, 2008; 77 FR 24419, Apr. 24, 2012; 81 FR 43109, July 1, 2016; 82 FR 16132, Apr. 3, 2017; 83 FR 60746, Nov. 27, 2018; 84 FR 23734, May 23, 2019; 84 FR 37072, July 31, 2019; 86 FR 1757, Jan. 11, 2021; 86 FR 23253, May 3, 2021; 87 FR 15867, Mar. 21, 2022; 88 FR 1126, Jan. 6, 2023; 88 FR 89561, Dec. 28, 2023; 89 FR 106295, Dec. 30, 2024; 90 FR 28136, July 1, 2025]

## § 213.17 Waivers.

- (a) Any owner of track to which this part applies, or other person subject to this part, may petition the Federal Railroad Administrator for a waiver from any or all requirements prescribed in this part. The filing of such a petition does not affect that person's responsibility for compliance with that requirement while the petition is being considered.
- (b) Each petition for a waiver under this section shall be filed in the manner and contain the information required by part 211 of this chapter.
- (c) If the Administrator finds that a waiver is in the public interest and is consistent with railroad safety, the Administrator may grant the exemption subject to any conditions the Administrator deems necessary. Where a waiver is granted, the Administrator publishes a notice containing the reasons for granting the waiver.

## Subpart B—Roadbed

### § 213.31 Scope.

This subpart prescribes minimum requirements for roadbed and areas immediately adjacent to roadbed.

### § 213.33 Drainage.

Each drainage or other water carrying facility under or immediately adjacent to the roadbed shall be maintained and kept free of obstruction, to accommodate expected water flow for the area concerned.

### § 213.37 Vegetation.

Vegetation on railroad property which is on or immediately adjacent to roadbed shall be controlled so that it does not—

- (a) Become a fire hazard to track-carrying structures;
- (b) Obstruct visibility of railroad signs and signals:
  - (1) Along the right-of-way, and
  - (2) At highway-rail crossings; (This paragraph (b)(2) is applicable September 21, 1999.)
- (c) Interfere with railroad employees performing normal trackside duties;
- (d) Prevent proper functioning of signal and communication lines; or
- (e) Prevent railroad employees from visually inspecting moving equipment from their normal duty stations.

## Subpart C—Track Geometry

### § 213.51 Scope.

This subpart prescribes requirements for the gage, alinement, and surface of track, and the elevation of outer rails and speed limitations for curved track.

### § 213.53 Gage.

- (a) Gage is measured between the heads of the rails at right-angles to the rails in a plane five-eighths of an inch below the top of the rail head.
- (b) Gage shall be within the limits prescribed in the following table—

| Class of track      | The gage must be at least— | But not more than—                  |
|---------------------|----------------------------|-------------------------------------|
| Excepted track      | N/A                        | 4'10 <sup>1</sup> / <sub>4</sub> ". |
| Class 1 track       | 4'8"                       | 4'10".                              |
| Class 2 and 3 track | 4'8"                       | 4'9 <sup>3</sup> / <sub>4</sub> ".  |
| Class 4 and 5 track | 4'8"                       | 4'9 <sup>1</sup> / <sub>2</sub> ".  |

### § 213.55 Track alinement.

- (a) Except as provided in paragraph (b) of this section, alinement may not deviate from uniformity more than the amount prescribed in the following table:

| Class of track | Tangent track                                                                                  | Curved track                                                                                      |                                                                                                   |
|----------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                | The deviation of the mid-offset from a 62-foot line <sup>1</sup> may not be more than—(inches) | The deviation of the mid-ordinate from a 31-foot chord <sup>2</sup> may not be more than—(inches) | The deviation of the mid-ordinate from a 62-foot chord <sup>2</sup> may not be more than—(inches) |
| Class 1 track  | 5                                                                                              | <sup>3</sup> N/A                                                                                  | 5                                                                                                 |
| Class 2 track  | 3                                                                                              | <sup>3</sup> N/A                                                                                  | 3                                                                                                 |
| Class 3 track  | 1 <sup>3</sup> / <sub>4</sub>                                                                  | 1 <sup>1</sup> / <sub>4</sub>                                                                     | 1 <sup>3</sup> / <sub>4</sub>                                                                     |
| Class 4 track  | 1 <sup>1</sup> / <sub>2</sub>                                                                  | 1                                                                                                 | 1 <sup>1</sup> / <sub>2</sub>                                                                     |

<sup>1</sup> The ends of the line shall be at points on the gage side of the line rail, five-eighths of an inch below the top of the railhead. Either rail may be used as the line rail; however, the same rail shall be used for the full length of that tangential segment of the track.

<sup>2</sup> The ends of the chord shall be at points on the gage side of the outer rail, five-eighths of an inch below the top of the railhead.

<sup>3</sup> N/A—Not Applicable

| Class of track | Tangent track                                                                                  | Curved track                                                                                      |                                                                                                   |
|----------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                | The deviation of the mid-offset from a 62-foot line <sup>1</sup> may not be more than—(inches) | The deviation of the mid-ordinate from a 31-foot chord <sup>2</sup> may not be more than—(inches) | The deviation of the mid-ordinate from a 62-foot chord <sup>2</sup> may not be more than—(inches) |
| Class 5 track  | <sup>3</sup> / <sub>4</sub>                                                                    | <sup>1</sup> / <sub>2</sub>                                                                       | <sup>5</sup> / <sub>8</sub>                                                                       |

<sup>1</sup> The ends of the line shall be at points on the gage side of the line rail, five-eighths of an inch below the top of the railhead. Either rail may be used as the line rail; however, the same rail shall be used for the full length of that tangential segment of the track.

<sup>2</sup> The ends of the chord shall be at points on the gage side of the outer rail, five-eighths of an inch below the top of the railhead.

<sup>3</sup> N/A—Not Applicable

- (b) For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches, the alinement of the outside rail of the curve may not deviate from uniformity more than the amount prescribed in the following table:

| Class of track | Curved track                                                                                      |                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                | The deviation of the mid-ordinate from a 31-foot chord <sup>1</sup> may not be more than—(inches) | The deviation of the mid-ordinate from a 62-foot chord <sup>1</sup> may not be more than—(inches) |
| Class 1 track  | <sup>3</sup> N/A                                                                                  | <sup>1</sup> <sup>1</sup> / <sub>4</sub>                                                          |
| Class 2 track  | <sup>3</sup> N/A                                                                                  | <sup>1</sup> <sup>1</sup> / <sub>4</sub>                                                          |
| Class 3 track  | <sup>3</sup> / <sub>4</sub>                                                                       | <sup>1</sup> <sup>1</sup> / <sub>4</sub>                                                          |
| Class 4 track  | <sup>3</sup> / <sub>4</sub>                                                                       | <sup>7</sup> / <sub>8</sub>                                                                       |

<sup>1</sup> The ends of the chord shall be at points on the gage side of the outer rail, five-eighths of an inch below the top of the railhead.

<sup>2</sup> Restraining rails or other systems may be required for derailment prevention.

<sup>3</sup> N/A—Not Applicable

| Class of track | Curved track                                                                                      |                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                | The deviation of the mid-ordinate from a 31-foot chord <sup>1</sup> may not be more than—(inches) | The deviation of the mid-ordinate from a 62-foot chord <sup>1</sup> may not be more than—(inches) |
| Class 5 track  | <sup>1</sup> / <sub>2</sub>                                                                       | <sup>5</sup> / <sub>8</sub>                                                                       |

<sup>1</sup> The ends of the chord shall be at points on the gage side of the outer rail, five-eighths of an inch below the top of the railhead.

<sup>2</sup> Restraining rails or other systems may be required for derailment prevention.

<sup>3</sup> N/A—Not Applicable

[78 FR 16100, Mar. 13, 2013]

## § 213.57 Curves; elevation and speed limitations.

- (a) The maximum elevation of the outside rail of a curve may not be more than 8 inches on track Classes 1 and 2, and 7 inches on track Classes 3 through 5. The outside rail of a curve may not be lower than the inside rail by design, except when engineered to address specific track or operating conditions; the limits in § 213.63 apply in all cases.
- (b) The maximum allowable posted timetable operating speed for each curve is determined by the following formula—

$$V_{\max} = \sqrt{\frac{E_a + E_u}{0.0007D}}$$

Where—

$V_{\max}$  = Maximum allowable posted timetable operating speed (m.p.h.).

$E_a$  = Actual elevation of the outside rail (inches).<sup>[1]</sup>

$E_u$  = Qualified cant deficiency<sup>[2]</sup> (inches) of the vehicle type.

$D$  = Degree of curvature (degrees).<sup>[3]</sup>

<sup>[1]</sup> Actual elevation,  $E_a$ , for each 155-foot track segment in the body of the curve is determined by averaging the elevation for 11 points through the segment at 15.5-foot spacing. If the curve length is less than 155 feet, the points are averaged through the full length of the body of the curve.

- (c) All vehicles are considered qualified for operating on track with a cant deficiency,  $E_u$ , not exceeding 3 inches. Table 1 of appendix A to this part is a table of speeds computed in accordance with the formula in paragraph (b) of this section, when  $E_u$  equals 3 inches, for various elevations and degrees of curvature.
- (d) Each vehicle type must be approved by FRA to operate on track with a qualified cant deficiency,  $E_u$ , greater than 3 inches. Each vehicle type must demonstrate, in a ready-for-service load condition, compliance with the requirements of either paragraph (d)(1) or (2) of this section.
  - (1) When positioned on a track with a uniform superelevation equal to the proposed cant deficiency:
    - (i) No wheel of the vehicle type unloads to a value less than 60 percent of its static value on perfectly level track; and
    - (ii) For passenger cars, the roll angle between the floor of the equipment and the horizontal does not exceed 8.6 degrees; or
  - (2) When operating through a constant radius curve at a constant speed corresponding to the proposed cant deficiency, and a test plan is submitted to and approved by FRA in accordance with § 213.345(e) and (f):
    - (i) The steady-state (average) load on any wheel, throughout the body of the curve, is not less than 60 percent of its static value on perfectly level track; and
    - (ii) For passenger cars, the steady-state (average) lateral acceleration measured on the floor of the carbody does not exceed 0.15g.
- (e) The track owner or railroad shall transmit the results of the testing specified in paragraph (d) of this section to FRA's Associate Administrator for Railroad Safety/Chief Safety Officer (FRA) requesting approval for the vehicle type to operate at the desired curving speeds allowed under the formula in paragraph (b) of this section. The request shall be made in writing and contain, at a minimum, the following information—
  - (1) A description of the vehicle type involved, including schematic diagrams of the suspension system(s) and the estimated location of the center of gravity above top of rail;
  - (2) The test procedure,<sup>[4]</sup> including the load condition under which the testing was performed, and description of the instrumentation used to qualify the vehicle type, as well as the maximum values for wheel unloading and roll angles or accelerations that were observed during testing; and

---

<sup>[2]</sup> If the actual elevation,  $E_a$ , and degree of curvature,  $D$ , change as a result of track degradation, then the actual cant deficiency for the maximum allowable posted timetable operating speed,  $V_{max}$ , may be greater than the qualified cant deficiency,  $E_u$ . This actual cant deficiency for each curve may not exceed the qualified cant deficiency,  $E_u$ , plus 1 inch.

<sup>[3]</sup> Degree of curvature,  $D$ , is determined by averaging the degree of curvature over the same track segment as the elevation.

<sup>[4]</sup> The test procedure may be conducted whereby all the wheels on one side (right or left) of the vehicle are raised to the proposed cant deficiency, the vertical wheel loads under each wheel are measured, and a level is used to record the angle through which the floor of the vehicle has been rotated.

- (3) For vehicle types not subject to parts 229 or 238 of this chapter, procedures or standards in effect that relate to the maintenance of all safety-critical components of the suspension system(s) for the particular vehicle type. Safety-critical components of the suspension system are those that impact or have significant influence on the roll of the carbody and the distribution of weight on the wheels.
- (f) In approving the request made pursuant to paragraph (e) of this section, FRA may impose conditions necessary for safely operating at the higher curving speeds. Upon FRA approval of the request, the track owner or railroad shall notify FRA in writing no less than 30 calendar days prior to the proposed implementation of the approved higher curving speeds allowed under the formula in paragraph (b) of this section. The notification shall contain, at a minimum, identification of the track segment(s) on which the higher curving speeds are to be implemented.
- (g) The documents required by this section must be provided to FRA by:
  - (1) The track owner; or
  - (2) A railroad that provides service with the same vehicle type over trackage of one or more track owner(s), with the written consent of each affected track owner.
- (h)
  - (1) Vehicle types permitted by FRA to operate at cant deficiencies,  $E_u$ , greater than 3 inches but not more than 5 inches shall be considered qualified under this section to operate at those permitted cant deficiencies for any track segment. The track owner or railroad shall notify FRA in writing no less than 30 calendar days prior to the proposed implementation of such curving speeds in accordance with paragraph (f) of this section.
  - (2) Vehicle types permitted by FRA to operate at cant deficiencies,  $E_u$ , greater than 5 inches shall be considered qualified under this section to operate at those permitted cant deficiencies only for the previously operated or identified track segments(s).
- (i) For vehicle types intended to operate at any curving speed producing more than 5 inches of cant deficiency, the following provisions of subpart G of this part shall apply: §§ 213.333(a) through (g), (j)(1), (k) and (m), 213.345, and 213.369(f).
- (j) As used in this section—
  - (1) **Vehicle** means a locomotive, as defined in § 229.5 of this chapter; a freight car, as defined in § 215.5 of this chapter; a passenger car, as defined in § 238.5 of this chapter; and any rail rolling equipment used in a train with either a freight car or a passenger car.
  - (2) **Vehicle type** means like vehicles with variations in their physical properties, such as suspension, mass, interior arrangements, and dimensions that do not result in significant changes to their dynamic characteristics.

[78 FR 16101, Mar. 13, 2013]

### § 213.59 Elevation of curved track; runoff.

- (a) If a curve is elevated, the full elevation shall be provided throughout the curve, unless physical conditions do not permit. If elevation runoff occurs in a curve, the actual minimum elevation shall be used in computing the maximum allowable posted timetable operating speed for that curve under § 213.57(b).

- (b) Elevation runoff shall be at a uniform rate, within the limits of track surface deviation prescribed in § 213.63, and it shall extend at least the full length of the spirals. If physical conditions do not permit a spiral long enough to accommodate the minimum length of runoff, part of the runoff may be on tangent track.

[63 FR 34029, June 22, 1998, as amended at 78 FR 16101, Mar. 13, 2013]

## § 213.63 Track surface.

- (a) Except as provided in paragraph (b) of this section, each track owner shall maintain the surface of its track within the limits prescribed in the following table:

| Track surface (inches)                                                                                                                                                                                                           | Class of track |                    |                    |                    |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|--------------------|--------------------|--------------------|
|                                                                                                                                                                                                                                  | 1              | 2                  | 3                  | 4                  | 5                  |
| The deviation from uniform profile on either rail at the mid-ordinate of a 62-foot chord may not be more than                                                                                                                    | 3              | 2<br>$\frac{3}{4}$ | 2<br>$\frac{1}{4}$ | 2                  | 1<br>$\frac{1}{4}$ |
| The deviation from zero crosslevel at any point on tangent or reverse crosslevel elevation on curves may not be more than                                                                                                        | 3              | 2<br>$\frac{3}{4}$ | 1<br>$\frac{1}{4}$ | 1                  | 1                  |
| The difference in crosslevel between any two points less than 62 feet apart may not be more than * <sup>1</sup> <sup>2</sup>                                                                                                     | 3              | 2<br>$\frac{1}{4}$ | 2                  | 1<br>$\frac{3}{4}$ | 1<br>$\frac{1}{2}$ |
| * Where determined by engineering decision prior to June 22, 1998, due to physical restrictions on spiral length and operating practices and experience, the variation in crosslevel on spirals per 31 feet may not be more than | 2              | 1<br>$\frac{3}{4}$ | 1<br>$\frac{1}{4}$ | 1                  | $\frac{3}{4}$      |

<sup>1</sup> Except as limited by § 213.57(a), where the elevation at any point in a curve equal or exceeds 6 inches, the difference in crosslevel within 62 feet between that point and a point with greater elevation may not be more than  $1\frac{1}{2}$  inches.

<sup>2</sup> However, to control harmonics on Class 2 through 5 jointed track with staggered joints, the crosslevel differences shall not exceed  $1\frac{1}{4}$  inches in all of six consecutive pairs of joints, as created by seven low joints. Track with joints staggered less than 10 feet apart shall not be considered as having staggered joints. Joints within the seven low joints outside of the regular joint spacing shall not be considered as joints for purposes of this footnote.

- (b) For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches, each track owner shall maintain the surface of the curve within the limits prescribed in the following table:

| Track surface (inches)                                                                                        | Class of track   |                  |   |   |   |
|---------------------------------------------------------------------------------------------------------------|------------------|------------------|---|---|---|
|                                                                                                               | 1                | 2                | 3 | 4 | 5 |
| The deviation from uniform profile on either rail at the mid-ordinate of a 31-foot chord may not be more than | N/A <sup>1</sup> | N/A <sup>1</sup> | 1 | 1 | 1 |

<sup>1</sup> N/A—Not Applicable.



| Track surface (inches)                                                                                          | Class of track                |                               |                               |                               |                               |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                                                                                                                 | 1                             | 2                             | 3                             | 4                             | 5                             |
| The deviation from uniform profile on either rail at the mid-ordinate of a 62-foot chord may not be more than   | 2 <sup>1</sup> / <sub>4</sub> | 2 <sup>1</sup> / <sub>4</sub> | 1 <sup>3</sup> / <sub>4</sub> | 1 <sup>1</sup> / <sub>4</sub> | 1                             |
| The difference in crosslevel between any two points less than 10 feet apart (short warp) shall not be more than | 2                             | 2                             | 1 <sup>3</sup> / <sub>4</sub> | 1 <sup>3</sup> / <sub>4</sub> | 1 <sup>1</sup> / <sub>2</sub> |

<sup>1</sup> N/A—Not Applicable.

[78 FR 16101, Mar. 13, 2013, as amended at 91 FR 22735, Apr. 28, 2026]

### § 213.65 Combined track alinement and surface deviations.

On any curved track where operations are conducted at a qualified cant deficiency,  $E_u$ , greater than 5 inches, the combination of alinement and surface deviations for the same chord length on the outside rail in the curve, as measured by a TGMS, shall comply with the following formula:

$$\frac{3}{4} \times \left| \frac{A_m}{A_L} + \frac{S_m}{S_L} \right| \leq 1$$

Where—

$A_m$  = measured alinement deviation from uniformity (outward is positive, inward is negative).

$A_L$  = allowable alinement limit as per § 213.55(b) (always positive) for the class of track.

$S_m$  = measured profile deviation from uniformity (down is positive, up is negative).

$S_L$  = allowable profile limit as per § 213.63(b) (always positive) for the class of track.

$$\left| \frac{A_m}{A_L} + \frac{S_m}{S_L} \right| = \text{the absolute (positive) value of the result of } \frac{A_m}{A_L} + \frac{S_m}{S_L} .$$

[78 FR 16102, Mar. 13, 2013]

## Subpart D—Track Structure

### § 213.101 Scope.

This subpart prescribes minimum requirements for ballast, crossties, track assembly fittings, and the physical conditions of rails.

### § 213.103 Ballast; general.

Unless it is otherwise structurally supported, all track shall be supported by material which will—

- (a) Transmit and distribute the load of the track and railroad rolling equipment to the subgrade;
- (b) Restrain the track laterally, longitudinally, and vertically under dynamic loads imposed by railroad rolling equipment and thermal stress exerted by the rails;
- (c) Provide adequate drainage for the track; and
- (d) Maintain proper track crosslevel, surface, and alinement.

### § 213.109 Crossties.

- (a) Crossties shall be made of a material to which rail can be securely fastened.
- (b) Each 39-foot segment of track shall have at a minimum—
  - (1) A sufficient number of crossties that in combination provide effective support that will—
    - (i) Hold gage within the limits prescribed in § 213.53(b);
    - (ii) Maintain surface within the limits prescribed in § 213.63; and
    - (iii) Maintain alinement within the limits prescribed in § 213.55;
  - (2) The minimum number and type of crossties specified in paragraph (b)(4) of this section and described in paragraph (c) or (d), as applicable, of this section effectively distributed to support the entire segment;
  - (3) At least one non-defective crosstie of the type specified in paragraphs (c) and (d) of this section that is located at a joint location as specified in paragraph (e) of this section; and
  - (4) The minimum number of crossties as indicated in the following table.

| FRA track class | Tangent track, turnouts, and curves                            |                                                  |
|-----------------|----------------------------------------------------------------|--------------------------------------------------|
|                 | Tangent track and curved track less than or equal to 2 degrees | Turnouts and curved track greater than 2 degrees |
| Class 1         | 5                                                              | 6                                                |
| Class 2         | 8                                                              | 9                                                |
| Class 3         | 8                                                              | 10                                               |
| Class 4 and 5   | 12                                                             | 14                                               |

- (c) Crossties, other than concrete, counted to satisfy the requirements set forth in paragraph (b)(4) of this section shall not be—
  - (1) Broken through;

- (2) Split or otherwise impaired to the extent the crosstie will allow the ballast to work through, or will not hold spikes or rail fasteners;
  - (3) So deteriorated that the crosstie plate or base of rail can move laterally  $\frac{1}{2}$  inch relative to the crosstie; or
  - (4) Cut by the crosstie plate through more than 40 percent of a crosstie's thickness.
- (d) Concrete crossties counted to satisfy the requirements set forth in paragraph (b)(4) of this section shall not be—
- (1) Broken through or deteriorated to the extent that prestressing material is visible;
  - (2) Deteriorated or broken off in the vicinity of the shoulder or insert so that the fastener assembly can either pull out or move laterally more than  $\frac{3}{8}$  inch relative to the crosstie;
  - (3) Deteriorated such that the base of either rail can move laterally more than  $\frac{3}{8}$  inch relative to the crosstie on curves of 2 degrees or greater; or can move laterally more than  $\frac{1}{2}$  inch relative to the crosstie on tangent track or curves of less than 2 degrees;
  - (4) Deteriorated or abraded at any point under the rail seat to a depth of  $\frac{1}{2}$  inch or more;
  - (5) Deteriorated such that the crosstie's fastening or anchoring system, including rail anchors (see § 213.127(b)), is unable to maintain longitudinal rail restraint, or maintain rail hold down, or maintain gage due to insufficient fastener toeload; or
  - (6) Configured with less than two fasteners on the same rail except as provided in § 213.127(c).
- (e) Class 1 and 2 track shall have one crosstie whose centerline is within 24 inches of each rail joint (end) location. Class 3, 4, and 5 track shall have either one crosstie whose centerline is within 18 inches of each rail joint location or two crossties whose centerlines are within 24 inches either side of each rail joint location. The relative position of these crossties is described in the following three diagrams:
- (1) Each rail joint in Class 1 and 2 track shall be supported by at least one crosstie specified in paragraphs (c) and (d) of this section whose centerline is within 48 inches as shown in Figure 1.

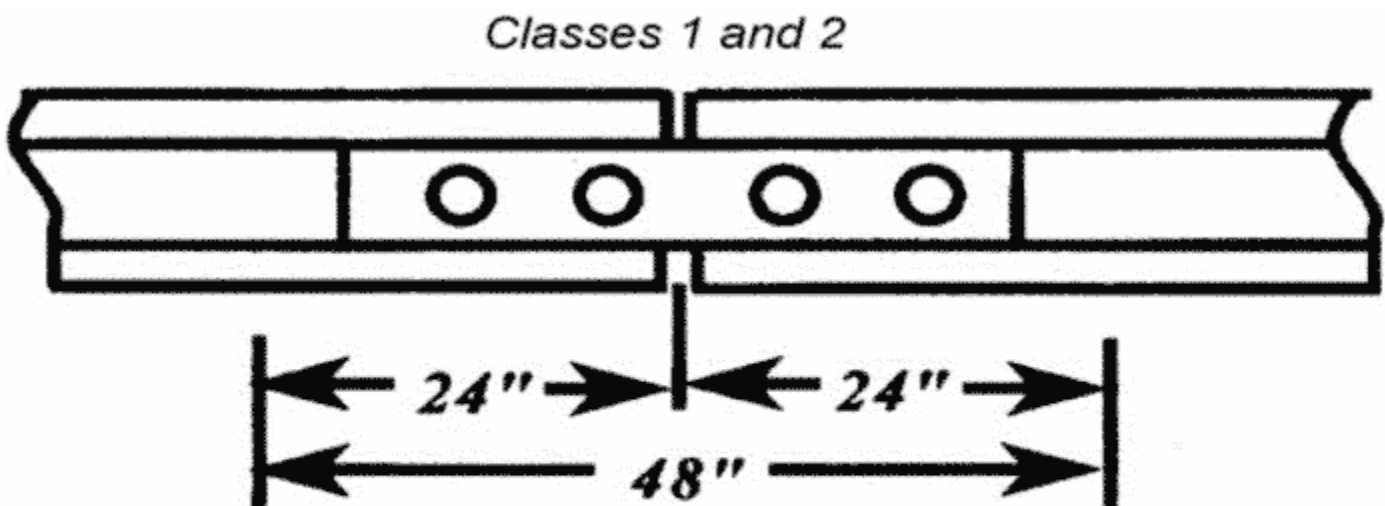


Figure 1

- (2) Each rail joint in Class 3, 4, and 5 track shall be supported by either at least one crosstie specified in paragraphs (c) and (d) of this section whose centerline is within 36 inches as shown in Figure 2, or:

***Classes 3 through 5***

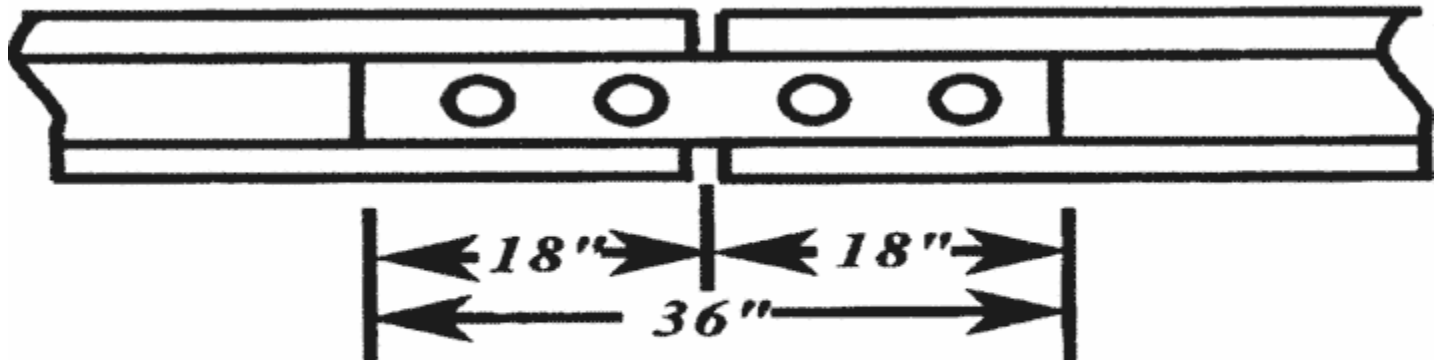


Figure 2

- (3) Two crossties, one on each side of the rail joint, whose centerlines are within 24 inches of the rail joint location as shown in Figure 3.

***Classes 3 through 5***

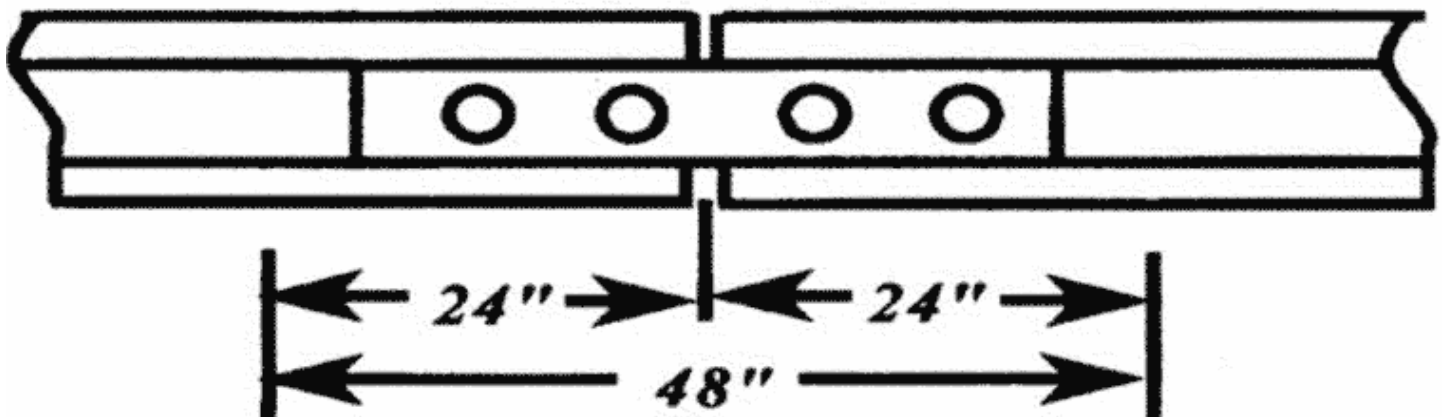


Figure 3

- (f) For track constructed without crossties, such as slab track, track connected directly to bridge structural components, track over servicing pits, etc., the track structure shall meet the requirements of paragraph (b)(1) of this section.

[76 FR 18084, Apr. 1, 2011]

## § 213.110 Gage restraint measurement systems.

- (a) A track owner may elect to implement a Gage Restraint Measurement System (GRMS), supplemented by the use of a Portable Track Loading Fixture (PTLF), to determine compliance with the crosstie and fastener requirements specified in §§ 213.109 and 213.127 provided that—
  - (1) The track owner notifies FRA's Track and Structures Division at least 30 days prior to the designation of any line segment on which GRMS technology will be implemented; and
  - (2) The track owner notifies FRA's Track and Structures Division at least 10 days prior to the removal of any line segment from GRMS designation.
- (b) Initial notification under paragraph (a)(1) of this section shall include—
  - (1) Identification of the line segment(s) by timetable designation, milepost limits, class of track, or other identifying criteria; and
  - (2) The most recent record of million gross tons of traffic per year over the identified segment(s).
- (c)
  - (1) The track owner shall also provide to FRA sufficient technical data to establish compliance with the following minimum design requirements of a GRMS vehicle:
  - (2) Gage restraint shall be measured between the heads of rail—
    - (i) At an interval not exceeding 16 inches;
    - (ii) Under an applied vertical load of no less than 10 kips per rail; and
    - (iii) Under an applied lateral load that provides for a lateral/vertical load ratio of between 0.5 and 1.25<sup>[5]</sup>, and a load severity greater than 3 kips but less than 8 kips per rail.

- (d) Load severity is defined by the formula:

$$S = L - cV$$

Where—

S = Load severity, defined as the lateral load applied to the fastener system (kips).

L = Actual lateral load applied (kips).

c = Coefficient of friction between rail/tie, which is assigned a nominal value of 0.4.

V = Actual vertical load applied (kips), or static vertical wheel load if vertical load is not measured.

- (e) The measured gage values shall be converted to a Projected Loaded Gage 24 (PLG24) as follows—

$$PLG24 = UTG + A \times (LTG - UTG)$$

Where—

---

<sup>[5]</sup> GRMS equipment using load combinations developing L/V ratios that exceed 0.8 shall be operated with caution to protect against the risk of wheel climb by the test wheelset.

UTG = Unloaded track gage measured by the GRMS vehicle at a point no less than 10 feet from any lateral or vertical load application.

LTG = Loaded track gage measured by the GRMS vehicle at a point no more than 12 inches from the lateral load application point.

A = The extrapolation factor used to convert the measured loaded gage to expected loaded gage under a 24-kip lateral load and a 33-kip vertical load.

For all track—

$$A = \frac{13.513}{(L - 0.258 \times V) - .009 \times (L - 0.258 \times V)^2}$$

Note: The A factor shall not exceed a value of 3.184 under any valid loading configuration.

L = Actual lateral load applied (kips).

V = Actual vertical load applied (kips), or static vertical wheel load if vertical load is not measured.

(f) The measured gage and load values shall be converted to a Gage Widening Projection (GWP) as follows:

$$GWP = (LTG - UTG) \times \frac{8.26}{L - 0.258 \times V}$$

- (g) The GRMS vehicle shall be capable of producing output reports that provide a trace, on a constant-distance scale, of all parameters specified in paragraph (l) of this section.
- (h) The GRMS vehicle shall be capable of providing an exception report containing a systematic listing of all exceptions, by magnitude and location, to all the parameters specified in paragraph (l) of this section.
- (i) The exception reports required by this section shall be provided to the appropriate person designated as fully qualified under § 213.7 prior to the next inspection required under § 213.233.
- (j) The track owner shall institute the necessary procedures for maintaining the integrity of the data collected by the GRMS and PTLF systems. At a minimum, the track owner shall—
- (1) Maintain and make available to the Federal Railroad Administration documented calibration procedures on each GRMS vehicle which, at a minimum, shall specify a daily instrument verification procedure that will ensure correlation between measurements made on the ground and those recorded by the instrumentation with respect to loaded and unloaded gage parameters; and
  - (2) Maintain each PTLF used for determining compliance with the requirements of this section such that the 4,000-pound reading is accurate to within five percent of that reading.
- (k) The track owner shall provide training in GRMS technology to all persons designated as fully qualified under § 213.7 and whose territories are subject to the requirements of this section. The training program shall be made available to the Federal Railroad Administration upon request. At a minimum, the training program shall address—

- (1) Basic GRMS procedures;
  - (2) Interpretation and handling of exception reports generated by the GRMS vehicle;
  - (3) Locating and verifying defects in the field;
  - (4) Remedial action requirements;
  - (5) Use and calibration of the PTLF; and
  - (6) Recordkeeping requirements.
- (l) The GRMS record of lateral restraint shall identify two exception levels. At a minimum, the track owner shall initiate the required remedial action at each exception level as defined in the following table—

| GRMS parameters <sup>1</sup> | If measurement value exceeds                                              | Remedial action required                                                                                                                                                                                                                                                                                          |
|------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIRST LEVEL EXCEPTION        |                                                                           |                                                                                                                                                                                                                                                                                                                   |
| UTG                          | 58 inches                                                                 | (1) Immediately protect the exception location with a 10 m.p.h. speed restriction, then verify location;<br>(2) Restore lateral restraint and maintain in compliance with PTLF criteria as described in paragraph (m) of this section; and<br>(3) Maintain compliance with § 213.53(b) as measured with the PTLF. |
| LTG                          | 58 inches                                                                 |                                                                                                                                                                                                                                                                                                                   |
| PLG24                        | 59 inches                                                                 |                                                                                                                                                                                                                                                                                                                   |
| GWP                          | 1 inch                                                                    |                                                                                                                                                                                                                                                                                                                   |
| SECOND LEVEL EXCEPTION       |                                                                           |                                                                                                                                                                                                                                                                                                                   |
| LTG                          | 57 <sup>3</sup> / <sub>4</sub> inches on Class 4 and 5 track <sup>2</sup> | (1) Limit operating speed to no more than the maximum allowable under § 213.9 for Class 3 track, then verify location;<br>(2) Maintain in compliance with PTLF criteria as described in paragraph (m) of this section; and<br>(3) Maintain compliance with § 213.53(b) as measured with the PTLF.                 |
| PLG24                        | 58 inches                                                                 |                                                                                                                                                                                                                                                                                                                   |

<sup>1</sup> Definitions for the GRMS parameters referenced in this table are found in paragraph (p) of this section.

<sup>2</sup> This note recognizes that good track will typically increase in total gage by as much as one-quarter of an inch due to outward rail rotation under GRMS loading conditions. For Class 2 and 3 track, the GRMS LTG values are also increased by one-quarter of inch to a maximum of 58 inches. However, for any class of track, GRMS LTG values in excess of 58 inches are considered First Level exceptions and the appropriate remedial action(s) must be taken by the track owner. This 1/4-inch increase in allowable gage applies only to GRMS LTG. For gage measured by traditional methods, or with the use of the PTLF, the table in § 213.53(b) applies.

| GRMS parameters <sup>1</sup> | If measurement value exceeds | Remedial action required |
|------------------------------|------------------------------|--------------------------|
| GWP                          | 0.75 inch                    |                          |

<sup>1</sup> Definitions for the GRMS parameters referenced in this table are found in paragraph (p) of this section.

<sup>2</sup> This note recognizes that good track will typically increase in total gage by as much as one-quarter of an inch due to outward rail rotation under GRMS loading conditions. For Class 2 and 3 track, the GRMS LTG values are also increased by one-quarter of inch to a maximum of 58 inches. However, for any class of track, GRMS LTG values in excess of 58 inches are considered First Level exceptions and the appropriate remedial action(s) must be taken by the track owner. This 1/4-inch increase in allowable gage applies only to GRMS LTG. For gage measured by traditional methods, or with the use of the PTLF, the table in § 213.53(b) applies.

- (m) Between GRMS inspections, the PTLF may be used as an additional analytical tool to assist fully qualified § 213.7 individuals in determining compliance with the crosstie and fastener requirements of §§ 213.109 and 213.127. When the PTLF is used, whether as an additional analytical tool or to fulfill the requirements of paragraph (l), it shall be used subject to the following criteria—
  - (1) At any location along the track that the PTLF is applied, that location will be deemed in compliance with the crosstie and fastener requirements specified in §§ 213.109 and 213.127 provided that—
    - (i) The total gage widening at that location does not exceed  $\frac{5}{8}$  inch when increasing the applied force from 0 to 4,000 pounds; and
    - (ii) The gage of the track under 4,000 pounds of applied force does not exceed the allowable gage prescribed in § 213.53(b) for the class of track.
  - (2) Gage widening in excess of  $\frac{5}{8}$  inch shall constitute a deviation from Class 1 standards.
  - (3) A person designated as fully qualified under § 213.7 retains the discretionary authority to prescribe additional remedial actions for those locations which comply with the requirements of paragraph (m)(1)(i) and (ii) of this section.
  - (4) When a functional PTLF is not available to a fully qualified person designated under § 213.7, the criteria for determining crosstie and fastener compliance shall be based solely on the requirements specified in §§ 213.109 and 213.127.
  - (5) If the PTLF becomes non-functional or is missing, the track owner will replace or repair it before the next inspection required under § 213.233.
  - (6) Where vertical loading of the track is necessary for contact with the lateral rail restraint components, a PTLF test will not be considered valid until contact with these components is restored under static loading conditions.
- (n) The track owner shall maintain a record of the two most recent GRMS inspections at locations which meet the requirements specified in § 213.241(b). At a minimum, records shall indicate the following—
  - (1) Location and nature of each First Level exception; and



- (2) Nature and date of remedial action, if any, for each exception identified in paragraph (n)(1) of this section.
- (o) The inspection interval for designated GRMS line segments shall be such that—
  - (1) On line segments where the annual tonnage exceeds two million gross tons, or where the maximum operating speeds for passenger trains exceeds 30 mph, GRMS inspections must be performed annually at an interval not to exceed 14 months; or
  - (2) On line segments where the annual tonnage is two million gross tons or less and the maximum operating speed for passenger trains does not exceed 30 mph, the interval between GRMS inspections must not exceed 24 months.
- (p) As used in this section—
  - (1) **Gage Restraint Measurement System (GRMS)** means a track loading vehicle meeting the minimum design requirements specified in this section.
  - (2) **Gage Widening Projection (GWP)** means the measured gage widening, which is the difference between loaded and unloaded gage, at the applied loads, projected to reference loads of 16 kips of lateral force and 33 kips of vertical force.
  - (3) **L/V ratio** means the numerical ratio of lateral load applied at a point on the rail to the vertical load applied at that same point. GRMS design requirements specify an L/V ratio of between 0.5 and 1.25.
  - (4) **Load severity** means the amount of lateral load applied to the fastener system after friction between rail and tie is overcome by any applied gage-widening lateral load.
  - (5) **Loaded Track Gage (LTG)** means the gage measured by the GRMS vehicle at a point no more than 12 inches from the lateral load application point.
  - (6) **Portable Track Loading Fixture (PTLF)** means a portable track loading device capable of applying an increasing lateral force from 0 to 4,000 pounds on the web/base fillet of each rail simultaneously.
  - (7) **Projected Loaded Gage (PLG)** means an extrapolated value for loaded gage calculated from actual measured loads and deflections. PLG 24 means the extrapolated value for loaded gage under a 24,000 pound lateral load and a 33,000 pound vertical load.
  - (8) **Unloaded Track Gage (UTG)** means the gage measured by the GRMS vehicle at a point no less than 10 feet from any lateral or vertical load.

[66 FR 1899, Jan. 10, 2001; 66 FR 8372, Jan. 31, 2001, as amended at 78 FR 16102, Mar. 13, 2013; 90 FR 28136, July 1, 2025]

## § 213.113 Defective rails.

- (a) When an owner of track learns that a rail in the track contains any of the defects listed in the table contained in paragraph (c) of this section, a person designated under § 213.7 shall determine whether the track may continue in use. If the designated person determines that the track may continue in use, operation over the defective rail is not permitted until—
  - (1) The rail is replaced or repaired; or
  - (2) The remedial action prescribed in the table contained in paragraph (c) of this section is initiated.

- (b) When an owner of track learns that a rail in the track contains an indication of any of the defects listed in the table contained in paragraph (c) of this section, the track owner shall verify the indication. Except as provided in § 213.240, the track owner must verify the indication within four hours, unless the track owner has an indication of the existence of a defect that requires remedial action A, A2, or B identified in the table contained in paragraph (c) of this section, in which case the track owner must immediately verify the indication. If the indication is verified, the track owner must—
  - (1) Replace or repair the rail; or
  - (2) Initiate the remedial action prescribed in the table contained in paragraph (c) of this section.
- (c) A track owner who learns that a rail contains one of the following defects shall prescribe the remedial action specified if the rail is not replaced or repaired, in accordance with this paragraph's table:

REMEDIAL ACTION TABLE

| Defect                                                                                                           | Length of defect (inch(es))                  |                                  | Percentage of existing rail head cross-sectional area weakened by defect |                                          | If the defective rail is not repaired, take the remedial action prescribed in note |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------|--------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------|
|                                                                                                                  | More than                                    | But not more than                | Less than                                                                | But not less than                        |                                                                                    |
| Compound Fissure                                                                                                 | .....                                        | .....                            | 70.....<br>100.....                                                      | 5.....<br>70.....<br>100.....            | B.<br>A2.<br>A.                                                                    |
| Transverse Fissure Detail Fracture Engine Burn Fracture Defective Weld                                           | .....<br>.....<br>.....<br>.....             | .....<br>.....<br>.....<br>..... | 25.....<br>60.....<br>100.....                                           | 5.....<br>25.....<br>60.....<br>100..... | C.<br>D.<br>A2, or [E and H].<br>A, or [E and H].                                  |
| Horizontal Split Head Vertical Split Head Split Web Piped Rail Head Web Separation Defective Weld (Longitudinal) | 1.....<br>2.....<br>4.....<br>(1).....       | 2.....<br>4.....<br>(1).....     | .....<br>.....<br>.....                                                  | .....<br>.....<br>.....                  | H and F.<br>I and G.<br>B.<br>A.                                                   |
| Bolt Hole Crack                                                                                                  | 1/2.....<br>1.....<br>1 1/2.....<br>(1)..... | 1.....<br>1 1/2.....<br>(1)..... | .....<br>.....<br>.....                                                  | .....<br>.....<br>.....                  | H and F.<br>H and G.<br>B.<br>A.                                                   |
| Broken Base                                                                                                      | 1.....<br>6 (1).....                         | .....<br>.....                   | .....<br>.....                                                           | .....<br>.....                           | D.<br>A, or [E and I].                                                             |
| Ordinary Break                                                                                                   | .....                                        | .....                            | .....                                                                    | .....                                    | A or E.                                                                            |
| Damaged Rail                                                                                                     | .....                                        | .....                            | .....                                                                    | .....                                    | C.                                                                                 |
| Flattened Rail                                                                                                   | Depth ≥ 3/8 and                              | .....                            | .....                                                                    | .....                                    | H.                                                                                 |
| Crushed Head                                                                                                     | Length ≥ 8.....                              | .....                            | .....                                                                    | .....                                    |                                                                                    |

(1) Break out in rail head.

(2) Remedial action D applies to a moon-shaped breakout, resulting from a derailment, with length greater than 6 inches but not exceeding 12 inches and width not exceeding one-third of the rail base width.

Notes:

A. Assign a person designated under § 213.7 to visually supervise each operation over the defective rail.

A2. Assign a person designated under § 213.7 to make a visual inspection. After a visual inspection, that person may authorize operation to continue without continuous visual supervision at a maximum of 10 m.p.h. for up to 24 hours prior to another such visual inspection or replacement or repair of the rail.

B. Limit operating speed over the defective rail to that as authorized by a person designated under § 213.7(a), who has at least one year of supervisory experience in railroad track maintenance. The operating speed cannot be over 30 m.p.h. or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower.

C. Apply joint bars bolted only through the outermost holes to the defect within 10 days after it is determined to continue the track in use. In the case of Class 3 through 5 track, limit the operating speed over the defective rail to 30 m.p.h. until joint bars are applied; thereafter, limit the speed to 50 m.p.h. or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower. When a search for internal rail defects is conducted under § 213.237, and defects are discovered in Class 3 through 5 track that require remedial action C, the operating speed shall be limited to 50 m.p.h. or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower, for a period not to exceed 4 days. If the defective rail has not been removed from the track or a permanent repair made within 4 days of the discovery, limit operating speed over the defective rail to 30 m.p.h. until joint bars are applied; thereafter, limit speed to 50 m.p.h. or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower. When joint bars have not been applied within 10 days, the speed must be limited to 10 m.p.h. until joint bars are applied.

D. Apply joint bars bolted only through the outermost holes to the defect within 7 days after it is determined to continue the track in use. In the case of Class 3 through 5 track, limit operating speed over the defective rail to 30 m.p.h. or less as authorized by a person designated under § 213.7(a), who has at least one year of supervisory experience in railroad track maintenance, until joint bars are applied; thereafter, limit speed to 50 m.p.h. or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower. When joint bars have not been applied within 7 days, the speed must be limited to 10 m.p.h. until the joint bars are applied.

E. Apply joint bars to the defect and bolt in accordance with § 213.121(d) and (e).

reinspection reveals the rail defect to have increased in size and therefore become subject to a more restrictive remedial action. This process continues indefinitely until the rail is removed from the track or repaired. If not inspected within 90 days, limit speed to that for Class 2 track or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower, until it is inspected.

G. Inspect rail within 30 days after it is determined to continue the track in use. If the rail remains in the track and is not replaced or repaired, the reinspection cycle starts over with each successive reinspection unless the reinspection reveals the rail defect to have increased in size and therefore become subject to a more restrictive remedial action. This process continues indefinitely until the rail is removed from the track or repaired. If not inspected within 30 days, limit speed to that for Class 2 track or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower, until it is inspected.

H. Limit operating speed over the defective rail to 50 m.p.h. or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower.

I. Limit operating speed over the defective rail to 30 m.p.h. or the maximum allowable speed under § 213.9 for the class of track concerned, whichever is lower.

(d) As used in this section—

- (1) **Bolt hole crack** means a crack across the web, originating from a bolt hole, and progressing on a path either inclined upward toward the rail head or inclined downward toward the base. Fully developed bolt hole cracks may continue horizontally along the head/web or base/web fillet, or they may progress into and through the head or base to separate a piece of the rail end from the rail. Multiple cracks occurring in one rail end are considered to be a single defect. However, bolt hole cracks occurring in adjacent rail ends within the same joint must be reported as separate defects.
- (2) **Broken base** means any break in the base of the rail.
- (3) **Compound fissure** means a progressive fracture originating from a horizontal split head that turns up or down, or in both directions, in the head of the rail. Transverse development normally progresses substantially at a right angle to the length of the rail.
- (4) **Crushed head** means a short length of rail, not at a joint, which has drooped or sagged across the width of the rail head to a depth of  $\frac{3}{8}$  inch or more below the rest of the rail head and 8 inches or more in length. Unlike flattened rail where the depression is visible on the rail head only, the sagging or drooping is also visible in the head/web fillet area.
- (5) **Damaged rail** means any rail broken or otherwise damaged by a derailment, broken, flat, or unbalanced wheel, wheel slipping, or similar causes.

- (6) **Defective weld** means a field or plant weld containing any discontinuities or pockets, exceeding 5 percent of the rail head area individually or 10 percent in the aggregate, oriented in or near the transverse plane, due to incomplete penetration of the weld metal between the rail ends, lack of fusion between weld and rail end metal, entrapment of slag or sand, under-bead or shrinkage cracking, or fatigue cracking. Weld defects may originate in the rail head, web, or base, and in some cases, cracks may progress from the defect into either or both adjoining rail ends. If the weld defect progresses longitudinally through the weld section, the defect is considered a split web for purposes of remedial action required by this section.
- (7) **Detail fracture** means a progressive fracture originating at or near the surface of the rail head. These fractures should not be confused with transverse fissures, compound fissures, or other defects which have internal origins. Detail fractures may arise from shelled spots, head checks, or flaking.
- (8) **Engine burn fracture** means a progressive fracture originating in spots where driving wheels have slipped on top of the rail head. In developing downward these fractures frequently resemble the compound or even transverse fissures with which they should not be confused or classified.
- (9) **Flattened rail** means a short length of rail, not at a joint, which has flattened out across the width of the rail head to a depth of  $\frac{3}{8}$  inch or more below the rest of the rail and 8 inches or more in length. Flattened rail occurrences have no repetitive regularity and thus do not include corrugations, and have no apparent localized cause such as a weld or engine burn. Their individual length is relatively short, as compared to a condition such as head flow on the low rail of curves.
- (10) **Head and web separation** means a progressive fracture, longitudinally separating the head from the web of the rail at the head fillet area.
- (11) **Horizontal split head** means a horizontal progressive defect originating inside of the rail head, usually  $\frac{1}{4}$  inch or more below the running surface and progressing horizontally in all directions, and generally accompanied by a flat spot on the running surface. The defect appears as a crack lengthwise of the rail when it reaches the side of the rail head.
- (12) **Ordinary break** means a partial or complete break in which there is no sign of a fissure, and in which none of the other defects described in this paragraph (d) is found.
- (13) **Piped rail** means a vertical split in a rail, usually in the web, due to failure of the shrinkage cavity in the ingot to unite in rolling.
- (14) **Split web** means a lengthwise crack along the side of the web and extending into or through it.
- (15) **Transverse fissure** means a progressive crosswise fracture starting from a crystalline center or nucleus inside the head from which it spreads outward as a smooth, bright, or dark round or oval surface substantially at a right angle to the length of the rail. The distinguishing features of a transverse fissure from other types of fractures or defects are the crystalline center or nucleus and the nearly smooth surface of the development which surrounds it.
- (16) **Vertical split head** means a vertical split through or near the middle of the head, and extending into or through it. A crack or rust streak may show under the head close to the web or pieces may be split off the side of the head.

[79 FR 4256, Jan. 24, 2014; 79 FR 4633, Jan. 29, 2014, as amended at 85 FR 63388, Oct. 7, 2020]

### § 213.115 Rail end mismatch.

Any mismatch of rails at joints may not be more than that prescribed by the following table—

| Class of track      | Any mismatch of rails at joints may not be more than the following— |                                          |
|---------------------|---------------------------------------------------------------------|------------------------------------------|
|                     | On the tread of the rail ends (inch)                                | On the gage side of the rail ends (inch) |
| Class 1 track       | $\frac{1}{4}$                                                       | $\frac{1}{4}$                            |
| Class 2 track       | $\frac{1}{4}$                                                       | $\frac{3}{16}$                           |
| Class 3 track       | $\frac{3}{16}$                                                      | $\frac{3}{16}$                           |
| Class 4 and 5 track | $\frac{1}{8}$                                                       | $\frac{1}{8}$                            |

### § 213.118 Continuous welded rail (CWR); plan review and approval.

- (a) Each track owner with track constructed of CWR shall have in effect and comply with a plan that contains written procedures which address: the installation, adjustment, maintenance, and inspection of CWR; inspection of CWR joints; and a training program for the application of those procedures.
- (b) The track owner shall file its CWR plan with the FRA Associate Administrator for Railroad Safety/Chief Safety Officer (Associate Administrator). Within 30 days of receipt of the submission, FRA will review the plan for compliance with this subpart. FRA will approve, disapprove or conditionally approve the submitted plan, and will provide written notice of its determination.
- (c) The track owner's existing plan shall remain in effect until the track owner's new plan is approved or conditionally approved and is effective pursuant to paragraph (d) of this section.
- (d) The track owner shall, upon receipt of FRA's approval or conditional approval, establish the plan's effective date. The track owner shall advise in writing FRA and all affected employees of the effective date.
- (e) FRA, for cause stated, may, subsequent to plan approval or conditional approval, require revisions to the plan to bring the plan into conformity with this subpart. Notice of a revision requirement shall be made in writing and specify the basis of FRA's requirement. The track owner may, within 30 days of the revision requirement, respond and provide written submissions in support of the original plan. FRA renders a final decision in writing. Not more than 30 days following any final decision requiring revisions to a CWR plan, the track owner shall amend the plan in accordance with FRA's decision and resubmit the conforming plan. The conforming plan becomes effective upon its submission to FRA.

[74 FR 43002, Aug. 25, 2009]

### § 213.119 Continuous welded rail (CWR); plan contents.

The track owner shall comply with the contents of the CWR plan approved or conditionally approved under § 213.118. The plan shall contain the following elements—

- (a) Procedures for the installation and adjustment of CWR which include—
  - (1) Designation of a desired rail installation temperature range for the geographic area in which the CWR is located; and

- (2) De-stressing procedures/methods which address proper attainment of the desired rail installation temperature range when adjusting CWR.
- (b) Rail anchoring or fastening requirements that will provide sufficient restraint to limit longitudinal rail and crosstie movement to the extent practical, and specifically addressing CWR rail anchoring or fastening patterns on bridges, bridge approaches, and at other locations where possible longitudinal rail and crosstie movement associated with normally expected train-induced forces, is restricted.
- (c) CWR joint installation and maintenance procedures which require that—
  - (1) Each rail shall be bolted with at least two bolts at each CWR joint;
  - (2) In the case of a bolted joint installed during CWR installation after October 21, 2009, the track owner shall either, within 60 days—
    - (i) Weld the joint;
    - (ii) Install a joint with six bolts; or
    - (iii) Anchor every tie 195 feet in both directions from the joint; and
  - (3) In the case of a bolted joint in CWR experiencing service failure or a failed bar with a rail gap present, the track owner shall either—
    - (i) Weld the joint;
    - (ii) Replace the broken bar(s), replace the broken bolts, adjust the anchors and, within 30 days, weld the joint;
    - (iii) Replace the broken bar(s), replace the broken bolts, install one additional bolt per rail end, and adjust anchors;
    - (iv) Replace the broken bar(s), replace the broken bolts, and anchor every tie 195 feet in both directions from the CWR joint; or
    - (v) Replace the broken bar(s), replace the broken bolts, add rail with provisions for later adjustment pursuant to paragraph (d)(2) of this section, and reapply the anchors.
- (d) Procedures which specifically address maintaining a desired rail installation temperature range when cutting CWR, including rail repairs, in-track welding, and in conjunction with adjustments made in the area of tight track, a track buckle, or a pull-apart. Rail repair practices shall take into consideration existing rail temperature so that—
  - (1) When rail is removed, the length installed shall be determined by taking into consideration the existing rail temperature and the desired rail installation temperature range; and
  - (2) Under no circumstances should rail be added when the rail temperature is below that designated by paragraph (a)(1) of this section, without provisions for later adjustment.
- (e) Procedures which address the monitoring of CWR in curved track for inward shifts of alignment toward the center of the curve as a result of disturbed track.
- (f) Procedures which govern train speed on CWR track when—
  - (1) Maintenance work, track rehabilitation, track construction, or any other event occurs which disturbs the roadbed or ballast section and reduces the lateral or longitudinal resistance of the track; and



- (2) The difference between the average rail temperature and the average rail neutral temperature is in a range that causes buckling-prone conditions to be present at a specific location; and
- (3) In formulating the procedures under paragraphs (f)(1) and (f)(2) of this section, the track owner shall—
  - (i) Determine the speed required, and the duration and subsequent removal of any speed restriction based on the restoration of the ballast, along with sufficient ballast re-consolidation to stabilize the track to a level that can accommodate expected train-induced forces. Ballast re-consolidation can be achieved through either the passage of train tonnage or mechanical stabilization procedures, or both; and
  - (ii) Take into consideration the type of crossties used.

(g) Procedures which prescribe when physical track inspections are to be performed.

- (1) At a minimum, these procedures shall address inspecting track to identify—
  - (i) Buckling-prone conditions in CWR track, including—
    - (A) Locations where tight or kinky rail conditions are likely to occur; and
    - (B) Locations where track work of the nature described in paragraph (f)(1)(i) of this section has recently been performed; and
  - (ii) Pull-apart prone conditions in CWR track, including locations where pull-apart or stripped-joint rail conditions are likely to occur; and
- (2) In formulating the procedures under paragraph (g)(1) of this section, the track owner shall—
  - (i) Specify when the inspections will be conducted; and
  - (ii) Specify the appropriate remedial actions to be taken when either buckling-prone or pull-apart prone conditions are found.

(h) Procedures which prescribe the scheduling and conduct of inspections to detect cracks and other indications of potential failures in CWR joints. In formulating the procedures under this paragraph, the track owner shall—

- (1) Address the inspection of joints and the track structure at joints, including, at a minimum, periodic on-foot inspections;
- (2) Identify joint bars with visible or otherwise detectable cracks and conduct remedial action pursuant to § 213.121;
- (3) Specify the conditions of actual or potential joint failure for which personnel must inspect, including, at a minimum, the following items:
  - (i) Loose, bent, or missing joint bolts;
  - (ii) Rail end batter or mismatch that contributes to instability of the joint; and
  - (iii) Evidence of excessive longitudinal rail movement in or near the joint, including, but not limited to; wide rail gap, defective joint bolts, disturbed ballast, surface deviations, gap between tie plates and rail, or displaced rail anchors;

- (4) Specify the procedures for the inspection of CWR joints that are imbedded in highway-rail crossings or in other structures that prevent a complete inspection of the joint, including procedures for the removal from the joint of loose material or other temporary material;
- (5) Specify the appropriate corrective actions to be taken when personnel find conditions of actual or potential joint failure, including on-foot follow-up inspections to monitor conditions of potential joint failure in any period prior to completion of repairs;
- (6) Specify the timing of periodic inspections, which shall be based on the configuration and condition of the joint:
  - (i) Except as provided in paragraphs (h)(6)(ii) through (h)(6)(iv) of this section, track owners must specify that all CWR joints are inspected, at a minimum, in accordance with the intervals identified in the following table:

### MINIMUM NUMBER OF INSPECTIONS PER CALENDAR YEAR <sup>1</sup>

|                   | Freight trains operating over track with<br>an annual tonnage of: |                 |                        | Passenger trains operating over track with<br>an annual tonnage of: |                                    |
|-------------------|-------------------------------------------------------------------|-----------------|------------------------|---------------------------------------------------------------------|------------------------------------|
|                   | Less than<br>40 mgt                                               | 40 to 60<br>mgt | Greater than<br>60 mgt | Less than 20<br>mgt                                                 | Greater than or equal to<br>20 mgt |
| Class 5 & above   | 2                                                                 | 3 <sup>2</sup>  | 4 <sup>2</sup>         | 3 <sup>2</sup>                                                      | 3 <sup>2</sup>                     |
| Class 4           | 2                                                                 | 3 <sup>2</sup>  | 4 <sup>2</sup>         | 2                                                                   | 3 <sup>2</sup>                     |
| Class 3           | 1                                                                 | 2               | 2                      | 2                                                                   | 2                                  |
| Class 2           | 0                                                                 | 0               | 0                      | 1                                                                   | 1                                  |
| Class 1           | 0                                                                 | 0               | 0                      | 0                                                                   | 0                                  |
| Excepted<br>Track | 0                                                                 | 0               | 0                      | n/a                                                                 | n/a                                |

4 = Four times per calendar year, with one inspection in each of the following periods: January to March, April to June, July to September, and October to December; and with consecutive inspections separated by at least 60 calendar days.

3 = Three times per calendar year, with one inspection in each of the following periods: January to April, May to August, and September to December; and with consecutive inspections separated by at least 90 calendar days.

2 = Twice per calendar year, with one inspection in each of the following periods: January to June and July to December; and with consecutive inspections separated by at least 120 calendar days.

<sup>1</sup> Where a track owner operates both freight and passenger trains over a given segment of track, and there are two different possible inspection interval requirements, the more frequent inspection interval applies.

<sup>2</sup> When extreme weather conditions prevent a track owner from conducting an inspection of a particular territory within the required interval, the track owner may extend the interval by up to 30 calendar days from the last day that the extreme weather condition prevented the required inspection.

|  | Freight trains operating over track with an annual tonnage of: |              |                     | Passenger trains operating over track with an annual tonnage of: |                                 |
|--|----------------------------------------------------------------|--------------|---------------------|------------------------------------------------------------------|---------------------------------|
|  | Less than 40 mgt                                               | 40 to 60 mgt | Greater than 60 mgt | Less than 20 mgt                                                 | Greater than or equal to 20 mgt |

1 = Once per calendar year, with consecutive inspections separated by at least 180 calendar days.

<sup>1</sup> Where a track owner operates both freight and passenger trains over a given segment of track, and there are two different possible inspection interval requirements, the more frequent inspection interval applies.

<sup>2</sup> When extreme weather conditions prevent a track owner from conducting an inspection of a particular territory within the required interval, the track owner may extend the interval by up to 30 calendar days from the last day that the extreme weather condition prevented the required inspection.

- (ii) Consistent with any limitations applied by the track owner, a passenger train conducting an unscheduled detour operation may proceed over track not normally used for passenger operations at a speed not to exceed the maximum authorized speed otherwise allowed, even though CWR joints have not been inspected in accordance with the frequency identified in paragraph (h)(6)(i) of this section, provided that:
    - (A) All CWR joints have been inspected consistent with requirements for freight service; and
    - (B) The unscheduled detour operation lasts no more than 14 consecutive calendar days. In order to continue operations beyond the 14-day period, the track owner must inspect the CWR joints in accordance with the requirements of paragraph (h)(6)(i) of this section.
  - (iii) Tourist, scenic, historic, or excursion operations, if limited to the maximum authorized speed for passenger trains over the next lower class of track, need not be considered in determining the frequency of inspections under paragraph (h)(6)(i) of this section.
  - (iv) All CWR joints that are located in switches, turnouts, track crossings, lift rail assemblies or other transition devices on moveable bridges must be inspected on foot at least monthly, consistent with the requirements in § 213.235; and all records of those inspections must be kept in accordance with the requirements in § 213.241. A track owner may include in its § 213.235 inspections, in lieu of the joint inspections required by paragraph (h)(6)(i) of this section, CWR joints that are located in track structure that is adjacent to switches and turnouts, provided that the track owner precisely defines the parameters of that arrangement in the CWR plans.
- (7) Specify the recordkeeping requirements related to joint bars in CWR, including the following:
- (i) The track owner shall keep a record of each periodic and follow-up inspection required to be performed by the track owner's CWR plan, except for those inspections conducted pursuant to § 213.235 for which track owners must maintain records pursuant to § 213.241. The record shall be prepared on the day the inspection is made and signed by the person making the inspection. The record shall include, at a minimum, the following items: the boundaries of the territory inspected; the nature and location of any deviations at the joint from the requirements of this part or of the track owner's CWR plan, with the location identified with sufficient

precision that personnel could return to the joint and identify it without ambiguity; the date of the inspection; the remedial action, corrective action, or both, that has been taken or will be taken; and the name or identification number of the person who made the inspection.

(ii) [Reserved]

(8) In lieu of the requirements for the inspection of rail joints contained in paragraphs (h)(1) through (h)(7) of this section, a track owner may seek approval from FRA to use alternate procedures.

(i) The track owner shall submit the proposed alternate procedures and a supporting statement of justification to the Associate Administrator.

(ii) If the Associate Administrator finds that the proposed alternate procedures provide an equivalent or higher level of safety than the requirements in paragraphs (h)(1) through (h)(7) of this section, the Associate Administrator will approve the alternate procedures by notifying the track owner in writing. The Associate Administrator will specify in the written notification the date on which the procedures will become effective, and after that date, the track owner shall comply with the procedures. If the Associate Administrator determines that the alternate procedures do not provide an equivalent level of safety, the Associate Administrator will disapprove the alternate procedures in writing, and the track owner shall continue to comply with the requirements in paragraphs (h)(1) through (h)(7) of this section.

(iii) While a determination is pending with the Associate Administrator on a request submitted pursuant to paragraph (h)(8) of this section, the track owner shall continue to comply with the requirements contained in paragraphs (h)(1) through (h)(7) of this section.

(i) The track owner shall have in effect a comprehensive training program for the application of these written CWR procedures, with provisions for annual re-training, for those individuals designated under § 213.7(c) as qualified to supervise the installation, adjustment, and maintenance of CWR track and to perform inspections of CWR track. The track owner shall make the training program available for review by FRA upon request.

(j) The track owner shall prescribe and comply with recordkeeping requirements necessary to provide an adequate history of track constructed with CWR. At a minimum, these records must include:

(1) Rail temperature, location, and date of CWR installations. Each record shall be retained for at least one year;

(2) A record of any CWR installation or maintenance work that does not conform to the written procedures. Such record shall include the location of the rail and be maintained until the CWR is brought into conformance with such procedures; and

(3) Information on inspection of rail joints as specified in paragraph (h)(7) of this section.

(k) The track owner shall make readily available, at every job site where personnel are assigned to install, inspect or maintain CWR, a copy of the track owner's CWR procedures and all revisions, appendices, updates, and referenced materials related thereto prior to their effective date. Such CWR procedures shall be issued and maintained in one CWR standards and procedures manual.

(l) As used in this section—

*Adjusting/de-stressing* means a procedure by which a rail's neutral temperature is re-adjusted to the desired value. It typically consists of cutting the rail and removing rail anchoring devices, which provides for the necessary expansion and contraction, and then re-assembling the track.

**Annual re-training** means training every calendar year.

**Buckling incident** means the formation of a lateral misalignment sufficient in magnitude to constitute a deviation from the Class 1 requirements specified in § 213.55. These normally occur when rail temperatures are relatively high and are caused by high longitudinal compressive forces.

**Buckling-prone condition** means a track condition that can result in the track being laterally displaced due to high compression forces caused by critical rail temperature combined with insufficient track strength and/or train dynamics.

**Continuous welded rail (CWR)** means rail that has been welded together into lengths exceeding 400 feet. Rail installed as CWR remains CWR, regardless of whether a joint or plug is installed into the rail at a later time.

**Corrective actions** mean those actions which track owners specify in their CWR plans to address conditions of actual or potential joint failure, including, as applicable, repair, restrictions on operations, and additional on-foot inspections.

**CWR joint** means any joint directly connected to CWR.

**Desired rail installation temperature range** means the rail temperature range, within a specific geographical area, at which forces in CWR should not cause a buckling incident in extreme heat, or a pull apart during extreme cold weather.

**Disturbed track** means the disturbance of the roadbed or ballast section, as a result of track maintenance or any other event, which reduces the lateral or longitudinal resistance of the track, or both.

**Mechanical stabilization** means a type of procedure used to restore track resistance to disturbed track following certain maintenance operations. This procedure may incorporate dynamic track stabilizers or ballast consolidators, which are units of work equipment that are used as a substitute for the stabilization action provided by the passage of tonnage trains.

**Pull apart or stripped joint** means a condition when no bolts are mounted through a joint on the rail end, rendering the joint bar ineffective due to excessive expansive or contractive forces.

**Pull-apart prone condition** means a condition when the actual rail temperature is below the rail neutral temperature at or near a joint where longitudinal tensile forces may affect the fastenings at the joint.

**Rail anchors** mean those devices which are attached to the rail and bear against the side of the crosstie to control longitudinal rail movement. Certain types of rail fasteners also act as rail anchors and control longitudinal rail movement by exerting a downward clamping force on the upper surface of the rail base.

**Rail neutral temperature** is the temperature at which the rail is neither in compression nor tension.

**Rail temperature** means the temperature of the rail, measured with a rail thermometer.

**Remedial actions** mean those actions which track owners are required to take as a result of requirements of this part to address a non-compliant condition.

**Tight/kinky rail** means CWR which exhibits minute alignment irregularities which indicate that the rail is in a considerable amount of compression.

**Tourist, scenic, historic, or excursion operations** mean railroad operations that carry passengers with the conveyance of the passengers to a particular destination not being the principal purpose.

*Track lateral resistance* means the resistance provided by the rail/crosstie structure against lateral displacement.

*Track longitudinal resistance* means the resistance provided by the rail anchors/rail fasteners and the ballast section to the rail/crosstie structure against longitudinal displacement.

*Train-induced forces* means the vertical, longitudinal, and lateral dynamic forces which are generated during train movement and which can contribute to the buckling potential of the rail.

*Unscheduled detour operation* means a short-term, unscheduled operation where a track owner has no more than 14 calendar days' notice that the operation is going to occur.

[74 FR 43002, Aug. 25, 2009, as amended at 74 FR 53889, Oct. 21, 2009; 75 FR 4705, Jan. 29, 2010; 79 FR 4258, Jan. 24, 2014]

## § 213.121 Rail joints.

- (a) Each rail joint, insulated joint, and compromise joint shall be of a structurally sound design and dimensions for the rail on which it is applied.
- (b) If a joint bar on Classes 3 through 5 track is cracked, broken, or because of wear allows excessive vertical movement of either rail when all bolts are tight, it shall be replaced.
- (c) If a joint bar is cracked or broken between the middle two bolt holes it shall be replaced.
- (d) In the case of conventional jointed track, each rail shall be bolted with at least two bolts at each joint in Classes 2 through 5 track, and with at least one bolt in Class 1 track.
- (e) In the case of continuous welded rail track, each rail shall be bolted with at least two bolts at each joint.
- (f) Each joint bar shall be held in position by track bolts tightened to allow the joint bar to firmly support the abutting rail ends and to allow longitudinal movement of the rail in the joint to accommodate expansion and contraction due to temperature variations. When no-slip, joint-to-rail contact exists by design, the requirements of this paragraph do not apply. Those locations when over 400 feet in length, are considered to be continuous welded rail track and shall meet all the requirements for continuous welded rail track prescribed in this part.
- (g) No rail shall have a bolt hole which is torch cut or burned in Classes 2 through 5 track. For Class 2 track, this paragraph (g) is applicable September 21, 1999.
- (h) No joint bar shall be reconfigured by torch cutting in Classes 3 through 5 track.

## § 213.122 Torch cut rail.

- (a) Except as a temporary repair in emergency situations no rail having a torch cut end shall be used in Classes 3 through 5 track. When a rail end is torch cut in emergency situations, train speed over that rail end shall not exceed the maximum allowable for Class 2 track. For existing torch cut rail ends in Classes 3 through 5 track the following shall apply—
  - (1) Within one year of September 21, 1998, all torch cut rail ends in Class 5 track shall be removed;
  - (2) Within two years of September 21, 1998, all torch cut rail ends in Class 4 track shall be removed; and
  - (3) Within one year of September 21, 1998, all torch cut rail ends in Class 3 track over which regularly scheduled passenger trains operate, shall be inventoried by the track owner.

- (b) Following the expiration of the time limits specified in paragraphs (a)(1), (2), and (3) of this section, any torch cut rail end not removed from Classes 4 and 5 track, or any torch cut rail end not inventoried in Class 3 track over which regularly scheduled passenger trains operate, shall be removed within 30 days of discovery. Train speed over that rail end shall not exceed the maximum allowable for Class 2 track until removed.

### § 213.123 Tie plates.

- (a) In Classes 3 through 5 track where timber crossties are in use there shall be tie plates under the running rails on at least eight of any 10 consecutive ties.
- (b) In Classes 3 through 5 track no metal object which causes a concentrated load by solely supporting a rail shall be allowed between the base of the rail and the bearing surface of the tie plate. This paragraph (b) is applicable September 21, 1999.)

### § 213.127 Rail fastening systems.

- (a) Track shall be fastened by a system of components that effectively maintains gage within the limits prescribed in § 213.53(b). Each component of each such system shall be evaluated to determine whether gage is effectively being maintained.
- (b) If rail anchors are applied to concrete crossties, the combination of the crossties, fasteners, and rail anchors must provide effective longitudinal restraint.
- (c) Where fastener placement impedes insulated joints from performing as intended, the fastener may be modified or removed, provided that the crosstie supports the rail.

[76 FR 18086, Apr. 1, 2011]

### § 213.133 Turnouts and track crossings generally.

- (a) In turnouts and track crossings, the fastenings shall be intact and maintained so as to keep the components securely in place. Also, each switch, frog, and guard rail shall be kept free of obstructions that may interfere with the passage of wheels.
- (b) Classes 3 through 5 track shall be equipped with rail anchoring through and on each side of track crossings and turnouts, to restrain rail movement affecting the position of switch points and frogs. For Class 3 track, this paragraph (b) is applicable September 21, 1999.)
- (c) Each flangeway at turnouts and track crossings shall be at least 1<sup>1</sup>/<sub>2</sub> inches wide.

### § 213.135 Switches.

- (a) Each stock rail must be securely seated in switch plates, but care shall be used to avoid canting the rail by overtightening the rail braces.
- (b) Each switch point shall fit its stock rail properly, with the switch stand in either of its closed positions to allow wheels to pass the switch point. Lateral and vertical movement of a stock rail in the switch plates or of a switch plate on a tie shall not adversely affect the fit of the switch point to the stock rail. Broken or cracked switch point rails will be subject to the requirements of § 213.113, except that where remedial actions C, D, or E require the use of joint bars, and joint bars cannot be placed due to the physical configuration of the switch, remedial action B will govern, taking into account any added safety provided by the presence of reinforcing bars on the switch points.

- (c) Each switch shall be maintained so that the outer edge of the wheel tread cannot contact the gage side of the stock rail.
- (d) The heel of each switch rail shall be secure and the bolts in each heel shall be kept tight.
- (e) Each switch stand and connecting rod shall be securely fastened and operable without excessive lost motion.
- (f) Each throw lever shall be maintained so that it cannot be operated with the lock or keeper in place.
- (g) Each switch position indicator shall be clearly visible at all times.
- (h) Unusually chipped or worn switch points shall be repaired or replaced. Metal flow shall be removed to insure proper closure.
- (i) Tongue & Plain Mate switches, which by design exceed Class 1 and excepted track maximum gage limits, are permitted in Class 1 and excepted track.

### § 213.137 Frogs.

- (a) Except as provided in paragraph (e) of this section, the flangeway depth measured from a plane across the wheel-bearing area of a frog on Class 1 track shall not be less than  $1\frac{3}{8}$  inches, or less than  $1\frac{1}{2}$  inches on Classes 2 through 5 track.
- (b) If a frog point is chipped, broken, or worn more than five-eighths inch down and 6 inches back, operating speed over the frog shall not be more than 10 m.p.h.
- (c) If the tread portion of a frog casting is worn down more than three-eighths inch below the original contour, operating speed over that frog shall not be more than 10 m.p.h.
- (d) Where frogs are designed as flange-bearing, flangeway depth may be less than that shown for Class 1 if operated at Class 1 speeds.
- (e) The flange depth requirements in paragraph (a) do not apply to a frog designed as a flange-bearing frog (FBF) used in a crossing diamond in Classes 2 through 5 track, provided that the crossing angle is greater than 20 degrees unless movable guard rails are used.

*[63 FR 34029, June 22, 1998, as amended at 85 FR 63388, Oct. 7, 2020]*

### § 213.139 Spring rail frogs.

- (a) The outer edge of a wheel tread shall not contact the gage side of a spring wing rail.
- (b) The toe of each wing rail shall be solidly tamped and fully and tightly bolted.
- (c) Each frog with a bolt hole defect or head-web separation shall be replaced.
- (d) Each spring shall have compression sufficient to hold the wing rail against the point rail.
- (e) The clearance between the holddown housing and the horn shall not be more than one-fourth of an inch.

### § 213.141 Self-guarded frogs.

- (a) The raised guard on a self-guarded frog shall not be worn more than three-eighths of an inch.
- (b) If repairs are made to a self-guarded frog without removing it from service, the guarding face shall be restored before rebuilding the point.



## § 213.143 Frog guard rails and guard faces; gage.

- (a) The guard check and guard face gages in frogs shall be within the following limits—

**TABLE 1 TO § 213.143(a)**

| Class of track      | Guard check gage                                                                                                                                                                                               | Guard face gage                                                                                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                     | The distance between the gage line of a frog to the guard line <sup>1</sup> of its guard rail or guarding face, measured across the track at right angles to the gage line, <sup>2</sup> may not be less than— | The distance between guard lines, <sup>1</sup> measured across the track at right angles to the gage line, <sup>2</sup> may not be more than— |
| Class 1 track       | 4'6 <sup>1</sup> / <sub>8</sub> "                                                                                                                                                                              | 4'5 <sup>1</sup> / <sub>4</sub> "                                                                                                             |
| Class 2 track       | 4'6 <sup>1</sup> / <sub>4</sub> "                                                                                                                                                                              | 4'5 <sup>1</sup> / <sub>8</sub> "                                                                                                             |
| Class 3 and 4 track | 4'6 <sup>3</sup> / <sub>8</sub> "                                                                                                                                                                              | 4'5 <sup>1</sup> / <sub>8</sub> "                                                                                                             |
| Class 5 track       | <sup>3</sup> 4'6 <sup>1</sup> / <sub>2</sub> "                                                                                                                                                                 | 4'5"                                                                                                                                          |

<sup>1</sup> A line along that side of the flangeway which is nearer to the center of the track and at the same elevation as the gage line.

<sup>2</sup> A line five-eighths of an inch below the top of the center line of the head of the running rail, or corresponding location of the tread portion of the track structure.

<sup>3</sup> See paragraph (b) of this section.

- (b) For any heavy-point frog (HPF) on Class 5 track, the guard check gage may be less than 4'6<sup>1</sup>/<sub>2</sub>" but not be less than 4'6<sup>3</sup>/<sub>8</sub>", provided that:
- (1) Each HPF and guard rails on both rails through the turnout are equipped with at least three serviceable through-gage plates with elastic rail fasteners and guard rail braces that permit adjustment of the guard check gage without removing spikes or other fasteners from the crossties; and
  - (2) Each HPF bears an identifying mark applied by either the track owner, railroad, or the frog manufacturer that identifies the frog as an HPF.

[85 FR 63388, Oct. 7, 2020]

## Subpart E—Track Appliances and Track-Related Devices

### § 213.201 Scope.

This subpart prescribes minimum requirements for certain track appliances and track-related devices.

### § 213.205 Derails.

- (a) Each derail shall be clearly visible.
- (b) When in a locked position, a derail shall be free of lost motion which would prevent it from performing its intended function.
- (c) Each derail shall be maintained to function as intended.
- (d) Each derail shall be properly installed for the rail to which it is applied. (This paragraph (d) is applicable September 21, 1999.)

## Subpart F—Inspection

### § 213.231 Scope.

This subpart prescribes requirements for the frequency and manner of inspecting track to detect deviations from the standards prescribed in this part.

### § 213.233 Visual track inspections.

- (a) All track shall be inspected in accordance with the schedule prescribed in paragraph (c) of this section by a person designated under § 213.7.
- (b) Each inspection shall be made on foot or by traversing the track in a vehicle at a speed that allows the person making the inspection to visually inspect the track structure for compliance with this part. However, mechanical, electrical, and other track inspection devices may be used to supplement visual inspection. If a vehicle is used for visual inspection, the speed of the vehicle may not be more than 5 m.p.h. when traversing track crossings and turnouts; otherwise, the inspection vehicle speed shall be at the sole discretion of the inspector, based on track conditions and inspection requirements. When traversing the track in a vehicle, the inspection will be subject to the following conditions—
  - (1) One inspector in a vehicle may inspect up to two tracks at one time provided that the inspector's visibility remains unobstructed by any cause and that the second track is not centered more than 30 feet from the track the inspector traverses;
  - (2) Two inspectors in one vehicle may inspect up to four tracks at a time provided that the inspectors' visibility remains unobstructed by any cause and that each track being inspected is centered within 39 feet from the track the inspectors traverse;
  - (3) Each main track must be traversed by the vehicle or inspected on foot at least once every two weeks, and each siding must be traversed by the vehicle or inspected on foot at least once every month; and
  - (4) Track inspection records shall indicate which track(s) are traversed by the vehicle or inspected on foot as outlined in paragraph (b)(3) of this section.

(c) Each track inspection shall be made in accordance with the following schedule—

| Class of track                              | Type of track                     | Required frequency                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excepted track, and Class 1, 2, and 3 track | Main track and sidings            | Weekly <sup>1</sup> with at least 3 calendar days' interval between inspections, or before use, if the track is used less than once a week, or twice weekly with at least 1 calendar day interval between inspections, if the track carries passenger trains <sup>2</sup> or more than 10 million gross tons of traffic during the preceding calendar year. |
| Excepted track and Class 1, 2, and 3 track  | Other than main track and sidings | Monthly with at least 20 calendar days interval between inspections.                                                                                                                                                                                                                                                                                        |
| Class 4 and 5 track                         |                                   | Twice weekly with at least 1 calendar day interval between inspections.                                                                                                                                                                                                                                                                                     |

<sup>1</sup> An inspection week is defined as a seven (7) day period beginning on Sunday and ending on Saturday.

<sup>2</sup> "Twice weekly" inspection requirement for track carrying regularly scheduled passenger trains does not apply where passenger train service consists solely of tourist, scenic, historic, or excursion operations as defined in 49 CFR 238.5 and the following conditions are met for an inspection week: (1) No passenger service is operated during the inspection week, or (2) if passenger service is operated during the inspection week: (i) The passenger service is operated only on a weekend or a 3-day extended weekend (weekend plus a contiguous Monday or Friday), and (ii) an inspection is conducted no more than 1 calendar day before a weekend or 3-day extended weekend on which passenger service is to be operated.

(d) If the § 213.7 qualified person making the inspection finds a deviation from the requirements of this part, the inspector shall immediately initiate remedial action. Any subsequent movements to facilitate repairs on track that is out of service must be authorized by a § 213.7 qualified person.

Note to § 213.233: Except as provided in paragraph (b) of this section, no part of this section will in any way be construed to limit the inspector's discretion as it involves inspection speed and sight distance.

[63 FR 34029, June 22, 1998, as amended at 85 FR 63389, Oct. 7, 2020]

## § 213.234 Automated inspection of track constructed with concrete crossties.

- (a) **General.** Except for track described in paragraph (c) of this section, the provisions in this section are applicable on and after July 1, 2012. In addition to the track inspection required under § 213.233, for Class 3 main track constructed with concrete crossties over which regularly scheduled passenger service trains operate, and for Class 4 and 5 main track constructed with concrete crossties, automated inspection technology shall be used as indicated in paragraph (b) of this section, as a supplement to visual inspection, by Class I railroads (including Amtrak), Class II railroads, other intercity passenger railroads, and commuter railroads or small governmental jurisdictions that serve populations greater than 50,000. Automated inspection shall identify and report exceptions to conditions described in § 213.109(d)(4).
- (b) **Frequency of automated inspections.** Automated inspections shall be conducted at the following frequencies:
- (1) If annual tonnage on Class 4 and 5 main track and Class 3 main track with regularly scheduled passenger service, exceeds 40 million gross tons (mgt) annually, at least twice each calendar year, with no less than 160 days between inspections.
  - (2) If annual tonnage on Class 4 and 5 main track and Class 3 main track with regularly scheduled passenger service is equal to or less than 40 mgt annually, at least once each calendar year.
  - (3) On Class 3, 4, and 5 main track with exclusively passenger service, either an automated inspection or walking inspection must be conducted once per calendar year.
  - (4) Track not inspected in accordance with paragraph (b)(1) or (b)(2) of this section because of train operation interruption shall be reinspected within 45 days of the resumption of train operations by a walking or automated inspection. If this inspection is conducted as a walking inspection, the next inspection shall be an automated inspection as prescribed in this paragraph.
- (c) **Nonapplication.** Sections of tangent track 600 feet or less constructed of concrete crossties, including, but not limited to, isolated track segments, experimental or test track segments, highway-rail crossings, and wayside detectors, are excluded from the requirements of this section.
- (d) **Performance standard for automated inspection measurement system.** The automated inspection measurement system must be capable of indicating and processing rail seat deterioration requirements that specify the following:
- (1) An accuracy, to within  $\frac{1}{8}$  of an inch;
  - (2) A distance-based sampling interval, which shall not exceed five feet; and
  - (3) Calibration procedures and parameters assigned to the system, which assure that indicated and recorded values accurately represent rail seat deterioration.
- (e) **Exception reports to be produced by system; duty to field-verify exceptions.** The automated inspection measurement system shall produce an exception report containing a systematic listing of all exceptions to § 213.109(d)(4), identified so that an appropriate person(s) designated as fully qualified under § 213.7 can field-verify each exception.
- (1) Exception reports must be provided to or be made available to all persons designated as fully qualified under § 213.7 and whose territories are subject to the requirements of § 213.234.

- (2) Each exception must be located and field-verified no later than 48 hours after the automated inspection.
- (3) All field-verified exceptions are subject to all the requirements of this part.
- (4) Exception reports must note areas identified between  $\frac{3}{8}$  of an inch and  $\frac{1}{2}$  of an inch as an "alert."
- (f) **Recordkeeping requirements.** The track owner shall maintain and make available to FRA a record of the inspection data and the exception record for the track inspected in accordance with this paragraph for a minimum of two years. The exception reports must include the following:
  - (1) Date and location of limits of the inspection;
  - (2) Type and location of each exception;
  - (3) Results of field verification; and
  - (4) Remedial action if required.
- (g) **Procedures for integrity of data.** The track owner shall institute the necessary procedures for maintaining the integrity of the data collected by the measurement system. At a minimum, the track owner shall do the following:
  - (1) Maintain and make available to FRA documented calibration procedures of the measurement system that, at a minimum, specify an instrument verification procedure that ensures correlation between measurements made on the ground and those recorded by the instrumentation; and
  - (2) Maintain each instrument used for determining compliance with this section such that it accurately provides an indication of the depth of rail seat deterioration in accordance with paragraph (d)(1) of this section.
- (h) **Training.** The track owner shall provide annual training in handling rail seat deterioration exceptions to all persons designated as fully qualified under § 213.7 and whose territories are subject to the requirements of § 213.234. At a minimum, the training shall address the following:
  - (1) Interpretation and handling of the exception reports generated by the automated inspection measurement system;
  - (2) Locating and verifying exceptions in the field and required remedial action; and
  - (3) Recordkeeping requirements.

[76 FR 18086, Apr. 1, 2011, as amended at 76 FR 55825, Sept. 9, 2011]

### **§ 213.235 Inspection of switches, track crossings, and lift rail assemblies or other transition devices on moveable bridges.**

- (a) Except as provided in paragraph (c) of this section, each switch, turnout, track crossing, and moveable bridge lift rail assembly or other transition device shall be inspected on foot at least monthly.
- (b) Each switch in Classes 3 through 5 track that is held in position only by the operating mechanism and one connecting rod shall be operated to all of its positions during one inspection in every 3 month period.
- (c) In the case of track that is used less than once a month, each switch, turnout, track crossing, and moveable bridge lift rail assembly or other transition device shall be inspected on foot before it is used.

## § 213.237 Inspection of rail.

- (a) In addition to the inspections required by § 213.233, each track owner shall conduct internal rail inspections sufficient to maintain service failure rates per rail inspection segment in accordance with this paragraph (a) for a 12-month period, as determined by the track owner and calculated within 45 days of the end of the period. These rates shall not include service failures that occur in rail that has been replaced through rail relay since the time of the service failure. Rail used to repair a service failure defect is not considered relayed rail. The service failure rates shall not exceed—
  - (1) 0.1 service failure per year per mile of track for all Class 4 and 5 track;
  - (2) 0.09 service failure per year per mile of track for all Class 3, 4, and 5 track that carries regularly-scheduled passenger trains or is a hazardous materials route; and
  - (3) 0.08 service failure per year per mile of track for all Class 3, 4, and 5 track that carries regularly-scheduled passenger trains and is a hazardous materials route.
- (b) Each rail inspection segment shall be designated by the track owner no later than March 25, 2014 for track that is Class 4 or 5 track, or Class 3 track that carries regularly-scheduled passenger trains or is a hazardous materials route and is used to determine the milepost limits for the individual rail inspection frequency.
  - (1) To change the designation of a rail inspection segment or to establish a new segment pursuant to this section, a track owner must submit a detailed request to the FRA Associate Administrator for Railroad Safety/Chief Safety Officer (Associate Administrator). Within 30 days of receipt of the submission, FRA will review the request. FRA will approve, disapprove, or conditionally approve the submitted request, and will provide written notice of its determination.
  - (2) The track owner's existing designation shall remain in effect until the track owner's new designation is approved or conditionally approved by FRA.
  - (3) The track owner shall, upon receipt of FRA's approval or conditional approval, establish the designation's effective date. The track owner shall advise in writing FRA and all affected railroad employees of the effective date.
- (c) Internal rail inspections on Class 4 and 5 track, or Class 3 track with regularly-scheduled passenger trains or that is a hazardous materials route, shall not exceed a time interval of 370 days between inspections or a tonnage interval of 30 million gross tons (mgt) between inspections, whichever is shorter. Internal rail inspections on Class 3 track that is without regularly-scheduled passenger trains and not a hazardous materials route must be inspected at least once each calendar year, with no more than 18 months between inspections, or at least once every 30 mgt, whichever interval is longer, but in no case may inspections be more than 5 years apart.
  - (1) Any rail used as a replacement plug rail in track that is required to be tested in accordance with this section must have been tested for internal rail flaws.
  - (2) The track owner must verify that any plug rail installed after March 25, 2014 has not accumulated more than a total of 30 mgt in previous and new locations since its last internal rail flaw test, before the next test on the rail required by this section is performed.
  - (3) If plug rail not in compliance with this paragraph (c) is in use after March 25, 2014, trains over that rail must not exceed Class 2 speeds until the rail is tested in accordance with this section.

- (d) If the service failure rate target identified in paragraph (a) of this section is not achieved, the track owner must inform FRA of this fact within 45 days of the end of the defined 12-month period in which the performance target is exceeded. In addition, the track owner may provide to FRA an explanation as to why the performance target was not achieved and provide a remedial action plan.
  - (1) If the performance target rate is not met for two consecutive years, then for the area where the greatest number of service failures is occurring, either:
    - (i) The inspection tonnage interval between tests must be reduced to 10 mgt; or
    - (ii) The class of track must be reduced to Class 2 until the target service failure rate is achieved.
  - (2) In cases where a single service failure would cause the rate to exceed the applicable service failure rate as designated in paragraph (a) of this section, the service failure rate will be considered to comply with paragraph (a) of this section unless a second such failure occurs within a designated 12-month period. For the purposes of this paragraph (d)(2), a period begins no earlier than January 24, 2014.
- (e) Each defective rail shall be marked with a highly visible marking on both sides of the web and base except that, where a side or sides of the web and base are inaccessible because of permanent features, the highly visible marking may be placed on or next to the head of the rail.
- (f) Inspection equipment shall be capable of detecting defects between joint bars, in the area enclosed by joint bars.
- (g) If the person assigned to operate the rail defect detection equipment (i.e., the qualified operator) determines that a valid search for internal defects could not be made over a particular length of track, that particular length of track may not be considered as internally inspected under paragraphs (a) and (c) of this section.
- (h) If a valid search for internal defects could not be conducted, the track owner shall, before expiration of the time or tonnage limits in paragraph (a) or (c) of this section—
  - (1) Conduct a valid search for internal defects;
  - (2) Reduce operating speed to a maximum of 25 m.p.h. until such time as a valid search can be made; or
  - (3) Replace the rail that had not been inspected.
- (i) The person assigned to operate the rail defect detection equipment must be a qualified operator as defined in § 213.238 and have demonstrated proficiency in the rail flaw detection process for each type of equipment the operator is assigned.
- (j) As used in this section—
  - (1) **Hazardous materials route** means track over which a minimum of 10,000 car loads or intermodal portable tank car loads of hazardous materials as defined in 49 CFR 171.8 travel over a period of one calendar year; or track over which a minimum of 4,000 car loads or intermodal portable tank car loads of the hazardous materials specified in 49 CFR 172.820 travel, in a period of one calendar year.
  - (2) **Plug rail** means a length of rail that has been removed from one track location and stored for future use as a replacement rail at another location.

- (3) **Service failure** means a broken rail occurrence, the cause of which is determined to be a compound fissure, transverse fissure, detail fracture, or vertical split head.
- (4) **Valid search** means a continuous inspection for internal rail defects where the equipment performs as intended and equipment responses are interpreted by a qualified operator as defined in § 213.238.

[79 FR 4258, Jan. 24, 2014]

### § 213.238 Qualified operator.

- (a) Each provider of rail flaw detection shall have a documented training program in place and shall identify the types of rail flaw detection equipment for which each equipment operator it employs has received training and is qualified. A provider of rail flaw detection may be the track owner. A track owner shall not utilize a provider of rail flaw detection that fails to comply with the requirements of this paragraph.
- (b) A qualified operator shall be trained and have written authorization from his or her employer to:
  - (1) Conduct a valid search for internal rail defects utilizing the specific type(s) of equipment for which he or she is authorized and qualified to operate;
  - (2) Determine that such equipment is performing as intended;
  - (3) Interpret equipment responses and institute appropriate action in accordance with the employer's procedures and instructions; and
  - (4) Determine that each valid search for an internal rail defect is continuous throughout the area inspected and has not been compromised due to environmental contamination, rail conditions, or equipment malfunction.
- (c) To be qualified, the operator must have received training in accordance with the documented training program and a minimum of 160 hours of rail flaw detection experience under direct supervision of a qualified operator or rail flaw detection equipment manufacturer's representative, or some combination of both. The operator must demonstrate proficiency in the rail defect detection process, including the equipment to be utilized, prior to initial qualification and authorization by the employer for each type of equipment.
- (d) Each employer shall reevaluate the qualifications of, and administer any necessary recurrent training for, the operator as determined by and in accordance with the employer's documented program. The reevaluation process shall require that the employee successfully complete a recorded examination and demonstrate proficiency to the employer on the specific equipment type(s) to be operated. Proficiency may be determined by a periodic review of test data submitted by the operator.
- (e) Each employer of a qualified operator shall maintain written or electronic records of each qualification in effect. Each record shall include the name of the employee, the equipment to which the qualification applies, date of qualification, and date of the most recent reevaluation, if any.
- (f) Any employee who has demonstrated proficiency in the operation of rail flaw detection equipment prior to January 24, 2014, is deemed a qualified operator, regardless of the previous training program under which the employee was qualified. Such an operator shall be subject to paragraph (d) of this section.



- (g) Records concerning the qualification of operators, including copies of equipment-specific training programs and materials, recorded examinations, demonstrated proficiency records, and authorization records, shall be kept at a location designated by the employer and available for inspection and copying by FRA during regular business hours.

[79 FR 4259, Jan. 24, 2014]

### § 213.239 Special inspections.

In the event of fire, flood, severe storm, or other occurrence which might have damaged track structure, a special inspection shall be made of the track involved as soon as possible after the occurrence and, if possible, before the operation of any train over that track.

### § 213.240 Continuous rail testing.

- (a) Track owners may elect to use continuous rail testing to satisfy the requirements for conducting internal rail inspections under § 213.237 or § 213.339. When a track owner utilizes the continuous rail test inspection process under the requirements of this section, the track owner is exempt from the requirements of § 213.113(b); all other requirements of § 213.113 apply.
- (b) Track owners shall adopt the necessary procedures for conducting continuous testing. At a minimum, the procedures must conform to the requirements of this section and ensure the following:
  - (1) Test data is timely and accurately transmitted and analyzed;
  - (2) Suspect locations are accurately identified for field verification;
  - (3) Suspect locations are categorized and prioritized according to their potential severity;
  - (4) Suspect locations are accurately field-verified; and
  - (5) Suspect locations will be designated following field verification.
- (c) The track owner must designate and record the type of rail test (continuous or stop-and-verify) to be conducted prior to commencing the test over a track segment and make those records available to FRA upon request during regular business hours following reasonable notice. If the type of rail test changes following commencement of the test, the change must be documented and include the time the test was started and when it was changed, and the milepost where the test started and where it was changed. If the track owner intends to conduct a continuous test, the track owner must designate and record whether the test is being conducted to satisfy the requirements for an internal rail inspection under § 213.237 or § 213.339. This documentation must be provided to FRA upon request during regular business hours following reasonable notice.
- (d)
  - (1) Continuous rail test inspection vehicle operators must be qualified under § 213.238, with the exception of § 213.238(b)(3).
  - (2) Internal rail inspection data collected during continuous rail tests must be reviewed and interpreted by a person qualified to interpret the equipment responses. Each employer of a person qualified to interpret equipment responses shall maintain written or electronic records of each qualification in effect, including the name of the employee, the equipment to which the qualification applies, the date of qualification, and the date of the most recent reevaluation of the qualification, if any. Records

concerning these qualifications, including copies of training programs, training materials, and recorded examinations shall be kept at a location designated by the employer and available for inspection and copying by FRA during regular business hours, following reasonable notice.

- (3) All suspect locations must be field-verified by a person qualified under § 213.238.
- (e) At a minimum, the continuous rail test process must produce a report containing a systematic listing of all suspected locations that may contain any of the defects listed in the table in § 213.113(c), identified so that a person qualified under § 213.238 can accurately locate and field-verify each suspected defect.
  - (1) Except as provided in paragraph (e)(6) of this section, and subject to the requirements of paragraphs (e)(2) and (3) of this section, if the continuous rail test inspection vehicle indicates a suspect location, field verification must be conducted within 84 hours of the indication of the suspect location.
  - (2) Except as provided in paragraph (e)(6) of this section, and subject to the requirements of paragraph (e)(3) of this section, if the continuous rail test inspection vehicle indicates a suspect location containing a suspected defect that, if verified, requires remedial action A, A2, or B identified in the table contained in § 213.113(c), the track owner must field-verify the suspect location no more than 36 hours from indication of the suspect location.
  - (3) If the continuous rail test inspection vehicle indicates a broken rail with rail separation, the track owner must have procedures to ensure that adequate protection is immediately implemented.
  - (4) A suspect location is not considered a defect under § 213.113(c) until it has been field-verified by a person qualified under § 213.238. After the suspect location is field-verified and determined to be a defect, the track owner must immediately perform all required remedial actions prescribed in § 213.113(a).
  - (5) Any suspected location not field-verified within the time required under paragraphs (e)(1) and (2) of this section must be protected by applying the most restrictive remedial action under § 213.113(c) for the suspected type and size of the suspected defect. The remedial action must be applied over a sufficient segment of track to assure coverage of the suspected defect location until field-verified.
  - (6) A continuous rail test that is not conducted to satisfy the requirements for an internal rail inspection under § 213.237 or § 213.339, and has been properly designated and recorded by the track owner under paragraph (c) of this section, is exempt from the requirements of paragraphs (e)(1), (2), and (5) of this section.
- (f) Each suspect location must be recorded with repeatable accuracy that allows for the location to be accurately located for subsequent verification and, as necessary, remedial action.
- (g) Within 45 days following the end of each calendar year, each track owner utilizing continuous rail testing must provide the FRA Associate Administrator for Railroad Safety/Chief Safety Officer with an annual report, in a reasonably usable format, or its native electronic format, containing at least the following information for each track segment requiring internal rail inspection under § 213.237 or § 213.339:
  - (1) The track owner's name;
  - (2) The railroad division and subdivision;
  - (3) The segment identifier, milepost limits, and length of each segment;
  - (4) The track number;

- (5) The class of track;
- (6) The annual million gross tons over the track;
- (7) The total number of stop-and-verify rail tests and the total number of continuous rail tests over each track segment;
- (8) The total number of defects identified over each track segment; and
- (9) The total number of service failures on each track segment.

[85 FR 63389, Oct. 7, 2020]

### § 213.241 Inspection records.

- (a) Each owner of track to which this part applies shall keep a record of each inspection required to be performed on that track under this subpart.
- (b) Each record of an inspection under §§ 213.4, 213.119, 213.233, and 213.235 shall be prepared on the day the inspection is made and signed or otherwise certified by the person making the inspection. Records shall specify the author of the record, the type of track inspected, date and location of inspection, location and nature of any deviation from the requirements of this part, and the remedial action taken by the person making the inspection. The track owner shall designate the location(s) where each original record shall be maintained for at least one year after the inspection covered by the record. The track owner shall also designate one location, within 100 miles of each State in which it conducts operations, where copies of records that apply to those operations are maintained or can be viewed following 10 days' notice by the Federal Railroad Administration.
- (c) Records of internal rail inspections required by § 213.237 shall specify the—
  - (1) Date of inspection;
  - (2) Track inspected, including beginning and end points;
  - (3) Location and type of defects found under § 213.113;
  - (4) Size of defects found under § 213.113, if not removed prior to the next train movement;
  - (5) Initial remedial action taken and the date thereof; and
  - (6) Location of any track not tested pursuant to § 213.237(g).
- (d) The track owner shall retain a rail inspection record under paragraph (c) of this section for at least two years after the inspection and for one year after initial remedial action is taken.
- (e) The track owner shall maintain records sufficient to demonstrate the means by which it computes the service failure rate on all track segments subject to the requirements of § 213.237(a) for the purpose of determining compliance with the applicable service failure rate target.
- (f) Records of continuous rail testing under § 213.240 shall—
  - (1) Include all information required under § 213.240(e);
  - (2) State whether the test is being conducted to satisfy the requirements for an internal rail inspection under § 213.237;

- (3) List the date(s) and time(s) of the continuous rail test data collection, including the date and time of the start and end of the test run, and the date and time each suspect location was identified and field-verified;
- (4) Include the determination made after field verification of each suspect location, including the:
  - (i) Location and type of defect found;
  - (ii) Size of defect; and
  - (iii) Initial remedial action taken, if required, and the date thereof; and
- (5) Be retained for at least two years after the inspection and for at least one year after initial remedial action is taken, whichever is later.
- (g) Track owners that elect to utilize continuous rail testing under § 213.240 shall maintain records of all continuous rail testing operations sufficient for monitoring and determining compliance with all applicable regulations and shall make those records available to FRA during regular business hours following reasonable notice.
- (h) Track inspection records shall be kept available to persons who performed the inspections and to persons performing subsequent inspections of the track segment.
- (i) Each track owner required to keep inspection records under this section shall make those records available for inspection and copying by FRA upon request during regular business hours following reasonable notice.
- (j) For purposes of complying with the requirements of this section, a track owner may create, retain, transmit, store, and retrieve records by electronic means provided that—
  - (1) The system used to generate the electronic record meets all requirements and contains the information required under this subpart;
  - (2) The track owner monitors its electronic records database to ensure record accuracy;
  - (3) The electronic system is designed to uniquely identify the author of the record. No two persons shall have the same electronic identity;
  - (4) The electronic system ensures that each record cannot be modified in any way, or replaced, once the record is completed;
  - (5) The electronic storage of each record shall be initiated by the person making the inspection within 72 hours following the completion of that inspection; and
  - (6) Any amendment to a record shall be electronically stored apart from the record which it amends. Each amendment to a record shall be uniquely identified as to the person making the amendment.

[63 FR 34029, June 22, 1998, as amended at 70 FR 66298, Nov. 2, 2005; 79 FR 4259, Jan. 24, 2014; 85 FR 63390, Oct. 7, 2020]

## Subpart G—Train Operations at Track Classes 6 and Higher

### § 213.301 Scope of subpart.

This subpart applies to all track used for the operation of trains at a speed greater than 90 m.p.h. for passenger equipment and greater than 80 m.p.h. for freight equipment.

### § 213.303 Responsibility for compliance.

- (a) Any owner of track to which this subpart applies who knows or has notice that the track does not comply with the requirements of this subpart, shall—
  - (1) Bring the track into compliance; or
  - (2) Halt operations over that track.
- (b) If an owner of track to which this subpart applies assigns responsibility for the track to another person (by lease or otherwise), notification of the assignment shall be provided to FRA's Track and Structures Division at least 30 days in advance of the assignment. The notification may be made by any party to that assignment, but shall be in writing and include the following—
  - (1) The name and address of the track owner;
  - (2) The name and address of the person to whom responsibility is assigned (assignee);
  - (3) A statement of the exact relationship between the track owner and the assignee;
  - (4) A precise identification of the track;
  - (5) A statement as to the competence and ability of the assignee to carry out the duties of the track owner under this subpart;
  - (6) A statement signed by the assignee acknowledging the assignment to that person of responsibility for purposes of compliance with this subpart.
- (c) The Administrator may hold the track owner or the assignee or both responsible for compliance with this subpart and subject to the penalties under § 213.15.
- (d) When any person, including a contractor for a railroad or track owner, performs any function required by this part, that person is required to perform that function in accordance with this part.

*[63 FR 34029, June 22, 1998, as amended at 90 FR 28136, July 1, 2025]*

### § 213.305 Designation of qualified individuals; general qualifications.

Each track owner to which this subpart applies shall designate qualified individuals responsible for the maintenance and inspection of track in compliance with the safety requirements prescribed in this subpart. Each individual, including a contractor or an employee of a contractor who is not a railroad employee, designated to:

- (a) Supervise restorations and renewals of track shall meet the following minimum requirements:
  - (1) At least;
    - (i) Five years of responsible supervisory experience in railroad track maintenance in track Class 4 or higher and the successful completion of a course offered by the employer or by a college level engineering program, supplemented by special on the job training emphasizing the techniques to be employed in the supervision, restoration, and renewal of high speed track; or

- (ii) A combination of at least one year of responsible supervisory experience in track maintenance in Class 4 or higher and the successful completion of a minimum of 80 hours of specialized training in the maintenance of high speed track provided by the employer or by a college level engineering program, supplemented by special on the job training provided by the employer with emphasis on the maintenance of high speed track; or
- (iii) A combination of at least two years of experience in track maintenance in track Class 4 or higher and the successful completion of a minimum of 120 hours of specialized training in the maintenance of high speed track provided by the employer or by a college level engineering program supplemented by special on the job training provided by the employer with emphasis on the maintenance of high speed track.

(2) Demonstrate to the track owner that the individual:

- (i) Knows and understands the requirements of this subpart that apply to the restoration and renewal of the track for which he or she is responsible;
- (ii) Can detect deviations from those requirements; and
- (iii) Can prescribe appropriate remedial action to correct or safely compensate for those deviations; and

(3) Be authorized by the track owner to prescribe remedial actions to correct or safely compensate for deviations from the requirements of this subpart and successfully completed a recorded examination on this subpart as part of the qualification process.

(b) Inspect track for defects shall meet the following minimum qualifications:

(1) At least:

- (i) Five years of responsible experience inspecting track in Class 4 or above and the successful completion of a course offered by the employer or by a college level engineering program, supplemented by special on the job training emphasizing the techniques to be employed in the inspection of high speed track; or
- (ii) A combination of at least one year of responsible experience in track inspection in Class 4 or above and the successful completion of a minimum of 80 hours of specialized training in the inspection of high speed track provided by the employer or by a college level engineering program, supplemented by special on the job training provided by the employer with emphasis on the inspection of high speed track; or
- (iii) A combination of at least two years of experience in track maintenance in Class 4 or above and the successful completion of a minimum of 120 hours of specialized training in the inspection of high speed track provided by the employer or from a college level engineering program, supplemented by special on the job training provided by the employer with emphasis on the inspection of high speed track.

(2) Demonstrate to the track owner that the individual:

- (i) Knows and understands the requirements of this subpart that apply to the inspection of the track for which he or she is responsible.
- (ii) Can detect deviations from those requirements; and

- (iii) Can prescribe appropriate remedial action to correct or safely compensate for those deviations; and
- (3) Be authorized by the track owner to prescribe remedial actions to correct or safely compensate for deviations from the requirements in this subpart and successfully completed a recorded examination on this subpart as part of the qualification process.
- (c) Individuals designated under paragraphs (a) or (b) of this section that inspect continuous welded rail (CWR) track or supervise the installation, adjustment, and maintenance of CWR in accordance with the written procedures established by the track owner shall have:
  - (1) Current qualifications under either paragraph (a) or (b) of this section;
  - (2) Successfully completed a training course of at least eight hours duration specifically developed for the application of written CWR procedures issued by the track owner; and
  - (3) Demonstrated to the track owner that the individual:
    - (i) Knows and understands the requirements of those written CWR procedures;
    - (ii) Can detect deviations from those requirements; and
    - (iii) Can prescribe appropriate remedial action to correct or safely compensate for those deviations; and
  - (4) Authorization from the track owner to prescribe remedial actions to correct or safely compensate for deviations from the requirements in those procedures and successfully completed a recorded examination on those procedures as part of the qualification process. The recorded examination may be written, or it may be a computer file with the results of an interactive training course.
- (d) Persons not fully qualified to supervise certain renewals and inspect track as outlined in paragraphs (a), (b) and (c) of this section, but with at least one year of maintenance of way or signal experience, may pass trains over broken rails and pull aparts provided that—
  - (1) The track owner determines the person to be qualified and, as part of doing so, trains, examines, and re-examines the person periodically within two years after each prior examination on the following topics as they relate to the safe passage of trains over broken rails or pull aparts: rail defect identification, crosstie condition, track surface and alignment, gage restraint, rail end mismatch, joint bars, and maximum distance between rail ends over which trains may be allowed to pass. The sole purpose of the examination is to ascertain the person's ability to effectively apply these requirements and the examination may not be used to disqualify the person from other duties. A minimum of four hours training is adequate for initial training;
  - (2) The person deems it safe, and train speeds are limited to a maximum of 10 m.p.h. over the broken rail or pull apart;
  - (3) The person shall watch all movements over the broken rail or pull apart and be prepared to stop the train if necessary; and
  - (4) Person(s) fully qualified under § 213.305 of this subpart are notified and dispatched to the location as soon as practicable for the purpose of authorizing movements and effectuating temporary or permanent repairs.
- (e) With respect to designations under paragraphs (a), (b), (c) and (d) of this section, each track owner shall maintain records of:

- (1) Each designation in effect;
- (2) The date each designation was made; and
- (3) The basis for each designation, including but not limited to:

(i) The exact nature of any training courses attended and the dates thereof; and

(ii) The manner in which the track owner has determined a successful completion of that training course, including test scores or other qualifying results.

(f) Each track owner shall keep these designation records readily available for inspection or copying by the Federal Railroad Administration during regular business hours, following reasonable notice.
- [63 FR 34029, June 22, 1998; 63 FR 45959, Aug. 28, 1998, as amended at 78 FR 16103, Mar. 13, 2013; 85 FR 63391, Oct. 7, 2020]
- § 213.307 Classes of track: operating speed limits.
- (a) Except as provided in paragraph (b) of this section and as otherwise provided in this subpart G, the following maximum allowable speeds apply:
- | Over track that meets all of the requirements prescribed in this subpart for— | The maximum allowable operating speed for trains is <sup>1</sup> |
|-------------------------------------------------------------------------------|------------------------------------------------------------------|
| Class 6 track                                                                 | 110 m.p.h.                                                       |
| Class 7 track                                                                 | 125 m.p.h.                                                       |
| Class 8 track                                                                 | 160 m.p.h. <sup>2</sup>                                          |
- <sup>1</sup> Freight may be transported at passenger train speeds if the following conditions are met:

(1) The vehicles utilized to carry such freight are of equal dynamic performance and have been qualified in accordance with § 213.329 and § 213.345.

(2) The load distribution and securement in the freight vehicle will not adversely affect the dynamic performance of the vehicle. The axle loading pattern is uniform and does not exceed the passenger locomotive axle loadings utilized in passenger service, if any, operating at the same maximum speed.

(3) No carrier may accept or transport a hazardous material, as defined at 49 CFR 171.8, except as provided in Column 9A of the Hazardous Materials Table (49 CFR 172.101) for movement in the same train as a passenger-carrying vehicle or in Column 9B of the Table for movement in a train with no passenger-carrying vehicles.

<sup>2</sup> Operating speeds in excess of 125 m.p.h. are authorized by this part only in conjunction with FRA regulatory approval addressing other safety issues presented by the railroad system. For operations on a dedicated right-of-way, FRA's regulatory approval may allow for the use of inspection and maintenance criteria and procedures in the alternative to those contained in this subpart, based upon a showing that at least an equivalent level of safety is provided.
- 49 CFR 213.307(a) (enhanced display)

page 56 of 110



| Over track that meets all of the requirements prescribed in this subpart for— | The maximum allowable operating speed for trains is <sup>1</sup> |
|-------------------------------------------------------------------------------|------------------------------------------------------------------|
| Class 9 track                                                                 | 220 m.p.h. <sup>2</sup>                                          |

<sup>1</sup> Freight may be transported at passenger train speeds if the following conditions are met:

- (1) The vehicles utilized to carry such freight are of equal dynamic performance and have been qualified in accordance with § 213.329 and § 213.345.
- (2) The load distribution and securement in the freight vehicle will not adversely affect the dynamic performance of the vehicle. The axle loading pattern is uniform and does not exceed the passenger locomotive axle loadings utilized in passenger service, if any, operating at the same maximum speed.
- (3) No carrier may accept or transport a hazardous material, as defined at 49 CFR 171.8, except as provided in Column 9A of the Hazardous Materials Table (49 CFR 172.101) for movement in the same train as a passenger-carrying vehicle or in Column 9B of the Table for movement in a train with no passenger-carrying vehicles.

<sup>2</sup> Operating speeds in excess of 125 m.p.h. are authorized by this part only in conjunction with FRA regulatory approval addressing other safety issues presented by the railroad system. For operations on a dedicated right-of-way, FRA's regulatory approval may allow for the use of inspection and maintenance criteria and procedures in the alternative to those contained in this subpart, based upon a showing that at least an equivalent level of safety is provided.

- (b) If a segment of track does not meet all of the requirements for its intended class, it is to be reclassified to the next lower class of track for which it does meet all of the requirements of this subpart. If a segment does not meet all of the requirements for Class 6, the requirements for Classes 1 through 5 apply.

[63 FR 34029, June 22, 1998, as amended at 78 FR 16104, Mar. 13, 2013]

### § 213.309 Restoration or renewal of track under traffic conditions.

- (a) Restoration or renewal of track under traffic conditions is limited to the replacement of worn, broken, or missing components or fastenings that do not affect the safe passage of trains.
- (b) The following activities are expressly prohibited under traffic conditions:
  - (1) Any work that interrupts rail continuity, e.g., as in joint bar replacement or rail replacement;
  - (2) Any work that adversely affects the lateral or vertical stability of the track with the exception of spot tamping an isolated condition where not more than 15 lineal feet of track are involved at any one time and the ambient air temperature is not above 95 degrees Fahrenheit; and
  - (3) Removal and replacement of the rail fastenings on more than one tie at a time within 15 feet.

### § 213.311 Measuring track not under load.

When unloaded track is measured to determine compliance with requirements of this subpart, evidence of rail movement, if any, that occurs while the track is loaded shall be added to the measurements of the unloaded track.

### **§ 213.313 Application of requirements to curved track.**

Unless otherwise provided in this part, requirements specified for curved track apply only to track having a curvature greater than 0.25 degree.

[78 FR 16104, Mar. 13, 2013]

### **§ 213.317 Waivers.**

- (a) Any owner of track to which this subpart applies may petition the Federal Railroad Administrator for a waiver from any or all requirements prescribed in this subpart.
- (b) Each petition for a waiver under this section shall be filed in the manner and contain the information required by §§ 211.7 and 211.9 of this chapter.
- (c) If the Administrator finds that a waiver is in the public interest and is consistent with railroad safety, the Administrator may grant the waiver subject to any conditions the Administrator deems necessary. Where a waiver is granted, the Administrator publishes a notice containing the reasons for granting the waiver.

### **§ 213.319 Drainage.**

Each drainage or other water carrying facility under or immediately adjacent to the roadbed shall be maintained and kept free of obstruction, to accommodate expected water flow for the area concerned.

### **§ 213.321 Vegetation.**

Vegetation on railroad property which is on or immediately adjacent to roadbed shall be controlled so that it does not—

- (a) Become a fire hazard to track-carrying structures;
- (b) Obstruct visibility of railroad signs and signals:
  - (1) Along the right of way, and
  - (2) At highway-rail crossings;
- (c) Interfere with railroad employees performing normal trackside duties;
- (d) Prevent proper functioning of signal and communication lines; or
- (e) Prevent railroad employees from visually inspecting moving equipment from their normal duty stations.

### **§ 213.323 Track gage.**

- (a) Gage is measured between the heads of the rails at right-angles to the rails in a plane five-eighths of an inch below the top of the rail head.

(b) Gage shall be within the limits prescribed in the following table:

| Class of track | The gage must be at least—        | But not more than—                | The change of gage within 31 feet must not be greater than— |
|----------------|-----------------------------------|-----------------------------------|-------------------------------------------------------------|
| Class 6 track  | 4'8"                              | 4'9 <sup>1</sup> / <sub>4</sub> " | 3/ <sub>4</sub> "                                           |
| Class 7 track  | 4'8"                              | 4'9 <sup>1</sup> / <sub>4</sub> " | 1/ <sub>2</sub> "                                           |
| Class 8 track  | 4'8"                              | 4'9 <sup>1</sup> / <sub>4</sub> " | 1/ <sub>2</sub> "                                           |
| Class 9 track  | 4'8 <sup>1</sup> / <sub>4</sub> " | 4'9 <sup>1</sup> / <sub>4</sub> " | 1/ <sub>2</sub> "                                           |

[63 FR 34029, June 22, 1998, as amended at 78 FR 16104, Mar. 13, 2013]

## § 213.327 Track alinement.

(a) Uniformity at any point along the track is established by averaging the measured mid-chord offset values for nine consecutive points that are centered around that point and spaced according to the following table:

| Chord length | Spacing |
|--------------|---------|
| 31'          | 7'9"    |
| 62'          | 15'6"   |
| 124'         | 31'0"   |

(b) Except as provided in paragraph (c) of this section, a single alinement deviation from uniformity may not be more than the amount prescribed in the following table:

| Class of track | Tangent/Curved track | The deviation from uniformity of the mid-chord offset for a 31-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 62-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 124-foot chord may not be more than—(inches) |
|----------------|----------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Class 6 track  | Tangent              | 1/ <sub>2</sub>                                                                                         | 3/ <sub>4</sub>                                                                                         | 1 1/ <sub>2</sub>                                                                                        |

| Class of track | Tangent/<br>Curved track | The deviation from uniformity of the mid-chord offset for a 31-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 62-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 124-foot chord may not be more than—(inches) |
|----------------|--------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Class 7 track  | Curved                   | $\frac{1}{2}$                                                                                           | $\frac{5}{8}$                                                                                           | $1\frac{1}{2}$                                                                                           |
|                | Tangent                  | $\frac{1}{2}$                                                                                           | $\frac{3}{4}$                                                                                           | $1\frac{1}{4}$                                                                                           |
| Class 8 track  | Curved                   | $\frac{1}{2}$                                                                                           | $\frac{1}{2}$                                                                                           | $1\frac{1}{4}$                                                                                           |
|                | Tangent                  | $\frac{1}{2}$                                                                                           | $\frac{3}{4}$                                                                                           | 1                                                                                                        |
| Class 9 track  | Curved                   | $\frac{1}{2}$                                                                                           | $\frac{1}{2}$                                                                                           | $\frac{3}{4}$                                                                                            |
|                | Tangent                  | $\frac{1}{2}$                                                                                           | $\frac{1}{2}$                                                                                           | $\frac{3}{4}$                                                                                            |
|                | Curved                   | $\frac{1}{2}$                                                                                           | $\frac{1}{2}$                                                                                           | $\frac{3}{4}$                                                                                            |

- (c) For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches, a single alignment deviation from uniformity of the outside rail of the curve may not be more than the amount prescribed in the following table:

| Class of track | Track type | The deviation from uniformity of the mid-chord offset for a 31-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 62-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 124-foot chord may not be more than—(inches) |
|----------------|------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Class 6 track  | Curved     | $\frac{1}{2}$                                                                                           | $\frac{5}{8}$                                                                                           | $1\frac{1}{4}$                                                                                           |
| Class 7 track  | Curved     | $\frac{1}{2}$                                                                                           | $\frac{1}{2}$                                                                                           | 1                                                                                                        |
| Class 8 track  | Curved     | $\frac{1}{2}$                                                                                           | $\frac{1}{2}$                                                                                           | $\frac{3}{4}$                                                                                            |
| Class 9 track  | Curved     | $\frac{1}{2}$                                                                                           | $\frac{1}{2}$                                                                                           | $\frac{3}{4}$                                                                                            |

| Class of track | Track type | The deviation from uniformity of the mid-chord offset for a 31-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 62-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 124-foot chord may not be more than—(inches) |
|----------------|------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| track          |            |                                                                                                         |                                                                                                         |                                                                                                          |

- (d) For three or more non-overlapping deviations from uniformity in track alinement occurring within a distance equal to five times the specified chord length, each of which exceeds the limits in the following table, each track owner shall maintain the alinement of the track within the limits prescribed for each deviation:

| Class of track | The deviation from uniformity of the mid-chord offset for a 31-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 62-foot chord may not be more than—(inches) | The deviation from uniformity of the mid-chord offset for a 124-foot chord may not be more than— (inches) |
|----------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Class 6 track  | $\frac{3}{8}$                                                                                           | $\frac{1}{2}$                                                                                           | 1                                                                                                         |
| Class 7 track  | $\frac{3}{8}$                                                                                           | $\frac{3}{8}$                                                                                           | $\frac{7}{8}$                                                                                             |
| Class 8 track  | $\frac{3}{8}$                                                                                           | $\frac{3}{8}$                                                                                           | $\frac{1}{2}$                                                                                             |
| Class 9 track  | $\frac{3}{8}$                                                                                           | $\frac{3}{8}$                                                                                           | $\frac{1}{2}$                                                                                             |

- (e) For purposes of complying with this section, the ends of the chord shall be at points on the gage side of the rail, five-eighths of an inch below the top of the railhead. On tangent track, either rail may be used as the line rail; however, the same rail shall be used for the full length of that tangential segment of the track. On curved track, the line rail is the outside rail of the curve.

[78 FR 16104, Mar. 13, 2013]

### § 213.329 Curves; elevation and speed limitations.

- (a) The maximum elevation of the outside rail of a curve may not be more than 7 inches. The outside rail of a curve may not be lower than the inside rail by design, except when engineered to address specific track or operating conditions; the limits in § 213.331 apply in all cases.
- (b) The maximum allowable posted timetable operating speed for each curve is determined by the following formula:

$$V_{\max} = \sqrt{\frac{E_a + E_u}{0.0007D}}$$

Where—

$V_{\max}$  = Maximum allowable posted timetable operating speed (m.p.h.).

$E_a$  = Actual elevation of the outside rail (inches).<sup>[6]</sup>

$E_u$  = Qualified cant deficiency<sup>[7]</sup> (inches) of the vehicle type.

$D$  = Degree of curvature (degrees).<sup>[8]</sup>

- (c) All vehicles are considered qualified for operating on track with a cant deficiency,  $E_u$ , not exceeding 3 inches. Table 1 of appendix A to this part is a table of speeds computed in accordance with the formula in paragraph (b) of this section, when  $E_u$  equals 3 inches, for various elevations and degrees of curvature.
- (d) Each vehicle type must be approved by FRA to operate on track with a qualified cant deficiency,  $E_u$ , greater than 3 inches. Each vehicle type must demonstrate, in a ready-for-service load condition, compliance with the requirements of either paragraph (d)(1) or (2) of this section.
  - (1) When positioned on a track with a uniform superelevation equal to the proposed cant deficiency:
    - (i) No wheel of the vehicle type unloads to a value less than 60 percent of its static value on perfectly level track; and

<sup>[6]</sup> Actual elevation,  $E_a$ , for each 155-foot track segment in the body of the curve is determined by averaging the elevation for 11 points through the segment at 15.5-foot spacing. If the curve length is less than 155 feet, the points are averaged through the full length of the body of the curve.

<sup>[7]</sup> If the actual elevation,  $E_a$ , and degree of curvature,  $D$ , change as a result of track degradation, then the actual cant deficiency for the maximum allowable posted timetable operating speed,  $V_{\max}$ , may be greater than the qualified cant deficiency,  $E_u$ . This actual cant deficiency for each curve may not exceed the qualified cant deficiency,  $E_u$ , plus one-half inch.

<sup>[8]</sup> Degree of curvature,  $D$ , is determined by averaging the degree of curvature over the same track segment as the elevation.

- (ii) For passenger cars, the roll angle between the floor of the equipment and the horizontal does not exceed 8.6 degrees; or
- (2) When operating through a constant radius curve at a constant speed corresponding to the proposed cant deficiency, and a test plan is submitted and approved by FRA in accordance with § 213.345(e) and (f):
  - (i) The steady-state (average) load on any wheel, throughout the body of the curve, is not less than 60 percent of its static value on perfectly level track; and
  - (ii) For passenger cars, the steady-state (average) lateral acceleration measured on the floor of the carbody does not exceed 0.15g.
- (e) The track owner or railroad shall transmit the results of the testing specified in paragraph (d) of this section to FRA's Associate Administrator for Railroad Safety/Chief Safety Officer (FRA) requesting approval for the vehicle type to operate at the desired curving speeds allowed under the formula in paragraph (b) of this section. The request shall be made in writing and contain, at a minimum, the following information—
  - (1) A description of the vehicle type involved, including schematic diagrams of the suspension system(s) and the estimated location of the center of gravity above top of rail;
  - (2) The test procedure,<sup>[9]</sup> including the load condition under which the testing was performed, and description of the instrumentation used to qualify the vehicle type, as well as the maximum values for wheel unloading and roll angles or accelerations that were observed during testing; and
  - (3) For vehicle types not subject to part 238 or part 229 of this chapter, procedures or standards in effect that relate to the maintenance of all safety-critical components of the suspension system(s) for the particular vehicle type. Safety-critical components of the suspension system are those that impact or have significant influence on the roll of the carbody and the distribution of weight on the wheels.
- (f) In approving the request made pursuant to paragraph (e) of this section, FRA may impose conditions necessary for safely operating at the higher curving speeds. Upon FRA approval of the request, the track owner or railroad shall notify FRA in writing no less than 30 calendar days prior to the proposed implementation of the approved higher curving speeds allowed under the formula in paragraph (b) of this section. The notification shall contain, at a minimum, identification of the track segment(s) on which the higher curving speeds are to be implemented.
- (g) The documents required by this section must be provided to FRA by:
  - (1) The track owner; or
  - (2) A railroad that provides service with the same vehicle type over trackage of one or more track owner(s), with the written consent of each affected track owner.
- (h)

---

<sup>[9]</sup> The test procedure may be conducted whereby all the wheels on one side (right or left) of the vehicle are raised to the proposed cant deficiency, the vertical wheel loads under each wheel are measured, and a level is used to record the angle through which the floor of the vehicle has been rotated.

- (1) Vehicle types permitted by FRA to operate at cant deficiencies,  $E_u$ , greater than 3 inches but not more than 5 inches shall be considered qualified under this section to operate at those permitted cant deficiencies for any Class 6 track segment. The track owner or railroad shall notify FRA in writing no less than 30 calendar days prior to the proposed implementation of such curving speeds in accordance with paragraph (f) of this section.
- (2) Vehicle types permitted by FRA to operate at cant deficiencies,  $E_u$ , greater than 5 inches on Class 6 track, or greater than 3 inches on Class 7 through 9 track, shall be considered qualified under this section to operate at those permitted cant deficiencies only for the previously operated or identified track segments(s). Operation of these vehicle types at such cant deficiencies and track class on any other track segment is permitted only in accordance with the qualification requirements in this subpart.

(i) As used in this section and in §§ 213.333 and 213.345—

- (1) **Vehicle** means a locomotive, as defined in § 229.5 of this chapter; a freight car, as defined in § 215.5 of this chapter; a passenger car, as defined in § 238.5 of this chapter; and any rail rolling equipment used in a train with either a freight car or a passenger car.
- (2) **Vehicle type** means like vehicles with variations in their physical properties, such as suspension, mass, interior arrangements, and dimensions that do not result in significant changes to their dynamic characteristics.

[78 FR 16105, Mar. 13, 2013]

### § 213.331 Track surface.

- (a) For a single deviation in track surface, each track owner shall maintain the surface of its track within the limits prescribed in the following table:

| Track surface (inches)                                                                                                     | Class of track |   |               |               |
|----------------------------------------------------------------------------------------------------------------------------|----------------|---|---------------|---------------|
|                                                                                                                            | 6              | 7 | 8             | 9             |
| The deviation from uniform <sup>1</sup> profile on either rail at the mid-ordinate of a 31-foot chord may not be more than | 1              | 1 | $\frac{3}{4}$ | $\frac{1}{2}$ |
| The deviation from uniform profile on either rail at the mid-ordinate of a 62-foot chord may not be more than              | 1              | 1 | 1             | $\frac{3}{4}$ |

<sup>1</sup> Uniformity for profile is established by placing the midpoint of the specified chord at the point of maximum measurement.

<sup>2</sup> If physical conditions do not permit a spiral long enough to accommodate the minimum length of runoff, part of the runoff may be on tangent track.

<sup>3</sup> However, to control harmonics on jointed track with staggered joints, the crosslevel differences shall not exceed 1 inch in all of six consecutive pairs of joints, as created by seven low joints. Track with joints staggered less than 10 feet apart shall not be considered as having staggered joints. Joints within the seven low joints outside of the regular joint spacing shall not be considered as joints for purposes of this footnote.



| Track surface (inches)                                                                                                                                              | Class of track   |                  |                  |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|
|                                                                                                                                                                     | 6                | 7                | 8                | 9                |
| Except as provided in paragraph (b) of this section, the deviation from uniform profile on either rail at the mid-ordinate of a 124-foot chord may not be more than | 1 <sup>3/4</sup> | 1 <sup>1/2</sup> | 1 <sup>1/4</sup> | 1                |
| The deviation from zero crosslevel at any point on tangent track may not be more than <sup>2</sup>                                                                  | 1                | 1                | 1                | 1                |
| Reverse elevation on curves may not be more than                                                                                                                    | 1 <sup>1/2</sup> | 1 <sup>1/2</sup> | 1 <sup>1/2</sup> | 1 <sup>1/2</sup> |
| The difference in crosslevel between any two points less than 62 feet apart may not be more than <sup>3</sup>                                                       | 1 <sup>1/2</sup> | 1 <sup>1/2</sup> | 1 <sup>1/4</sup> | 1                |
| On curved track, the difference in crosslevel between any two points less than 10 feet apart (short warp) may not be more than                                      | 1 <sup>1/4</sup> | 1 <sup>1/8</sup> | 1                | 3/4              |

<sup>1</sup> Uniformity for profile is established by placing the midpoint of the specified chord at the point of maximum measurement.

<sup>2</sup> If physical conditions do not permit a spiral long enough to accommodate the minimum length of runoff, part of the runoff may be on tangent track.

<sup>3</sup> However, to control harmonics on jointed track with staggered joints, the crosslevel differences shall not exceed 1 inch in all of six consecutive pairs of joints, as created by seven low joints. Track with joints staggered less than 10 feet apart shall not be considered as having staggered joints. Joints within the seven low joints outside of the regular joint spacing shall not be considered as joints for purposes of this footnote.

- (b) For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches, a single deviation in track surface shall be within the limits prescribed in the following table:

| Track surface (inches)                                                                                        | Class of track   |                  |                  |     |
|---------------------------------------------------------------------------------------------------------------|------------------|------------------|------------------|-----|
|                                                                                                               | 6                | 7                | 8                | 9   |
| The difference in crosslevel between any two points less than 10 feet apart (short warp) may not be more than | 1 <sup>1/4</sup> | 1                | 1 <sup>1</sup>   | 3/4 |
| The deviation from uniform profile on either rail at the mid-ordinate of a                                    | 1 <sup>1/2</sup> | 1 <sup>1/4</sup> | 1 <sup>1/4</sup> | 1   |

<sup>1</sup> For curves with a qualified cant deficiency,  $E_u$ , of more than 7 inches, the difference in crosslevel between any two points less than 10 feet apart (short warp) may not be more than three-quarters of an inch.

| Track surface (inches)              | Class of track |   |   |   |
|-------------------------------------|----------------|---|---|---|
|                                     | 6              | 7 | 8 | 9 |
| 124-foot chord may not be more than |                |   |   |   |

<sup>1</sup> For curves with a qualified cant deficiency,  $E_u$ , of more than 7 inches, the difference in crosslevel between any two points less than 10 feet apart (short warp) may not be more than three-quarters of an inch.

- (c) For three or more non-overlapping deviations in track surface occurring within a distance equal to five times the specified chord length, each of which exceeds the limits in the following table, each track owner shall maintain the surface of the track within the limits prescribed for each deviation:

| Track surface (inches)                                                                                         | Class of track |               |               |               |
|----------------------------------------------------------------------------------------------------------------|----------------|---------------|---------------|---------------|
|                                                                                                                | 6              | 7             | 8             | 9             |
| The deviation from uniform profile on either rail at the mid-ordinate of a 31-foot chord may not be more than  | $\frac{3}{4}$  | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{3}{8}$ |
| The deviation from uniform profile on either rail at the mid-ordinate of a 62-foot chord may not be more than  | $\frac{3}{4}$  | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{1}{2}$ |
| The deviation from uniform profile on either rail at the mid-ordinate of a 124-foot chord may not be more than | $1\frac{1}{4}$ | 1             | $\frac{7}{8}$ | $\frac{5}{8}$ |

[78 FR 16106, Mar. 13, 2013]

### § 213.332 Combined track alinement and surface deviations.

- (a) This section applies to any curved track where operations are conducted at a qualified cant deficiency,  $E_u$ , greater than 5 inches, and to all Class 9 track, either curved or tangent.
- (b) For the conditions defined in paragraph (a) of this section, the combination of alinement and surface deviations for the same chord length on the outside rail in a curve and on any of the two rails of a tangent section, as measured by a TGMS, shall comply with the following formula:

$$\frac{3}{4} \times \left| \frac{A_m}{A_L} + \frac{S_m}{S_L} \right| \leq 1$$

Where—

$A_m$  = measured alinement deviation from uniformity (outward is positive, inward is negative).

$A_L$  = allowable alinement limit as per § 213.327(c) (always positive) for the class of track.

$S_m$  = measured profile deviation from uniformity (down is positive, up is negative).

$S_L$  = allowable profile limit as per § 213.331(a) and § 213.331(b) (always positive) for the class of track.

$$\left| \frac{A_m}{A_L} + \frac{S_m}{S_L} \right| = \text{the absolute (positive) value of the result of } \frac{A_m}{A_L} + \frac{S_m}{S_L} .$$

[78 FR 16107, Mar. 13, 2013]

### § 213.333 Automated vehicle-based inspection systems.

- (a) A qualifying Track Geometry Measurement System (TGMS) shall be operated at the following frequency:
  - (1) For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches on track Classes 1 through 5, at least twice per calendar year with not less than 120 days between inspections.
  - (2) For track Class 6, at least once per calendar year with not less than 170 days between inspections. For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches on track Class 6, at least twice per calendar year with not less than 120 days between inspections.
  - (3) For track Class 7, at least twice within any 120-day period with not less than 25 days between inspections.
  - (4) For track Classes 8 and 9, at least twice within any 60-day period with not less than 12 days between inspections.
- (b) A qualifying TGMS shall meet or exceed minimum design requirements which specify that—
  - (1) Track geometry measurements shall be taken no more than 3 feet away from the contact point of wheels carrying a vertical load of no less than 10 kips per wheel, unless otherwise approved by FRA;
  - (2) Track geometry measurements shall be taken and recorded on a distance-based sampling interval preferably at 1 foot not exceeding 2 feet; and
  - (3) Calibration procedures and parameters are assigned to the system which assure that measured and recorded values accurately represent track conditions. Track geometry measurements recorded by the system shall not differ on repeated runs at the same site at the same speed more than  $\frac{1}{8}$  inch.
- (c) A qualifying TGMS shall be capable of measuring and processing the necessary track geometry parameters to determine compliance with—
  - (1) For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches on track Classes 1 through 5: § 213.53, Track gage; § 213.55(b), Track alignment; § 213.57, Curves; elevation and speed limitations; § 213.63, Track surface; and § 213.65, Combined track alignment and surface deviations.
  - (2) For track Classes 6 through 9: § 213.323, Track gage; § 213.327, Track alignment; § 213.329, Curves; elevation and speed limitations; § 213.331, Track surface; and for operations at a cant deficiency of more than 5 inches § 213.332, Combined track alignment and surface deviations.
- (d) A qualifying TGMS shall be capable of producing, within 24 hours of the inspection, output reports that—

- (1) Provide a continuous plot, on a constant-distance axis, of all measured track geometry parameters required in paragraph (c) of this section;
  - (2) Provide an exception report containing a systematic listing of all track geometry conditions which constitute an exception to the class of track over the segment surveyed.
- (e) The output reports required under paragraph (c) of this section shall contain sufficient location identification information which enable field forces to easily locate indicated exceptions.
- (f) Following a track inspection performed by a qualifying TGMS, the track owner shall, within two days after the inspection, field verify and institute remedial action for all exceptions to the class of track.
- (g) The track owner or railroad shall maintain for a period of one year following an inspection performed by a qualifying TGMS, a copy of the plot and the exception report for the track segment involved, and additional records which:
- (1) Specify the date the inspection was made and the track segment involved; and
  - (2) Specify the location, remedial action taken, and the date thereof, for all listed exceptions to the class.
- (h) For track Classes 8 and 9, a qualifying Gage Restraint Measurement System (GRMS) shall be operated at least once per calendar year with at least 170 days between inspections. The lateral capacity of the track structure shall not permit a Gage Widening Projection (GWP) greater than 0.5 inch.
- (i) A GRMS shall meet or exceed minimum design requirements specifying that—
- (1) Gage restraint shall be measured between the heads of the rail:
    - (i) At an interval not exceeding 16 inches;
    - (ii) Under an applied vertical load of no less than 10 kips per rail; and
    - (iii) Under an applied lateral load that provides a lateral/vertical load ratio of between 0.5 and 1.25,<sup>[10]</sup> and a load severity greater than 3 kips but less than 8 kips per rail. Load severity is defined by the formula:

$$S = L - cV$$

Where—

S = Load severity, defined as the lateral load applied to the fastener system (kips).

L = Actual lateral load applied (kips).

c = Coefficient of friction between rail/tie, which is assigned a nominal value of 0.4.

V = Actual vertical load applied (kips), or static vertical wheel load if vertical load is not measured.

- (2) The measured gage and load values shall be converted to a GWP as follows:

---

<sup>[10]</sup> GRMS equipment using load combinations developing L/V ratios that exceed 0.8 shall be operated with caution to protect against the risk of wheel climb by the test wheelset.

$$GWP = (LTG - UTG) \times \frac{8.26}{L - 0.258 \times V}$$

Where—

UTG = Unloaded track gage measured by the GRMS vehicle at a point no less than 10 feet from any lateral or vertical load application.

LTG = Loaded track gage measured by the GRMS vehicle at a point no more than 12 inches from the lateral load application.

L = Actual lateral load applied (kips).

V = Actual vertical load applied (kips), or static vertical wheel load if vertical load is not measured.

GWP = Gage Widening Projection, which means the measured gage widening, which is the difference between loaded and unloaded gage, at the applied loads, projected to reference loads of 16 kips of lateral force and 33 kips of vertical force.

- (j) As further specified for the combination of track class, cant deficiencies, and vehicles subject to paragraphs (j)(1) through (3) of this section, a vehicle having dynamic response characteristics that are representative of other vehicles assigned to the service shall be operated over the route at the revenue speed profile. The vehicle shall either be instrumented or equipped with a portable device that monitors onboard instrumentation on trains. Track personnel shall be notified when onboard accelerometers indicate a possible track-related problem. Testing shall be conducted at the frequencies specified in paragraphs (j)(1) through (3) of this section, unless otherwise determined by FRA after reviewing the test data required by this subpart.
  - (1) For operations at a qualified cant deficiency,  $E_u$ , of more than 5 inches on track Classes 1 through 6, carbody acceleration shall be monitored at least once each calendar quarter with not less than 25 days between inspections on at least one passenger car of each type that is assigned to the service; and
  - (2) For operations at track Class 7 speeds, carbody and truck accelerations shall be monitored at least twice within any 60-day period with not less than 12 days between inspections on at least one passenger car of each type that is assigned to the service; and
  - (3) For operations at track Class 8 or 9 speeds, carbody acceleration shall be monitored at least four times within any 7-day period with not more than 3 days between inspections on at least one non-passenger and one passenger carrying vehicle of each type that is assigned to the service, as appropriate. Truck acceleration shall be monitored at least twice within any 60-day period with not less than 12 days between inspections on at least one passenger carrying vehicle of each type that is assigned to the service, as appropriate.
- (k)
  - (1) The instrumented vehicle or the portable device, as required in paragraph (j) of this section, shall monitor lateral and vertical accelerations of the carbody. The accelerometers shall be attached to the carbody on or under the floor of the vehicle, as near the center of a truck as practicable.

- (2) In addition, a device for measuring lateral accelerations shall be mounted on a truck frame at a longitudinal location as close as practicable to an axle's centerline (either outside axle for trucks containing more than 2 axles), or, if approved by FRA, at an alternate location. After monitoring this data for 2 years, or 1 million miles, whichever occurs first, the track owner or railroad may petition FRA for exemption from this requirement.
- (3) If any of the carbody lateral, carbody vertical, or truck frame lateral acceleration safety limits in this section's table of vehicle/track interaction safety limits is exceeded, corrective action shall be taken as necessary. Track personnel shall be notified when the accelerometers indicate a possible track-related problem.
- (l) For track Classes 8 and 9, the track owner or railroad shall submit a report to FRA, once each calendar year, which provides an analysis of the monitoring data collected in accordance with paragraphs (j) and (k) of this section. Based on a review of the report, FRA may require that an instrumented vehicle having dynamic response characteristics that are representative of other vehicles assigned to the service be operated over the track at the revenue speed profile. The instrumented vehicle shall be equipped to measure wheel/rail forces. If any of the wheel/rail force limits in this section's table of vehicle/track interaction safety limits is exceeded, appropriate speed restrictions shall be applied until corrective action is taken.
- (m) The track owner or railroad shall maintain a copy of the most recent exception records for the inspections required under paragraphs (j), (k), and (l) of this section, as appropriate.

## Vehicle/Track Interaction Safety Limits

| Wheel-Rail Forces <sup>1</sup>         |                                                                                           |                                                                                           |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                              | Safety Limit                                                                              | Filter/<br>Window                                                                         | Requirements                                                                                                                                                                                                                                                          |
| Single Wheel<br>Vertical Load<br>Ratio | $\geq 0.15$                                                                               | 5 ft                                                                                      | No wheel of the vehicle shall be permitted to unload to less than 15 percent of the static vertical wheel load for 5 or more continuous feet. The static vertical wheel load is defined as the load that the wheel would carry when stationary on level track.        |
| Single Wheel<br>L/V Ratio              | $\leq \frac{\tan(\delta) - 0.5}{1 + 0.5 \tan(\delta)}$                                    | 5 ft                                                                                      | The ratio of the lateral force that any wheel exerts on an individual rail to the vertical force exerted by the same wheel on the rail shall not be greater than the safety limit calculated for the wheel's flange angle ( $\delta$ ) for 5 or more continuous feet. |
| Net Axle Lateral<br>L/V Ratio          | $\leq 0.4 + \frac{5.0}{Va}$                                                               | 5 ft                                                                                      | The net axle lateral force, in kips, exerted by any axle on the track shall not exceed a total of 5 kips plus 40 percent of the static vertical load that the axle exerts on the track for 5 or more continuous feet.<br><i>Va</i> = static vertical axle load (kips) |
| Truck Side L/V<br>Ratio                | $\leq 0.6$                                                                                | 5 ft                                                                                      | The ratio of the lateral forces that the wheels on one side of any truck exert on an individual rail to the vertical forces exerted by the same wheels on that rail shall not be greater than 0.6 for 5 or more continuous feet.                                      |
| Carbody Accelerations <sup>2</sup>     |                                                                                           |                                                                                           |                                                                                                                                                                                                                                                                       |
| Parameter                              | Passenger<br>Cars                                                                         | Other Vehicles                                                                            | Requirements                                                                                                                                                                                                                                                          |
| Carbody Lateral<br>(Transient)         | $\leq 0.65g$ peak-<br>to-peak<br>1 sec window <sup>3</sup><br>excludes peaks<br>< 50 msec | $\leq 0.75g$ peak-<br>to-peak<br>1 sec window <sup>3</sup><br>excludes peaks<br>< 50 msec | The peak-to-peak accelerations, measured as the algebraic difference between the two extreme values of measured acceleration in any 1-second time period, excluding any peak lasting less than 50 milliseconds, shall not                                             |

|                                             |                                                                                           |                                                                                            |                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             |                                                                                           |                                                                                            | exceed 0.65g and 0.75g for passenger cars and other vehicles, respectively.                                                                                                                                                                                                                                                                               |
| Carbody Lateral<br>(Sustained Oscillatory)  | $\leq 0.10g \text{ RMS}_t^4$<br>4 sec window <sup>3</sup><br>4 sec sustained              | $\leq 0.12g \text{ RMS}_t^4$<br>4 sec window <sup>3</sup><br>4 sec sustained               | Sustained oscillatory lateral acceleration of the carbody shall not exceed the prescribed (root mean squared) safety limits of 0.10g and 0.12g for passenger cars and other vehicles, respectively. Root mean squared values shall be determined over a sliding 4-second window with linear trend removed and shall be sustained for more than 4 seconds. |
| Carbody Vertical<br>(Transient)             | $\leq 1.0g \text{ peak-to-peak}$<br>1 sec window <sup>3</sup><br>excludes peaks < 50 msec | $\leq 1.25g \text{ peak-to-peak}$<br>1 sec window <sup>3</sup><br>excludes peaks < 50 msec | The peak-to-peak accelerations, measured as the algebraic difference between the two extreme values of measured acceleration in any one second time period, excluding any peak lasting less than 50 milliseconds, shall not exceed 1.0g, or 1.25g, as specified.                                                                                          |
| Carbody Vertical<br>(Sustained Oscillatory) | $\leq 0.25g \text{ RMS}_t^4$<br>4 sec window <sup>3</sup><br>4 sec sustained              | $\leq 0.25g \text{ RMS}_t^4$<br>4 sec window <sup>3</sup><br>4 sec sustained               | Sustained oscillatory vertical acceleration of the carbody shall not exceed the prescribed (root mean squared) safety limit of 0.25g. Root mean squared values shall be determined over a sliding 4-second window with linear trend removed and shall be sustained for more than 4 seconds.                                                               |

### Truck Lateral Acceleration<sup>5</sup>

| Parameter                           | Safety Limit                 | Filter/<br>Window                            | Requirements                                                                                                                                                                                                                                                               |
|-------------------------------------|------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Truck Lateral<br>(enhanced display) | $\leq 0.30g \text{ RMS}_t^4$ | 2 sec window <sup>3</sup><br>2 sec sustained | Truck hunting shall not develop below the maximum authorized speed. Truck hunting is defined as a sustained cyclic oscillation of the truck evidenced by lateral accelerations exceeding 0.3g root mean squared for more than 2 seconds. Root mean squared values shall be |



<sup>1</sup> The lateral and vertical wheel forces shall be measured and processed through a low pass filter (LPF) with a minimum cut-off frequency of 25 Hz. The sample rate for wheel force data shall be at least 250 samples per second.

<sup>2</sup> Carbody accelerations in the vertical and lateral directions shall be measured by accelerometers oriented and located in accordance with § 213.333(k).

<sup>3</sup> Acceleration measurements shall be processed through an LPF with a minimum cut-off frequency of 10 Hz. The sample rate for acceleration data shall be at least 100 samples per second.

<sup>4</sup>  $RMS_t = RMS$  with linear trend removed.

<sup>5</sup> Truck lateral acceleration shall be measured on the truck frame by accelerometers oriented and located in accordance with § 213.333(k).

[63 FR 34029, June 22, 1998; 63 FR 46102, Aug. 28, 1998, as amended at 78 FR 16107, Mar. 13, 2013]

### § 213.334 Ballast; general.

Unless it is otherwise structurally supported, all track shall be supported by material which will—

- (a) Transmit and distribute the load of the track and railroad rolling equipment to the subgrade;
- (b) Restrain the track laterally, longitudinally, and vertically under dynamic loads imposed by railroad rolling equipment and thermal stress exerted by the rails;
- (c) Provide adequate drainage for the track; and
- (d) Maintain proper track crosslevel, surface, and alignment.

### § 213.335 Crossties.

- (a) Crossties shall be made of a material to which rail can be securely fastened.
- (b) Each 39 foot segment of track shall have—
  - (1) A sufficient number of crossties which in combination provide effective support that will—
    - (i) Hold gage within the limits prescribed in § 213.323(b);
    - (ii) Maintain surface within the limits prescribed in § 213.331; and
    - (iii) Maintain alignment within the limits prescribed in § 213.327.
  - (2) The minimum number and type of crossties specified in paragraph (c) of this section effectively distributed to support the entire segment; and
  - (3) Crossties of the type specified in paragraph (c) of this section that are(is) located at a joint location as specified in paragraph (e) of this section.
- (c) For non-concrete tie construction, each 39 foot segment of Class 6 track shall have fourteen crossties; Classes 7, 8 and 9 shall have 18 crossties which are not—
  - (1) Broken through;

- (2) Split or otherwise impaired to the extent the crossties will allow the ballast to work through, or will not hold spikes or rail fasteners;
  - (3) So deteriorated that the tie plate or base of rail can move laterally  $\frac{3}{8}$  inch relative to the crossties;
  - (4) Cut by the tie plate through more than 40 percent of a crosstie's thickness;
  - (5) Configured with less than 2 rail holding spikes or fasteners per tie plate; or
  - (6) So unable, due to insufficient fastener toeload, to maintain longitudinal restraint and maintain rail hold down and gage.
- (d) For concrete tie construction, each 39 foot segment of Class 6 track shall have fourteen crossties, Classes 7, 8 and 9 shall have 16 crossties which are not—
- (1) So deteriorated that the prestress strands are ineffective or withdrawn into the tie at one end and the tie exhibits structural cracks in the rail seat or in the gage of track;
  - (2) Configured with less than 2 fasteners on the same rail;
  - (3) So deteriorated in the vicinity of the rail fastener such that the fastener assembly may pull out or move laterally more than  $\frac{3}{8}$  inch relative to the crosstie;
  - (4) So deteriorated that the fastener base plate or base of rail can move laterally more than  $\frac{3}{8}$  inch relative to the crossties;
  - (5) So deteriorated that rail seat abrasion is sufficiently deep so as to cause loss of rail fastener toeload;
  - (6) Completely broken through; or
  - (7) So unable, due to insufficient fastener toeload, to maintain longitudinal restraint and maintain rail hold down and gage.
- (e) Class 6 track shall have one non-defective crosstie whose centerline is within 18 inches of the rail joint location or two crossties whose center lines are within 24 inches either side of the rail joint location. Class 7, 8, and 9 track shall have two non-defective ties within 24 inches each side of the rail joint.
- (f) For track constructed without crossties, such as slab track and track connected directly to bridge structural components, the track structure shall meet the requirements of paragraphs (b)(1)(i), (ii), and (iii) of this section.
- (g) In Classes 7, 8 and 9 there shall be at least three non-defective ties each side of a defective tie.
- (h) Where timber crossties are in use there shall be tie plates under the running rails on at least nine of 10 consecutive ties.
- (i) No metal object which causes a concentrated load by solely supporting a rail shall be allowed between the base of the rail and the bearing surface of the tie plate.

### § 213.337 Defective rails.

- (a) When an owner of track to which this part applies learns, through inspection or otherwise, that a rail in that track contains any of the defects listed in the following table, a person designated under § 213.305 shall determine whether or not the track may continue in use. If the person determines that the track may continue in use, operation over the defective rail is not permitted until—
- (1) The rail is replaced; or

(2) The remedial action prescribed in the table is initiated—

# REMEDIAL ACTION

| Defect                                                                                         | Length of defect (inch)                         |                                    | Percent of rail head cross-sectional area weakened by defect |                                             | If defective rail is not replaced, take the remedial action prescribed in note |
|------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------|--------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------|
|                                                                                                | More than                                       | But not more than                  | Less than                                                    | But not less than                           |                                                                                |
| Transverse fissure                                                                             |                                                 |                                    | 70 .....<br>100.....                                         | 5 .....<br>70 .....<br>100.....             | B.<br>A2.<br>A.                                                                |
| Compound fissure                                                                               |                                                 |                                    | 70 .....<br>100.....                                         | 5 .....<br>70 .....<br>100.....             | B.<br>A2.<br>A.                                                                |
| Detail fracture<br>Engine burn fracture<br>Defective weld                                      |                                                 |                                    | 25 .....<br>80 .....<br>100.....                             | 5 .....<br>25 .....<br>80 .....<br>100..... | C.<br>D.<br>[A2] or [E and H].<br>[A] or [E and H].                            |
| Horizontal split head<br>Vertical split head<br>Split web<br>Piped rail<br>Head web separation | 1 .....<br>2 .....<br>4 .....<br>(1).....       | 2 .....<br>4 .....<br>(1).....     | .....<br>.....<br>(1).....                                   | .....<br>.....<br>.....<br>.....            | H and F.<br>I and G.<br>B.<br>A.                                               |
| Bolt hole crack                                                                                | 1/2 .....<br>1 .....<br>1 1/2 .....<br>(1)..... | 1 .....<br>1 1/4 .....<br>(1)..... | .....<br>.....<br>(1).....                                   | .....<br>.....<br>.....<br>.....            | H and F.<br>H and G.<br>B.<br>A.                                               |
| Broken base                                                                                    | 1 .....<br>6.....                               | 6.....                             | .....                                                        | .....                                       | D.<br>[A] or [E and I].                                                        |
| Ordinary break                                                                                 | .....                                           | .....                              | .....                                                        | .....                                       | A or E.                                                                        |
| Damaged rail                                                                                   | .....                                           | .....                              | .....                                                        | .....                                       | D.                                                                             |
| Flattened rail                                                                                 | Depth > 3/8 and<br>Length > 8                   | .....                              | .....                                                        | .....                                       | H.                                                                             |

(1) Break out in rail head.

Notes: A. Assign person designated under § 213.305 to visually supervise each operation over defective rail.

A2. Assign person designated under § 213.305 to make visual inspection. That person may authorize operation to continue without visual supervision at a maximum of 10 m.p.h. for up to 24 hours prior to another such visual inspection or replacement or repair of the rail.

B. Limit operating speed over defective rail to that as authorized by a person designated under § 213.305(a)(1)(i) or (ii). The operating speed cannot be over 30 m.p.h.

C. Apply joint bars bolted only through the outermost holes to defect within 20 days after it is determined to continue the track in use. Limit operating speed over defective rail to 30 m.p.h. until joint bars are applied; thereafter, limit speed to 50 m.p.h. When a search for internal rail defects is conducted under § 213.339 and defects are discovered which require remedial action C, the operating speed shall be limited to 50 m.p.h., for a period not to exceed 4 days. If the defective rail has not been removed from the track or a permanent repair made within 4 days of the discovery, limit operating speed over the defective rail to 30 m.p.h. until joint bars are applied; thereafter, limit speed to 50 m.p.h.

D. Apply joint bars bolted only through the outermost holes to defect within 10 days after it is determined to continue the track in use. Limit operating speed over the defective rail to 30 m.p.h. or less as authorized by a person designated under § 213.305(a)(1)(i) or (ii) until joint bars are applied; thereafter, limit speed to 50 m.p.h.

E. Apply joint bars to defect and bolt in accordance with § 213.351(d) and (e).

F. Inspect rail 90 days after it is determined to continue the track in use.

G. Inspect rail 30 days after it is determined to continue the track in use.

H. Limit operating speed over defective rail to 50 m.p.h.

I. Limit operating speed over defective rail to 30 m.p.h.

(b) As used in this section—

- (1) **Transverse fissure** means a progressive crosswise fracture starting from a crystalline center or nucleus inside the head from which it spreads outward as a smooth, bright, or dark, round or oval surface substantially at a right angle to the length of the rail. The distinguishing features of a transverse fissure from other types of fractures or defects are the crystalline center or nucleus and the nearly smooth surface of the development which surrounds it.
- (2) **Compound fissure** means a progressive fracture originating in a horizontal split head which turns up or down in the head of the rail as a smooth, bright, or dark surface progressing until substantially at a right angle to the length of the rail. Compound fissures require examination of both faces of the fracture to locate the horizontal split head from which they originate.

- (3) **Horizontal split head** means a horizontal progressive defect originating inside of the rail head, usually one-quarter inch or more below the running surface and progressing horizontally in all directions, and generally accompanied by a flat spot on the running surface. The defect appears as a crack lengthwise of the rail when it reaches the side of the rail head.
- (4) **Vertical split head** means a vertical split through or near the middle of the head, and extending into or through it. A crack or rust streak may show under the head close to the web or pieces may be split off the side of the head.
- (5) **Split web** means a lengthwise crack along the side of the web and extending into or through it.
- (6) **Piped rail** means a vertical split in a rail, usually in the web, due to failure of the shrinkage cavity in the ingot to unite in rolling.
- (7) **Broken base** means any break in the base of the rail.
- (8) **Detail fracture** means a progressive fracture originating at or near the surface of the rail head. These fractures should not be confused with transverse fissures, compound fissures, or other defects which have internal origins. Detail fractures may arise from shelly spots, head checks, or flaking.
- (9) **Engine burn fracture** means a progressive fracture originating in spots where driving wheels have slipped on top of the rail head. In developing downward they frequently resemble the compound or even transverse fissures with which they should not be confused or classified.
- (10) **Ordinary break** means a partial or complete break in which there is no sign of a fissure, and in which none of the other defects described in this paragraph (b) are found.
- (11) **Damaged rail** means any rail broken or injured by wrecks, broken, flat, or unbalanced wheels, slipping, or similar causes.
- (12) **Flattened rail** means a short length of rail, not a joint, which has flattened out across the width of the rail head to a depth of  $\frac{3}{8}$  inch or more below the rest of the rail. Flattened rail occurrences have no repetitive regularity and thus do not include corrugations, and have no apparent localized cause such as a weld or engine burn. Their individual length is relatively short, as compared to a condition such as head flow on the low rail of curves.
- (13) **Bolt hole crack** means a crack across the web, originating from a bolt hole, and progressing on a path either inclined upward toward the rail head or inclined downward toward the base. Fully developed bolt hole cracks may continue horizontally along the head/web or base/web fillet, or they may progress into and through the head or base to separate a piece of the rail end from the rail. Multiple cracks occurring in one rail end are considered to be a single defect. However, bolt hole cracks occurring in adjacent rail ends within the same joint shall be reported as separate defects.
- (14) **Defective weld** means a field or plant weld containing any discontinuities or pockets, exceeding 5 percent of the rail head area individually or 10 percent in the aggregate, oriented in or near the transverse plane, due to incomplete penetration of the weld metal between the rail ends, lack of fusion between weld and rail end metal, entrapment of slag or sand, under-bead or other shrinkage cracking, or fatigue cracking. Weld defects may originate in the rail head, web, or base, and in some cases, cracks may progress from the defect into either or both adjoining rail ends.
- (15) **Head and web separation** means a progressive fracture, longitudinally separating the head from the web of the rail at the head fillet area.

[63 FR 34029, June 22, 1998; 63 FR 51638, Sept. 28, 1998]

### § 213.339 Inspection of rail in service.

- (a) A continuous search for internal defects shall be made of all rail in track at least twice annually with not less than 120 days between inspections.
- (b) Inspection equipment shall be capable of detecting defects between joint bars, in the area enclosed by joint bars.
- (c) Each defective rail shall be marked with a highly visible marking on both sides of the web and base.
- (d) If the person assigned to operate the rail defect detection equipment being used determines that, due to rail surface conditions, a valid search for internal defects could not be made over a particular length of track, the test on that particular length of track cannot be considered as a search for internal defects under § 213.337(a).
- (e) If a valid search for internal defects cannot be conducted for reasons described in paragraph (d) of this section, the track owner shall, before the expiration of time limits—
  - (1) Conduct a valid search for internal defects;
  - (2) Reduce operating speed to a maximum of 25 miles per hour until such time as a valid search for internal defects can be made; or
  - (3) Remove the rail from service.

### § 213.341 Initial inspection of new rail and welds.

The track owner shall provide for the initial inspection of newly manufactured rail, and for initial inspection of new welds made in either new or used rail. A track owner may demonstrate compliance with this section by providing for:

- (a) ***In-service inspection.*** A scheduled periodic inspection of rail and welds that have been placed in service, if conducted in accordance with the provisions of § 213.339, and if conducted not later than 90 days after installation, shall constitute compliance with paragraphs (b) and (c) of this section;
- (b) ***Mill inspection.*** A continuous inspection at the rail manufacturer's mill shall constitute compliance with the requirement for initial inspection of new rail, provided that the inspection equipment meets the applicable requirements specified in § 213.339. The track owner shall obtain a copy of the manufacturer's report of inspection and retain it as a record until the rail receives its first scheduled inspection under § 213.339;
- (c) ***Welding plant inspection.*** A continuous inspection at a welding plant, if conducted in accordance with the provisions of paragraph (b) of this section, and accompanied by a plant operator's report of inspection which is retained as a record by the track owner, shall constitute compliance with the requirements for initial inspection of new rail and plant welds, or of new plant welds made in used rail; and
- (d) ***Inspection of field welds.*** An initial inspection of field welds, either those joining the ends of CWR strings or those made for isolated repairs, shall be conducted not less than one day and not more than 30 days after the welds have been made. The initial inspection may be conducted by means of portable test equipment. The track owner shall retain a record of such inspections until the welds receive their first scheduled inspection under § 213.339.

- (e) Each defective rail found during inspections conducted under paragraph (a) or (d) of this section shall be marked with highly visible markings on both sides of the web and base and the remedial action as appropriate under § 213.337 will apply.

### § 213.343 Continuous welded rail (CWR).

Each track owner with track constructed of CWR shall have in effect and comply with written procedures which address the installation, adjustment, maintenance and inspection of CWR, and a training program for the application of those procedures, which shall be submitted to the Federal Railroad Administration by March 21, 1999. FRA reviews each plan for compliance with the following—

- (a) Procedures for the installation and adjustment of CWR which include—
  - (1) Designation of a desired rail installation temperature range for the geographic area in which the CWR is located; and
  - (2) De-stressing procedures/methods which address proper attainment of the desired rail installation temperature range when adjusting CWR.
- (b) Rail anchoring or fastening requirements that will provide sufficient restraint to limit longitudinal rail and crosstie movement to the extent practical, and specifically addressing CWR rail anchoring or fastening patterns on bridges, bridge approaches, and at other locations where possible longitudinal rail and crosstie movement associated with normally expected train-induced forces, is restricted.
- (c) Procedures which specifically address maintaining a desired rail installation temperature range when cutting CWR including rail repairs, in-track welding, and in conjunction with adjustments made in the area of tight track, a track buckle, or a pull-apart. Rail repair practices shall take into consideration existing rail temperature so that—
  - (1) When rail is removed, the length installed shall be determined by taking into consideration the existing rail temperature and the desired rail installation temperature range; and
  - (2) Under no circumstances should rail be added when the rail temperature is below that designated by paragraph (a)(1) of this section, without provisions for later adjustment.
- (d) Procedures which address the monitoring of CWR in curved track for inward shifts of alignment toward the center of the curve as a result of disturbed track.
- (e) Procedures which control train speed on CWR track when—
  - (1) Maintenance work, track rehabilitation, track construction, or any other event occurs which disturbs the roadbed or ballast section and reduces the lateral and/or longitudinal resistance of the track; and
  - (2) In formulating the procedures under this paragraph (e), the track owner shall—
    - (i) Determine the speed required, and the duration and subsequent removal of any speed restriction based on the restoration of the ballast, along with sufficient ballast re-consolidation to stabilize the track to a level that can accommodate expected train-induced forces. Ballast re-consolidation can be achieved through either the passage of train tonnage or mechanical stabilization procedures, or both; and
    - (ii) Take into consideration the type of crossties used.
- (f) Procedures which prescribe when physical track inspections are to be performed to detect buckling prone conditions in CWR track. At a minimum, these procedures shall address inspecting track to identify—



- (1) Locations where tight or kinky rail conditions are likely to occur;
- (2) Locations where track work of the nature described in paragraph (e)(1) of this section have recently been performed; and
- (3) In formulating the procedures under this paragraph (f), the track owner shall—
  - (i) Specify the timing of the inspection; and
  - (ii) Specify the appropriate remedial actions to be taken when buckling prone conditions are found.
- (g) The track owner shall have in effect a comprehensive training program for the application of these written CWR procedures, with provisions for periodic re-training, for those individuals designated under § 213.305(c) of this part as qualified to supervise the installation, adjustment, and maintenance of CWR track and to perform inspections of CWR track.
- (h) The track owner shall prescribe recordkeeping requirements necessary to provide an adequate history of track constructed with CWR. At a minimum, these records shall include:
  - (1) Rail temperature, location and date of CWR installations. This record shall be retained for at least one year; and
  - (2) A record of any CWR installation or maintenance work that does not conform with the written procedures. Such record shall include the location of the rail and be maintained until the CWR is brought into conformance with such procedures.
- (i) As used in this section—
  - (1) **Adjusting/de-stressing** means the procedure by which a rail's temperature is re-adjusted to the desired value. It typically consists of cutting the rail and removing rail anchoring devices, which provides for the necessary expansion and contraction, and then re-assembling the track.
  - (2) **Buckling incident** means the formation of a lateral mis-alignment sufficient in magnitude to constitute a deviation of 5 inches measured with a 62-foot chord. These normally occur when rail temperatures are relatively high and are caused by high longitudinal compressive forces.
  - (3) **Continuous welded rail (CWR)** means rail that has been welded together into lengths exceeding 400 feet.
  - (4) **Desired rail installation temperature range** means the rail temperature range, within a specific geographical area, at which forces in CWR should not cause a buckling incident in extreme heat, or a pull-apart during extreme cold weather.
  - (5) **Disturbed track** means the disturbance of the roadbed or ballast section, as a result of track maintenance or any other event, which reduces the lateral or longitudinal resistance of the track, or both.
  - (6) **Mechanical stabilization** means a type of procedure used to restore track resistance to disturbed track following certain maintenance operations. This procedure may incorporate dynamic track stabilizers or ballast consolidators, which are units of work equipment that are used as a substitute for the stabilization action provided by the passage of tonnage trains.

- (7) **Rail anchors** means those devices which are attached to the rail and bear against the side of the crosstie to control longitudinal rail movement. Certain types of rail fasteners also act as rail anchors and control longitudinal rail movement by exerting a downward clamping force on the upper surface of the rail base.
  - (8) **Rail temperature** means the temperature of the rail, measured with a rail thermometer.
  - (9) **Tight/kinky rail** means CWR which exhibits minute alignment irregularities which indicate that the rail is in a considerable amount of compression.
  - (10) **Train-induced forces** means the vertical, longitudinal, and lateral dynamic forces which are generated during train movement and which can contribute to the buckling potential.
  - (11) **Track lateral resistance** means the resistance provided to the rail/crosstie structure against lateral displacement.
  - (12) **Track longitudinal resistance** means the resistance provided by the rail anchors/rail fasteners and the ballast section to the rail/crosstie structure against longitudinal displacement.
- (j) Track owners shall revise their CWR plans to include provisions for the inspection of joint bars in accordance with §§ 213.119(g) and (i)(3).

[63 FR 34029, June 22, 1998; 63 FR 45959, Aug. 28, 1998, as amended at 70 FR 66298, Nov. 2, 2005]

### § 213.345 Vehicle/track system qualification.

- (a) **General.** All vehicle types intended to operate at track Class 6 speeds or above, or at any curving speed producing more than 5 inches of cant deficiency, shall be qualified for operation for their intended track classes in accordance with this subpart. A qualification program shall be used to demonstrate that the vehicle/track system will not exceed the wheel/rail force safety limits and the carbody and truck acceleration criteria specified in § 213.333—
  - (1) At any speed up to and including 5 m.p.h. above the proposed maximum operating speed; and
  - (2) On track meeting the requirements for the class of track associated with the proposed maximum operating speed. For purposes of qualification testing, speeds may exceed the maximum allowable operating speed for the class of track in accordance with the test plan approved by FRA.
- (b) **Existing vehicle type qualification.** Vehicle types previously qualified or permitted to operate at track Class 6 speeds or above or at any curving speeds producing more than 5 inches of cant deficiency prior to March 13, 2013, shall be considered as being successfully qualified under the requirements of this section for operation at the previously operated speeds and cant deficiencies over the previously operated track segment(s).
- (c) **New vehicle type qualification.** Vehicle types not previously qualified under this subpart shall be qualified in accordance with the requirements of this paragraph (c).
  - (1) **Simulations or measurement of wheel/rail forces.** For vehicle types intended to operate at track Class 6 speeds, simulations or measurement of wheel/rail forces during qualification testing shall demonstrate that the vehicle type will not exceed the wheel/rail force safety limits specified in § 213.333. Simulations, if conducted, shall be in accordance with paragraph (c)(2) of this section. Measurement of wheel/rail forces, if conducted, shall be performed over a representative segment of the full route on which the vehicle type is intended to operate.

- (2) **Simulations.** For vehicle types intended to operate at track Class 7 speeds or above, or at any curving speed producing more than 6 inches of cant deficiency, analysis of vehicle/track performance (computer simulations) shall be conducted using an industry recognized methodology on:
  - (i) An analytically defined track segment representative of minimally compliant track conditions (MCAT—Minimally Compliant Analytical Track) for the respective track class(es) as specified in appendix D to this part; and
  - (ii) A track segment representative of the full route on which the vehicle type is intended to operate. Both simulations and physical examinations of the route's track geometry shall be used to determine a track segment representative of the route.
- (3) **Carbody acceleration.** For vehicle types intended to operate at track Class 6 speeds or above, or at any curving speed producing more than 5 inches of cant deficiency, qualification testing conducted over a representative segment of the route shall demonstrate that the vehicle type will not exceed the carbody lateral and vertical acceleration safety limits specified in § 213.333.
- (4) **Truck lateral acceleration.** For vehicle types intended to operate at track Class 6 speeds or above, qualification testing conducted over a representative segment of the route shall demonstrate that the vehicle type will not exceed the truck lateral acceleration safety limit specified in § 213.333.
- (5) **Measurement of wheel/rail forces.** For vehicle types intended to operate at track Class 7 speeds or above, or at any curving speed producing more than 6 inches of cant deficiency, qualification testing conducted over a representative segment of the route shall demonstrate that the vehicle type will not exceed the wheel/rail force safety limits specified in § 213.333.
- (d) **Previously qualified vehicle types.** Vehicle types previously qualified under this subpart for a track class and cant deficiency on one route may be qualified for operation at the same class and cant deficiency on another route through analysis or testing, or both, to demonstrate compliance with paragraph (a) of this section in accordance with the following:
  - (1) **Simulations or measurement of wheel/rail forces.** For vehicle types intended to operate at any curving speed producing more than 6 inches of cant deficiency, or at curving speeds that both correspond to track Class 7 speeds or above and produce more than 5 inches of cant deficiency, simulations or measurement of wheel/rail forces during qualification testing shall demonstrate that the vehicle type will not exceed the wheel/rail force safety limits specified in § 213.333. Simulations, if conducted, shall be in accordance with paragraph (c)(2) of this section. Measurement of wheel/rail forces, if conducted, shall be performed over a representative segment of the new route.
  - (2) **Carbody acceleration.** For vehicle types intended to operate at any curving speed producing more than 5 inches of cant deficiency, or at track Class 7 speeds and above, qualification testing conducted over a representative segment of the new route shall demonstrate that the vehicle type will not exceed the carbody lateral and vertical acceleration safety limits specified in § 213.333.
  - (3) **Truck lateral acceleration.** For vehicle types intended to operate at track Class 7 speeds or above, measurement of truck lateral acceleration during qualification testing shall demonstrate that the vehicle type will not exceed the truck lateral acceleration safety limits specified in § 213.333. Measurement of truck lateral acceleration, if conducted, shall be performed over a representative segment of the new route.

- (e) **Qualification testing plan.** To obtain the data required to support the qualification program outlined in paragraphs (c) and (d) of this section, the track owner or railroad shall submit a qualification testing plan to FRA's Associate Administrator for Railroad Safety/Chief Safety Officer (FRA) at least 60 days prior to testing, requesting approval to conduct the testing at the desired speeds and cant deficiencies. This test plan shall provide for a test program sufficient to evaluate the operating limits of the track and vehicle type and shall include:
- (1) Identification of the representative segment of the route for qualification testing;
  - (2) Consideration of the operating environment during qualification testing, including operating practices and conditions, the signal system, highway-rail grade crossings, and trains on adjacent tracks;
  - (3) The maximum angle found on the gage face of the designed (newly-profiled) wheel flange referenced with respect to the axis of the wheelset that will be used for the determination of the Single Wheel L/V Ratio safety limit specified in § 213.333;
  - (4) A target maximum testing speed in accordance with paragraph (a) of this section and the maximum testing cant deficiency;
  - (5) An analysis and description of the signal system and operating practices to govern operations in track Classes 7 through 9, which shall include a statement of sufficiency in these areas for the class of operation; and
  - (6) The results of vehicle/track performance simulations that are required by this section.
- (f) **Qualification testing.** Upon FRA approval of the qualification testing plan, qualification testing shall be conducted in two sequential stages as required in this subpart.
- (1) Stage-one testing shall include demonstration of acceptable vehicle dynamic response of the subject vehicle as speeds are incrementally increased—
    - (i) On a segment of tangent track, from acceptable track Class 5 speeds to the target maximum test speed (when the target speed corresponds to track Class 6 and above operations); and
    - (ii) On a segment of curved track, from the speeds corresponding to 3 inches of cant deficiency to the maximum testing cant deficiency.
  - (2) When stage-one testing has successfully demonstrated a maximum safe operating speed and cant deficiency, stage-two testing shall commence with the subject equipment over a representative segment of the route as identified in paragraph (e)(1) of this section.
    - (i) A test run shall be conducted over the route segment at the speed the railroad will request FRA to approve for such service.
    - (ii) An additional test run shall be conducted at 5 m.p.h. above this speed.
  - (3) When conducting stage-one and stage-two testing, if any of the monitored safety limits is exceeded on any segment of track intended for operation at track Class 6 speeds or greater, or on any segment of track intended for operation at more than 5 inches of cant deficiency, testing may continue provided that the track location(s) where any of the limits is exceeded be identified and test speeds be limited at the track location(s) until corrective action is taken. Corrective action may include making an adjustment in the track, in the vehicle, or both of these system components. Measurements taken on track segments intended for operations below track Class 6 speeds and at 5 inches of cant deficiency, or less, are not required to be reported.

- (4) Prior to the start of the qualification testing program, a qualifying TGMS specified in § 213.333 shall be operated over the intended route within 30 calendar days prior to the start of the qualification testing program.
- (g) **Qualification testing results.** The track owner or railroad shall submit a report to FRA detailing all the results of the qualification program. When simulations are required as part of vehicle qualification, this report shall include a comparison of simulation predictions to the actual wheel/rail force or acceleration data, or both, recorded during full-scale testing. The report shall be submitted at least 60 days prior to the intended operation of the equipment in revenue service over the route.
- (h) Based on the test results and all other required submissions, FRA will approve a maximum train speed and value of cant deficiency for revenue service, normally within 45 days of receipt of all the required information. FRA may impose conditions necessary for safely operating at the maximum approved train speed and cant deficiency.
- (i) The documents required by this section must be provided to FRA by:
  - (1) The track owner; or
  - (2) A railroad that provides service with the same vehicle type over trackage of one or more track owner(s), with the written consent of each affected track owner.

[78 FR 16111, Mar. 13, 2013]

## § 213.347 Automotive or railroad crossings at grade.

- (a) There shall be no at-grade (level) highway crossings, public or private, or rail-to-rail crossings at-grade on Class 8 and 9 track.
- (b) If train operation is projected at Class 7 speed for a track segment that will include rail-highway grade crossings, the track owner shall submit for FRA's approval a complete description of the proposed warning/barrier system to address the protection of highway traffic and high speed trains. Trains shall not operate at Class 7 speeds over any track segment having highway-rail grade crossings unless:
  - (1) An FRA-approved warning/barrier system exists on that track segment; and
  - (2) All elements of that warning/barrier system are functioning.

## § 213.349 Rail end mismatch.

Any mismatch of rails at joints may not be more than that prescribed by the following table—

| Class of track      | Any mismatch of rails at joints may not be more than the following— |                                          |
|---------------------|---------------------------------------------------------------------|------------------------------------------|
|                     | On the tread of the rail ends (inch)                                | On the gage side of the rail ends (inch) |
| Class 6, 7, 8 and 9 | $\frac{1}{8}$                                                       | $\frac{1}{8}$                            |

### § 213.351 Rail joints.

- (a) Each rail joint, insulated joint, and compromise joint shall be of a structurally sound design and dimensions for the rail on which it is applied.
- (b) If a joint bar is cracked, broken, or because of wear allows excessive vertical movement of either rail when all bolts are tight, it shall be replaced.
- (c) If a joint bar is cracked or broken between the middle two bolt holes it shall be replaced.
- (d) Each rail shall be bolted with at least two bolts at each joint.
- (e) Each joint bar shall be held in position by track bolts tightened to allow the joint bar to firmly support the abutting rail ends and to allow longitudinal movement of the rail in the joint to accommodate expansion and contraction due to temperature variations. When no-slip, joint-to-rail contact exists by design, the requirements of this section do not apply. Those locations, when over 400 feet long, are considered to be continuous welded rail track and shall meet all the requirements for continuous welded rail track prescribed in this subpart.
- (f) No rail shall have a bolt hole which is torch cut or burned.
- (g) No joint bar shall be reconfigured by torch cutting.

### § 213.352 Torch cut rail.

- (a) Except as a temporary repair in emergency situations no rail having a torch cut end shall be used. When a rail end with a torch cut is used in emergency situations, train speed over that rail shall not exceed the maximum allowable for Class 2 track. All torch cut rail ends in Class 6 shall be removed within six months of September 21, 1998.
- (b) Following the expiration of the time limits specified in paragraph (a) of this section, any torch cut rail end not removed shall be removed within 30 days of discovery. Train speed over that rail shall not exceed the maximum allowable for Class 2 track until removed.

### § 213.353 Turnouts, crossovers, and lift rail assemblies or other transition devices on moveable bridges.

- (a) In turnouts and track crossings, the fastenings must be intact and maintained so as to keep the components securely in place. Also, each switch, frog, and guard rail shall be kept free of obstructions that may interfere with the passage of wheels. Use of rigid rail crossings at grade is limited per § 213.347.
- (b) Track shall be equipped with rail anchoring through and on each side of track crossings and turnouts, to restrain rail movement affecting the position of switch points and frogs. Elastic fasteners designed to restrict longitudinal rail movement are considered rail anchoring.
- (c) Each flangeway at turnouts and track crossings shall be at least 1<sup>1</sup>/<sub>2</sub> inches wide.
- (d) For all turnouts and crossovers, and lift rail assemblies or other transition devices on moveable bridges, the track owner shall prepare an inspection and maintenance Guidebook for use by railroad employees which shall be submitted to the Federal Railroad Administration. The Guidebook shall contain at a minimum—
  - (1) Inspection frequency and methodology including limiting measurement values for all components subject to wear or requiring adjustment.

(2) Maintenance techniques.

- (e) Each hand operated switch shall be equipped with a redundant operating mechanism for maintaining the security of switch point position.

### § 213.355 Frog guard rails and guard faces; gage.

The guard check and guard face gages in frogs shall be within the limits prescribed in the following table—

| Class of track            | Guard check gage                                                                                                                                                                                               | Guard face gage                                                                                                                               |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                           | The distance between the gage line of a frog to the guard line <sup>1</sup> of its guard rail or guarding face, measured across the track at right angles to the gage line, <sup>2</sup> may not be less than— | The distance between guard lines, <sup>1</sup> measured across the track at right angles to the gage line, <sup>2</sup> may not be more than— |
| Class 6, 7, 8 and 9 track | 4'6 <sup>1</sup> / <sub>2</sub> "                                                                                                                                                                              | 4'5"                                                                                                                                          |

<sup>1</sup> A line along that side of the flangeway which is nearer to the center of the track and at the same elevation as the gage line.

<sup>2</sup> A line five-eighths of an inch below the top of the center line of the head of the running rail, or corresponding location of the tread portion of the track structure.

[78 FR 16112, Mar. 13, 2013]

### § 213.357 Derails.

- (a) Each track, other than a main track, which connects with a Class 7, 8 or 9 main track shall be equipped with a functioning derail of the correct size and type, unless railroad equipment on the track, because of grade characteristics cannot move to foul the main track.
- (b) For the purposes of this section, a derail is a device which will physically stop or divert movement of railroad rolling stock or other railroad on-track equipment past the location of the device.
- (c) Each derail shall be clearly visible. When in a locked position, a derail shall be free of any lost motion which would prevent it from performing its intended function.
- (d) Each derail shall be maintained to function as intended.
- (e) Each derail shall be properly installed for the rail to which it is applied.
- (f) If a track protected by a derail is occupied by standing railroad rolling stock, the derail shall be in derailing position.
- (g) Each derail on a track which is connected to a Class 7, 8 or 9 main track shall be interconnected with the signal system.

### § 213.359 Track stiffness.

- (a) Track shall have a sufficient vertical strength to withstand the maximum vehicle loads generated at maximum permissible train speeds, cant deficiencies and surface defects. For purposes of this section, vertical track strength is defined as the track capacity to constrain vertical deformations so that the track shall return following maximum load to a configuration in compliance with the vehicle/track interaction safety limits and geometry requirements of this subpart.
- (b) Track shall have sufficient lateral strength to withstand the maximum thermal and vehicle loads generated at maximum permissible train speeds, cant deficiencies and lateral alignment defects. For purposes of this section lateral track strength is defined as the track capacity to constrain lateral deformations so that track shall return following maximum load to a configuration in compliance with the vehicle/track interaction safety limits and geometry requirements of this subpart.

### § 213.361 Right of way.

The track owner in Class 8 and 9 shall submit a barrier plan, termed a “right-of-way plan,” to the Federal Railroad Administration for approval. At a minimum, the plan will contain provisions in areas of demonstrated need for the prevention of—

- (a) Vandalism;
- (b) Launching of objects from overhead bridges or structures into the path of trains; and
- (c) Intrusion of vehicles from adjacent rights of way.

### § 213.365 Visual track inspections.

- (a) All track shall be visually inspected in accordance with the schedule prescribed in paragraph (c) of this section by a person designated under § 213.305.
- (b) Each inspection shall be made on foot or by traversing the track in a vehicle at a speed that allows the person making the inspection to visually inspect the track structure for compliance with this part. However, mechanical, electrical, and other track inspection devices may be used to supplement visual inspection. If a vehicle is used for visual inspection, the speed of the vehicle may not be more than 5 m.p.h. when traversing track crossings and turnouts; otherwise, the inspection vehicle speed shall be at the sole discretion of the inspector, based on track conditions and inspection requirements. When traversing the track in a vehicle, the inspection will be subject to the following conditions—
  - (1) One inspector in a vehicle may inspect up to two tracks at one time provided that the inspector's visibility remains unobstructed by any cause and that the second track is not centered more than 30 feet from the track upon which the inspector traverses;
  - (2) Two inspectors in one vehicle may inspect up to four tracks at a time provided that the inspectors' visibility remains unobstructed by any cause and that each track being inspected is centered within 39 feet from the track upon which the inspectors traverse;
  - (3) Each main track must be traversed by a vehicle or inspected on foot at least once every two weeks, and each siding must be traversed by a vehicle or inspected on foot at least once every month; and
  - (4) Track inspection records shall indicate which track(s) are traversed by the vehicle or inspected on foot as outlined in paragraph (b)(3) of this section.



- (c) Each visual track inspection shall be made in accordance with the following schedule—

**TABLE 1 TO § 213.365(c)**

| <b>Class of track</b> | <b>Required frequency</b>                                                                |
|-----------------------|------------------------------------------------------------------------------------------|
| 6, 7, and 8           | Twice weekly <sup>1</sup> with at least a 2 calendar day's interval between inspections. |
| 9                     | Three times per week.                                                                    |

<sup>1</sup> An inspection week is defined as a seven (7) day period beginning on Sunday and ending on Saturday.

- (d) If the § 213.305 qualified person making the inspection finds a deviation from the requirements of this part, the person shall immediately initiate remedial action. Any subsequent movements to facilitate repairs on track that is out of service must be authorized by a § 213.305 qualified person.
- (e) Each switch, turnout, track crossing, and lift rail assemblies on moveable bridges shall be inspected on foot at least weekly. The inspection shall be accomplished in accordance with the Guidebook required under § 213.353.
- (f) In track Classes 8 and 9, if no train traffic operates for a period of eight hours, a train shall be operated at a speed not to exceed 100 miles per hour over the track before the resumption of operations at the maximum authorized speed.

[63 FR 34029, June 22, 1998; 63 FR 45959, Aug. 28, 1998, as amended at 85 FR 63391, Oct. 7, 2020]

### **§ 213.367 Special inspections.**

In the event of fire, flood, severe storm, temperature extremes or other occurrence which might have damaged track structure, a special inspection shall be made of the track involved as soon as possible after the occurrence and, if possible, before the operation of any train over that track.

### **§ 213.369 Inspection records.**

- (a) Each owner of track to which this part applies shall keep a record of each inspection required to be performed on that track under this subpart.
- (b) Except as provided in paragraph (e) of this section, each record of an inspection under § 213.365 shall be prepared on the day the inspection is made and signed or otherwise certified by the person making the inspection. Records shall specify the author of the record, the type of track inspected, date of inspection, location of inspection, nature of any deviation from the requirements of this part, and the remedial action taken by the person making the inspection. The track owner shall designate the location(s) where each original record shall be maintained for at least one year after the inspection covered by the record. The track owner shall also designate one location, within 100 miles of each State in which it conducts operations, where copies of records that apply to those operations are maintained or can be viewed following 10 days' notice by the Federal Railroad Administration.

- (c) Rail inspection records shall specify the date of inspection, the location and nature of any internal defects found, the remedial action taken and the date thereof, and the location of any intervals of track not tested per § 213.339(d). The owner shall retain a rail inspection record for at least two years after the inspection and for one year after remedial action is taken.
- (d) Records of continuous rail testing under § 213.240 shall—
  - (1) Include all information required under § 213.240(e);
  - (2) State whether the test is being conducted to satisfy the requirements for an internal rail inspection under § 213.339;
  - (3) List the date(s) and time(s) of the continuous rail test data collection, including the date and time of the start and end of the test run, and the date and time each suspect location was identified and field-verified;
  - (4) Include the determination made after field verification of each suspect location, including the:
    - (i) Location and type of defect found;
    - (ii) Size of defect; and
    - (iii) Initial remedial action taken, if required, and the date thereof; and
  - (5) Be retained for at least two years after the inspection and for at least one year after initial remedial action is taken, whichever is later.
- (e) Track owners that elect to utilize continuous rail testing under § 213.240 shall maintain records of all continuous rail testing operations sufficient for monitoring and determining compliance with all applicable regulations and shall make those records available to FRA during regular business hours following reasonable notice.
- (f) Track inspection records shall be kept available to persons who perform the inspections and to persons performing subsequent inspections.
- (g) Each track owner required to keep inspection records under this section shall make those records available for inspection and copying by the Federal Railroad Administration upon request during regular business hours following reasonable notice.
- (h) For purposes of compliance with the requirements of this section, a track owner may create, retain, transmit, store, and retrieve records by electronic means provided that—
  - (1) The system used to generate the electronic record meets all requirements and contains the information required under this subpart;
  - (2) The track owner monitors its electronic records database to ensure record accuracy;
  - (3) The electronic system is designed to uniquely identify the author of the record. No two persons shall have the same electronic identity;
  - (4) The electronic system ensures that each record cannot be modified in any way, or replaced, once the record is completed;
  - (5) The electronic storage of each record shall be initiated by the person making the inspection within 72 hours following the completion of that inspection; and

(6) Any amendment to a record shall be electronically stored apart from the record which it amends.  
Each amendment to a record shall be uniquely identified as to the person making the amendment.

(i) Each vehicle/track interaction safety record required under § 213.333(g) and (m) shall be made available for inspection and copying by the FRA at the locations specified in paragraph (b) of this section.

*[63 FR 34029, June 22, 1998, as amended at 85 FR 63391, Oct. 7, 2020]*

## **Appendix A to Part 213—Maximum Allowable Curving Speeds**

This appendix contains four tables identifying maximum allowing curving speeds based on 3, 4, 5, and 6 inches of unbalance (cant deficiency), respectively.

**TABLE 1—THREE INCHES UNBALANCE**

|                     | Elevation of outer rail (inches)           |     |     |       |     |       |     |       |     |       |     |       |     |
|---------------------|--------------------------------------------|-----|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|
|                     | 0                                          | 1/2 | 1   | 1 1/2 | 2   | 2 1/2 | 3   | 3 1/2 | 4   | 4 1/2 | 5   | 5 1/2 | 6   |
| Degree of curvature | Maximum allowable operating speed (m.p.h.) |     |     |       |     |       |     |       |     |       |     |       |     |
| 0°30'               | 93                                         | 100 | 107 | 113   | 120 | 125   | 131 | 136   | 141 | 146   | 151 | 156   | 160 |
| 0°40'               | 80                                         | 87  | 93  | 98    | 104 | 109   | 113 | 118   | 122 | 127   | 131 | 135   | 139 |
| 0°50'               | 72                                         | 77  | 83  | 88    | 93  | 97    | 101 | 106   | 110 | 113   | 117 | 121   | 124 |
| 1°00'               | 65                                         | 71  | 76  | 80    | 85  | 89    | 93  | 96    | 100 | 104   | 107 | 110   | 113 |
| 1°15'               | 59                                         | 63  | 68  | 72    | 76  | 79    | 83  | 86    | 89  | 93    | 96  | 99    | 101 |
| 1°30'               | 53                                         | 58  | 62  | 65    | 69  | 72    | 76  | 79    | 82  | 85    | 87  | 90    | 93  |
| 1°45'               | 49                                         | 53  | 57  | 61    | 64  | 67    | 70  | 73    | 76  | 78    | 81  | 83    | 86  |
| 2°00'               | 46                                         | 50  | 53  | 57    | 60  | 63    | 65  | 68    | 71  | 73    | 76  | 78    | 80  |
| 2°15'               | 44                                         | 47  | 50  | 53    | 56  | 59    | 62  | 64    | 67  | 69    | 71  | 73    | 76  |
| 2°30'               | 41                                         | 45  | 48  | 51    | 53  | 56    | 59  | 61    | 63  | 65    | 68  | 70    | 72  |
| 2°45'               | 39                                         | 43  | 46  | 48    | 51  | 53    | 56  | 58    | 60  | 62    | 64  | 66    | 68  |
| 3°00'               | 38                                         | 41  | 44  | 46    | 49  | 51    | 53  | 56    | 58  | 60    | 62  | 64    | 65  |
| 3°15'               | 36                                         | 39  | 42  | 44    | 47  | 49    | 51  | 53    | 55  | 57    | 59  | 61    | 63  |
| 3°30'               | 35                                         | 38  | 40  | 43    | 45  | 47    | 49  | 52    | 53  | 55    | 57  | 59    | 61  |
| 3°45'               | 34                                         | 37  | 39  | 41    | 44  | 46    | 48  | 50    | 52  | 53    | 55  | 57    | 59  |
| 4°00'               | 33                                         | 35  | 38  | 40    | 42  | 44    | 46  | 48    | 50  | 52    | 53  | 55    | 57  |
| 4°30'               | 31                                         | 33  | 36  | 38    | 40  | 42    | 44  | 45    | 47  | 49    | 50  | 52    | 53  |
| 5°00'               | 29                                         | 32  | 34  | 36    | 38  | 40    | 41  | 43    | 45  | 46    | 48  | 49    | 51  |
| 5°30'               | 28                                         | 30  | 32  | 34    | 36  | 38    | 39  | 41    | 43  | 44    | 46  | 47    | 48  |
| 6°00'               | 27                                         | 29  | 31  | 33    | 35  | 36    | 38  | 39    | 41  | 42    | 44  | 45    | 46  |
| 6°30'               | 26                                         | 28  | 30  | 31    | 33  | 35    | 36  | 38    | 39  | 41    | 42  | 43    | 44  |
| 7°00'               | 25                                         | 27  | 29  | 30    | 32  | 34    | 35  | 36    | 38  | 39    | 40  | 42    | 43  |
| 8°00'               | 23                                         | 25  | 27  | 28    | 30  | 31    | 33  | 34    | 35  | 37    | 38  | 39    | 40  |
| 9°00'               | 22                                         | 24  | 25  | 27    | 28  | 30    | 31  | 32    | 33  | 35    | 36  | 37    | 38  |
| 10°00'              | 21                                         | 22  | 24  | 25    | 27  | 28    | 29  | 30    | 32  | 33    | 34  | 35    | 36  |
| 11°00'              | 20                                         | 21  | 23  | 24    | 25  | 27    | 28  | 29    | 30  | 31    | 32  | 33    | 34  |
| 12°00'              | 19                                         | 20  | 22  | 23    | 24  | 26    | 27  | 28    | 29  | 30    | 31  | 32    | 33  |

**TABLE 2—FOUR INCHES UNBALANCE**

|                     | Elevation of outer rail (inches)           |     |     |       |     |       |     |       |     |       |     |       |     |
|---------------------|--------------------------------------------|-----|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|
|                     | 0                                          | 1/2 | 1   | 1 1/2 | 2   | 2 1/2 | 3   | 3 1/2 | 4   | 4 1/2 | 5   | 5 1/2 | 6   |
| Degree of curvature | Maximum allowable operating speed (m.p.h.) |     |     |       |     |       |     |       |     |       |     |       |     |
| 0°30'               | 107                                        | 113 | 120 | 125   | 131 | 136   | 141 | 146   | 151 | 156   | 160 | 165   | 169 |
| 0°40'               | 93                                         | 98  | 104 | 109   | 113 | 118   | 122 | 127   | 131 | 135   | 139 | 143   | 146 |
| 0°50'               | 83                                         | 88  | 93  | 97    | 101 | 106   | 110 | 113   | 117 | 121   | 124 | 128   | 131 |
| 1°00'               | 76                                         | 80  | 85  | 89    | 93  | 96    | 100 | 104   | 107 | 110   | 113 | 116   | 120 |
| 1°15'               | 68                                         | 72  | 76  | 79    | 83  | 86    | 89  | 93    | 96  | 99    | 101 | 104   | 107 |
| 1°30'               | 62                                         | 65  | 69  | 72    | 76  | 79    | 82  | 85    | 87  | 90    | 93  | 95    | 98  |
| 1°45'               | 57                                         | 61  | 64  | 67    | 70  | 73    | 76  | 78    | 81  | 83    | 86  | 88    | 90  |
| 2°00'               | 53                                         | 57  | 60  | 63    | 65  | 68    | 71  | 73    | 76  | 78    | 80  | 82    | 85  |
| 2°15'               | 50                                         | 53  | 56  | 59    | 62  | 64    | 67  | 69    | 71  | 73    | 76  | 78    | 80  |
| 2°30'               | 48                                         | 51  | 53  | 56    | 59  | 61    | 63  | 65    | 68  | 70    | 72  | 74    | 76  |
| 2°45'               | 46                                         | 48  | 51  | 53    | 56  | 58    | 60  | 62    | 64  | 66    | 68  | 70    | 72  |
| 3°00'               | 44                                         | 46  | 49  | 51    | 53  | 56    | 58  | 60    | 62  | 64    | 65  | 67    | 69  |
| 3°15'               | 42                                         | 44  | 47  | 49    | 51  | 53    | 55  | 57    | 59  | 61    | 63  | 65    | 66  |
| 3°30'               | 40                                         | 43  | 45  | 47    | 49  | 52    | 53  | 55    | 57  | 59    | 61  | 62    | 64  |
| 3°45'               | 39                                         | 41  | 44  | 46    | 48  | 50    | 52  | 53    | 55  | 57    | 59  | 60    | 62  |
| 4°00'               | 38                                         | 40  | 42  | 44    | 46  | 48    | 50  | 52    | 53  | 55    | 57  | 58    | 60  |
| 4°30'               | 36                                         | 38  | 40  | 42    | 44  | 45    | 47  | 49    | 50  | 52    | 53  | 55    | 56  |
| 5°00'               | 34                                         | 36  | 38  | 40    | 41  | 43    | 45  | 46    | 48  | 49    | 51  | 52    | 53  |
| 5°30'               | 32                                         | 34  | 36  | 38    | 39  | 41    | 43  | 44    | 46  | 47    | 48  | 50    | 51  |
| 6°00'               | 31                                         | 33  | 35  | 36    | 38  | 39    | 41  | 42    | 44  | 45    | 46  | 48    | 49  |
| 6°30'               | 30                                         | 31  | 33  | 35    | 36  | 38    | 39  | 41    | 42  | 43    | 44  | 46    | 47  |
| 7°00'               | 29                                         | 30  | 32  | 34    | 35  | 36    | 38  | 39    | 40  | 42    | 43  | 44    | 45  |
| 8°00'               | 27                                         | 28  | 30  | 31    | 33  | 34    | 35  | 37    | 38  | 39    | 40  | 41    | 42  |
| 9°00'               | 25                                         | 27  | 28  | 30    | 31  | 32    | 33  | 35    | 36  | 37    | 38  | 39    | 40  |
| 10°00'              | 24                                         | 25  | 27  | 28    | 29  | 30    | 32  | 33    | 34  | 35    | 36  | 37    | 38  |
| 11°00'              | 23                                         | 24  | 25  | 27    | 28  | 29    | 30  | 31    | 32  | 33    | 34  | 35    | 36  |
| 12°00'              | 22                                         | 23  | 24  | 26    | 27  | 28    | 29  | 30    | 31  | 32    | 33  | 34    | 35  |

**TABLE 3—FIVE INCHES UNBALANCE**

|                     | Elevation of outer rail (inches)           |     |     |       |     |       |     |       |     |       |     |       |     |
|---------------------|--------------------------------------------|-----|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|
|                     | 0                                          | 1/2 | 1   | 1 1/2 | 2   | 2 1/2 | 3   | 3 1/2 | 4   | 4 1/2 | 5   | 5 1/2 | 6   |
| Degree of curvature | Maximum allowable operating speed (m.p.h.) |     |     |       |     |       |     |       |     |       |     |       |     |
| 0°30'               | 120                                        | 125 | 131 | 136   | 141 | 146   | 151 | 156   | 160 | 165   | 169 | 173   | 177 |
| 0°40'               | 104                                        | 109 | 113 | 118   | 122 | 127   | 131 | 135   | 139 | 143   | 146 | 150   | 150 |
| 0°50'               | 93                                         | 97  | 101 | 106   | 110 | 113   | 117 | 121   | 124 | 128   | 131 | 134   | 137 |
| 1°00'               | 85                                         | 89  | 93  | 96    | 100 | 104   | 107 | 110   | 113 | 116   | 120 | 122   | 125 |
| 1°15'               | 76                                         | 79  | 83  | 86    | 89  | 93    | 96  | 99    | 101 | 104   | 107 | 110   | 112 |
| 1°30'               | 69                                         | 72  | 76  | 79    | 82  | 85    | 87  | 90    | 93  | 95    | 98  | 100   | 102 |
| 1°45'               | 64                                         | 67  | 70  | 73    | 76  | 78    | 81  | 83    | 86  | 88    | 90  | 93    | 95  |
| 2°00'               | 60                                         | 63  | 65  | 68    | 71  | 73    | 76  | 78    | 80  | 82    | 85  | 87    | 89  |
| 2°15'               | 56                                         | 59  | 62  | 64    | 67  | 69    | 71  | 73    | 76  | 78    | 80  | 82    | 84  |
| 2°30'               | 53                                         | 56  | 59  | 61    | 63  | 65    | 68  | 70    | 72  | 74    | 76  | 77    | 79  |
| 2°45'               | 51                                         | 53  | 56  | 58    | 60  | 62    | 64  | 66    | 68  | 70    | 72  | 74    | 76  |
| 3°00'               | 49                                         | 51  | 53  | 56    | 58  | 60    | 62  | 64    | 65  | 67    | 69  | 71    | 72  |
| 3°15'               | 47                                         | 49  | 51  | 53    | 55  | 57    | 59  | 61    | 63  | 65    | 66  | 68    | 70  |
| 3°30'               | 45                                         | 47  | 49  | 52    | 53  | 55    | 57  | 59    | 61  | 62    | 64  | 65    | 67  |
| 3°45'               | 44                                         | 46  | 48  | 50    | 52  | 53    | 55  | 57    | 59  | 60    | 62  | 63    | 65  |
| 4°00'               | 42                                         | 44  | 46  | 48    | 50  | 52    | 53  | 55    | 57  | 58    | 60  | 61    | 63  |
| 4°30'               | 40                                         | 42  | 44  | 45    | 47  | 49    | 50  | 52    | 53  | 55    | 56  | 58    | 59  |
| 5°00'               | 38                                         | 40  | 41  | 43    | 45  | 46    | 48  | 49    | 51  | 52    | 53  | 55    | 56  |
| 5°30'               | 36                                         | 38  | 39  | 41    | 43  | 44    | 46  | 47    | 48  | 50    | 51  | 52    | 53  |
| 6°00'               | 35                                         | 36  | 38  | 39    | 41  | 42    | 44  | 45    | 46  | 48    | 49  | 50    | 51  |
| 6°30'               | 33                                         | 35  | 36  | 38    | 39  | 41    | 42  | 43    | 44  | 46    | 47  | 48    | 49  |
| 7°00'               | 32                                         | 34  | 35  | 36    | 38  | 39    | 40  | 42    | 43  | 44    | 45  | 46    | 47  |
| 8°00'               | 30                                         | 31  | 33  | 34    | 35  | 37    | 38  | 39    | 40  | 41    | 42  | 43    | 44  |
| 9°00'               | 28                                         | 30  | 31  | 32    | 33  | 35    | 36  | 37    | 38  | 39    | 40  | 41    | 42  |
| 10°00'              | 27                                         | 28  | 29  | 30    | 32  | 33    | 34  | 35    | 36  | 37    | 38  | 39    | 40  |
| 11°00'              | 25                                         | 27  | 28  | 29    | 30  | 31    | 32  | 33    | 34  | 35    | 36  | 37    | 38  |
| 12°00'              | 24                                         | 26  | 27  | 28    | 29  | 30    | 31  | 32    | 33  | 34    | 35  | 35    | 36  |

**TABLE 4—SIX INCHES UNBALANCE**

|                     | Elevation of outer rail (inches)           |     |     |       |     |       |     |       |     |       |     |       |     |
|---------------------|--------------------------------------------|-----|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|
|                     | 0                                          | 1/2 | 1   | 1 1/2 | 2   | 2 1/2 | 3   | 3 1/2 | 4   | 4 1/2 | 5   | 5 1/2 | 6   |
| Degree of curvature | Maximum allowable operating speed (m.p.h.) |     |     |       |     |       |     |       |     |       |     |       |     |
| 0°30'               | 131                                        | 136 | 141 | 146   | 151 | 156   | 160 | 165   | 169 | 173   | 177 | 181   | 185 |
| 0°40'               | 113                                        | 118 | 122 | 127   | 131 | 135   | 139 | 143   | 146 | 150   | 154 | 157   | 160 |
| 0°50'               | 101                                        | 106 | 110 | 113   | 117 | 121   | 124 | 128   | 131 | 134   | 137 | 140   | 143 |
| 1°00'               | 93                                         | 96  | 100 | 104   | 107 | 110   | 113 | 116   | 120 | 122   | 125 | 128   | 131 |
| 1°15'               | 83                                         | 86  | 89  | 93    | 96  | 99    | 101 | 104   | 107 | 110   | 112 | 115   | 117 |
| 1°30'               | 76                                         | 79  | 82  | 85    | 87  | 90    | 93  | 95    | 98  | 100   | 102 | 105   | 107 |
| 1°45'               | 70                                         | 73  | 76  | 78    | 81  | 83    | 86  | 88    | 90  | 93    | 95  | 97    | 99  |
| 2°00'               | 65                                         | 68  | 71  | 73    | 76  | 78    | 80  | 82    | 85  | 87    | 89  | 91    | 93  |
| 2°15'               | 62                                         | 64  | 67  | 69    | 71  | 73    | 76  | 78    | 80  | 82    | 84  | 85    | 87  |
| 2°30'               | 59                                         | 61  | 63  | 65    | 68  | 70    | 72  | 74    | 76  | 77    | 79  | 81    | 83  |
| 2°45'               | 56                                         | 58  | 60  | 62    | 64  | 66    | 68  | 70    | 72  | 74    | 76  | 77    | 79  |
| 3°00'               | 53                                         | 56  | 58  | 60    | 62  | 64    | 65  | 67    | 69  | 71    | 72  | 74    | 76  |
| 3°15'               | 51                                         | 53  | 55  | 57    | 59  | 61    | 63  | 65    | 66  | 68    | 70  | 71    | 73  |
| 3°30'               | 49                                         | 52  | 53  | 55    | 57  | 59    | 61  | 62    | 64  | 65    | 67  | 69    | 70  |
| 3°45'               | 48                                         | 50  | 52  | 53    | 55  | 57    | 59  | 60    | 62  | 63    | 65  | 66    | 68  |
| 4°00'               | 46                                         | 48  | 50  | 52    | 53  | 55    | 57  | 58    | 60  | 61    | 63  | 64    | 65  |
| 4°30'               | 44                                         | 45  | 47  | 49    | 50  | 52    | 53  | 55    | 56  | 58    | 59  | 60    | 62  |
| 5°00'               | 41                                         | 43  | 45  | 46    | 48  | 49    | 51  | 52    | 53  | 55    | 56  | 57    | 59  |
| 5°30'               | 39                                         | 41  | 43  | 44    | 46  | 47    | 48  | 50    | 51  | 52    | 53  | 55    | 56  |
| 6°00'               | 38                                         | 39  | 41  | 42    | 44  | 45    | 46  | 48    | 49  | 50    | 51  | 52    | 53  |
| 6°30'               | 36                                         | 38  | 39  | 41    | 42  | 43    | 44  | 46    | 47  | 48    | 49  | 50    | 51  |
| 7°00'               | 35                                         | 36  | 38  | 39    | 40  | 42    | 43  | 44    | 45  | 46    | 47  | 48    | 49  |
| 8°00'               | 33                                         | 34  | 35  | 37    | 38  | 39    | 40  | 41    | 42  | 43    | 44  | 45    | 46  |
| 9°00'               | 31                                         | 32  | 33  | 35    | 36  | 37    | 38  | 39    | 40  | 41    | 42  | 43    | 44  |
| 10°00'              | 29                                         | 30  | 32  | 33    | 34  | 35    | 36  | 37    | 38  | 39    | 40  | 41    | 41  |
| 11°00'              | 28                                         | 29  | 30  | 31    | 32  | 33    | 34  | 35    | 36  | 37    | 38  | 39    | 39  |
| 12°00'              | 27                                         | 28  | 29  | 30    | 31  | 32    | 33  | 34    | 35  | 35    | 36  | 37    | 38  |

[78 FR 16113, Mar. 13, 2013]

## Appendixes B-C to Part 213 [Reserved]

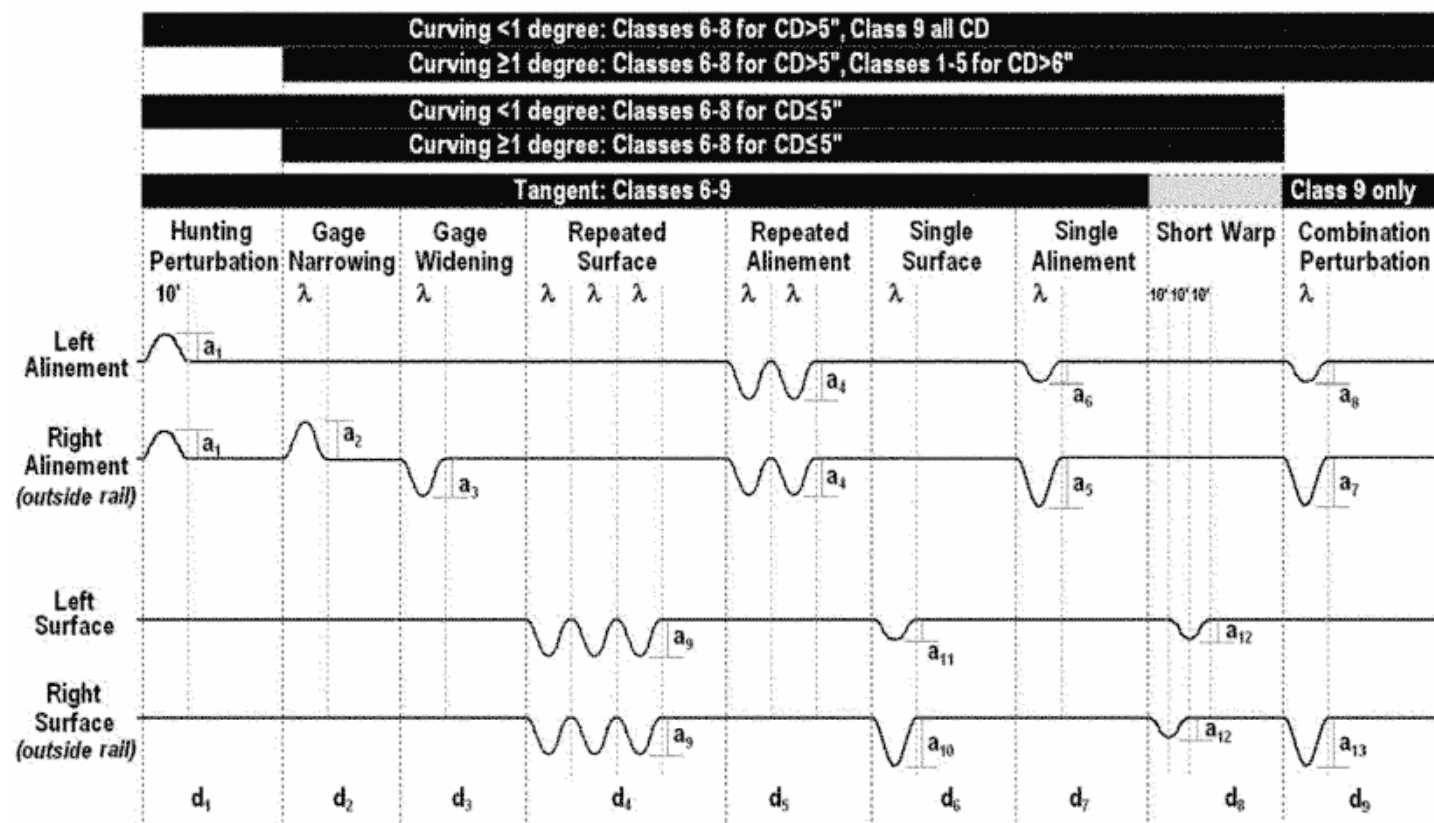
### Appendix D to Part 213—Minimally Compliant Analytical Track (MCAT) Simulations Used for Qualifying Vehicles To Operate at High Speeds and at High Cant Deficiencies

1. This appendix contains requirements for using computer simulations to comply with the vehicle/track system qualification testing requirements specified in subpart G of this part. These simulations shall be performed using a track model containing defined geometry perturbations at the limits that are permitted for a specific class of track and level of cant deficiency. This track model is known as MCAT, Minimally Compliant Analytical Track. These simulations shall be used to identify vehicle dynamic performance issues prior to service or, as appropriate, a change in service, and demonstrate that a vehicle type is suitable for operation on the track over which it is intended to operate.
2. As specified in § 213.345(c)(2), MCAT shall be used for the qualification of new vehicle types intended to operate at track Class 7 speeds or above, or at any curving speed producing more than 6 inches of cant deficiency. MCAT may also be used for the qualification of new vehicle types intended to operate at speeds corresponding to Class 6 track, as specified in § 213.345(c)(1). In addition, as specified in § 213.345(d)(1), MCAT may be used to qualify on new routes vehicle types that have previously been qualified on other routes and are intended to operate at any curving speed producing more than 6 inches of cant deficiency, or at curving speeds that both correspond to track Class 7 speeds or above and produce more than 5 inches of cant deficiency.
  - (a) **Validation.** To validate the vehicle model used for simulations under this part, the track owner or railroad shall obtain vehicle simulation predictions using measured track geometry data, chosen from the same track section over which testing shall be performed as specified in § 213.345(c)(2)(ii). These predictions shall be submitted to FRA in support of the request for approval of the qualification testing plan. Full validation of the vehicle model used for simulations under this part shall be determined when the results of the simulations demonstrate that they replicate all key responses observed during qualification testing.
  - (b) **MCAT layout.** MCAT consists of nine segments, each designed to test a vehicle's performance in response to a specific type of track perturbation. The basic layout of MCAT is shown in figure 1 of this appendix, by type of track (curving or tangent), class of track, and cant deficiency (CD). The values for wavelength,  $\lambda$ , amplitude of perturbation,  $a$ , and segment length,  $d$ , are specified in this appendix. The bars at the top of figure 1 show which segments are required depending on the speed and degree of curvature. For example, the hunting perturbation section is not required for simulation of curves greater than or equal to 1 degree.



Figure 1 of Appendix D to Part 213

### Basic MCAT Layout



(1) **MCAT segments.** MCAT's nine segments contain different types of track deviations in which the shape of each deviation is a versine having wavelength and amplitude varied for each simulation speed as further specified. The nine MCAT segments are defined as follows:

- (i) **Hunting perturbation ( $a_1$ ):** This segment contains an alinement deviation having a wavelength,  $\lambda$ , of 10 feet and amplitude of 0.25 inch on both rails to test vehicle stability on tangent track and on track that is curved less than 1 degree.
- (ii) **Gage narrowing ( $a_2$ ):** This segment contains an alinement deviation on one rail to reduce the gage from the nominal value to the minimum permissible gage or maximum alinement (whichever comes first).
- (iii) **Gage widening ( $a_3$ ):** This segment contains an alinement deviation on one rail to increase the gage from the nominal value to the maximum permissible gage or maximum alinement (whichever comes first).
- (iv) **Repeated surface ( $a_9$ ):** This segment contains three consecutive maximum permissible profile variations on each rail.
- (v) **Repeated alinement ( $a_4$ ):** This segment contains two consecutive maximum permissible alinement variations on each rail.

- (vi) **Single surface** ( $a_{10}$ ,  $a_{11}$ ): This segment contains a maximum permissible profile variation on one rail. If the maximum permissible profile variation alone produces a condition which exceeds the maximum allowed warp condition, a second profile variation is also placed on the opposite rail to limit the warp to the maximum permissible value.
  - (vii) **Single alinement** ( $a_5$ ,  $a_6$ ): This segment contains a maximum permissible alinement variation on one rail. If the maximum permissible alinement variation alone produces a condition which exceeds the maximum allowed gage condition, a second alinement variation is also placed on the opposite rail to limit the gage to the maximum permissible value.
  - (viii) **Short warp** ( $a_{12}$ ): This segment contains a pair of profile deviations to produce a maximum permissible 10-foot warp perturbation. The first is on the outside rail, and the second follows 10 feet farther on the inside rail. Each deviation has a wavelength,  $\lambda$ , of 20 feet and variable amplitude for each simulation speed as described below. This segment is to be used only on curved track simulations.
  - (ix) **Combined perturbation** ( $a_7$ ,  $a_8$ ,  $a_{13}$ ): This segment contains a maximum permissible down and out combined geometry condition on the outside rail in the body of the curve. If the maximum permissible variations produce a condition which exceeds the maximum allowed gage condition, a second variation is also placed on the opposite rail as for the MCAT segments described in paragraphs (b)(1)(vi) and (vii) of this appendix. This segment is to be used for all simulations on Class 9 track, and only for curved track simulations at speeds producing more than 5 inches of cant deficiency on track Classes 6 through 8, and at speeds producing more than 6 inches of cant deficiency on track Classes 1 through 5.
- (2) **Segment lengths:** Each MCAT segment shall be long enough to allow the vehicle's response to the track deviation(s) to damp out. Each segment shall also have a minimum length as specified in table 1 of this appendix, which references the distances in figure 1 of this appendix. For curved track segments, the perturbations shall be placed far enough in the body of the curve to allow for any spiral effects to damp out.

TABLE 1 OF APPENDIX D TO PART 213 MINIMUM  
LENGTHS OF MCAT SEGMENTS

| Distances (ft) |                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| d <sub>1</sub> | d <sub>2</sub> | d <sub>3</sub> | d <sub>4</sub> | d <sub>5</sub> | d <sub>6</sub> | d <sub>7</sub> | d <sub>8</sub> | d <sub>9</sub> |
| 1000           | 1000           | 1000           | 1500           | 1000           | 1000           | 1000           | 1000           | 1000           |

(3) *Degree of curvature.*

- (i) For each simulation involving assessment of curving performance, the degree of curvature,  $D$ , which generates a particular level of cant deficiency,  $E_u$ , for a given speed,  $V$ , shall be calculated using the following equation, which assumes a curve with 6 inches of superelevation:

$$D = \frac{6 + E_u}{0.0007 \times V^2}$$

---

Where—

$D$  = Degree of curvature (degrees).

$V$  = Simulation speed (m.p.h.).

$E_u$  = Cant deficiency (inches).

- (ii) Table 2 of this appendix depicts the degree of curvature for use in MCAT simulations of both passenger and freight equipment performance on Class 2 through 9 track, based on the equation in paragraph (b)(3)(i) of this appendix. The degree of curvature for use in MCAT simulations of equipment performance on Class 1 track is not depicted; it would be based on the same equation using an appropriate superelevation. The degree of curvature for use in MCAT simulations of freight equipment performance on Class 6 (freight) track is shown in italics for cant deficiencies not exceeding 6 inches, to emphasize that the values apply to freight equipment only.

**Table 2 of Appendix D to Part 213**  
**Degree of Curvature for Use in MCAT Simulations (Track Classes 2 through 9)**

|           |        | Tangent                                 | Cant Deficiency (inches) |      |      |      |      |      |      |        |         |      |      |      |      |    |
|-----------|--------|-----------------------------------------|--------------------------|------|------|------|------|------|------|--------|---------|------|------|------|------|----|
|           |        |                                         | 3                        | 4    | 5    | 6    | 7    | 8    | 9    |        |         |      |      |      |      |    |
| Passenger | m.p.h. | Degree of curvature used in simulations |                          |      |      |      |      |      |      | m.p.h. | Freight |      |      |      |      |    |
| Class 2   | 20     |                                         |                          |      |      |      | 46.4 | 50.0 | 53.6 | 20     | Class 2 |      |      |      |      |    |
|           | 25     |                                         |                          |      |      |      | 29.7 | 32.0 | 34.3 | 25     |         |      |      |      |      |    |
|           | 30     |                                         |                          |      |      |      | 20.6 | 22.2 | 23.8 | 30     |         |      |      |      |      |    |
| Class 3   | 35     |                                         |                          |      |      |      | 15.2 | 16.3 | 17.5 | 35     | Class 3 |      |      |      |      |    |
|           | 40     |                                         |                          |      |      |      | 11.6 | 12.5 | 13.4 | 40     |         |      |      |      |      |    |
|           | 45     |                                         |                          |      |      |      | 9.17 | 9.88 | 10.6 | 45     | Class 4 |      |      |      |      |    |
|           | 50     |                                         |                          |      |      |      | 7.43 | 8.00 | 8.57 | 50     |         |      |      |      |      |    |
|           | 55     |                                         |                          |      |      |      | 6.14 | 6.61 | 7.08 | 55     |         |      |      |      |      |    |
| Class 4   | 60     |                                         |                          |      |      |      | 5.16 | 5.56 | 5.95 | 60     | Class 5 |      |      |      |      |    |
|           | 65     |                                         |                          |      |      |      | 4.40 | 4.73 | 5.07 | 65     |         |      |      |      |      |    |
|           | 70     |                                         |                          |      |      |      | 3.79 | 4.08 | 4.37 | 70     |         |      |      |      |      |    |
|           | 75     |                                         |                          |      |      |      | 3.30 | 3.56 | 3.81 | 75     |         |      |      |      |      |    |
| Class 5   | 80     |                                         |                          |      |      |      | 2.90 | 3.13 | 3.35 | 80     | Class 6 |      |      |      |      |    |
|           | 85     |                                         |                          |      |      |      | 0.00 | 1.78 | 1.98 | 2.18   |         | 2.37 | 2.57 | 2.77 | 2.97 | 85 |
|           | 90     |                                         |                          |      |      |      | 0.00 | 1.59 | 1.76 | 1.94   |         | 2.12 | 2.29 | 2.47 | 2.65 | 90 |
|           | 95     |                                         |                          |      |      |      | 0.00 | 1.42 | 1.58 | 1.74   |         | 1.90 | 2.06 | 2.22 | 2.37 | 95 |
| Class 6   | 100    | 0.00                                    | 1.29                     | 1.43 | 1.57 | 1.71 | 1.86 | 2.00 | 2.14 | 100    | Class 7 |      |      |      |      |    |
|           | 105    | 0.00                                    | 1.17                     | 1.30 | 1.43 | 1.55 | 1.68 | 1.81 | 1.94 | 105    |         |      |      |      |      |    |
|           | 110    | 0.00                                    | 1.06                     | 1.18 | 1.30 | 1.42 | 1.53 | 1.65 | 1.77 | 110    |         |      |      |      |      |    |
| Class 7   | 115    | 0.00                                    | 0.97                     | 1.08 | 1.19 | 1.30 | 1.40 | 1.51 | 1.62 | 115    | Class 8 |      |      |      |      |    |
|           | 120    | 0.00                                    | 0.89                     | 0.99 | 1.09 | 1.19 | 1.29 | 1.39 | 1.49 | 120    |         |      |      |      |      |    |
|           | 125    | 0.00                                    | 0.82                     | 0.91 | 1.01 | 1.10 | 1.19 | 1.28 | 1.37 | 125    |         |      |      |      |      |    |
| Class 8   | 130    | 0.00                                    | 0.76                     | 0.85 | 0.93 | 1.01 | 1.10 | 1.18 | 1.27 | 130    | Class 9 |      |      |      |      |    |
|           | 135    | 0.00                                    | 0.71                     | 0.78 | 0.86 | 0.94 | 1.02 | 1.10 | 1.18 | 135    |         |      |      |      |      |    |
|           | 140    | 0.00                                    | 0.66                     | 0.73 | 0.80 | 0.87 | 0.95 | 1.02 | 1.09 | 140    |         |      |      |      |      |    |
|           | 145    | 0.00                                    | 0.61                     | 0.68 | 0.75 | 0.82 | 0.88 | 0.95 | 1.02 | 145    |         |      |      |      |      |    |
|           | 150    | 0.00                                    | 0.57                     | 0.63 | 0.70 | 0.76 | 0.83 | 0.89 | 0.95 | 150    |         |      |      |      |      |    |
|           | 155    | 0.00                                    | 0.54                     | 0.59 | 0.65 | 0.71 | 0.77 | 0.83 | 0.89 | 155    |         |      |      |      |      |    |
|           | 160    | 0.00                                    | 0.50                     | 0.56 | 0.61 | 0.67 | 0.73 | 0.78 | 0.84 | 160    |         |      |      |      |      |    |
| Class 9   | 165    | 0.00                                    | 0.47                     | 0.52 | 0.58 | 0.63 | 0.68 | 0.73 | 0.79 | 165    | Class 9 |      |      |      |      |    |
|           | 170    | 0.00                                    | 0.44                     | 0.49 | 0.54 | 0.59 | 0.64 | 0.69 | 0.74 | 170    |         |      |      |      |      |    |
|           | 175    | 0.00                                    | 0.42                     | 0.47 | 0.51 | 0.56 | 0.61 | 0.65 | 0.70 | 175    |         |      |      |      |      |    |
|           | 180    | 0.00                                    | 0.40                     | 0.44 | 0.49 | 0.53 | 0.57 | 0.62 | 0.66 | 180    |         |      |      |      |      |    |
|           | 185    | 0.00                                    | 0.38                     | 0.42 | 0.46 | 0.50 | 0.54 | 0.58 | 0.63 | 185    |         |      |      |      |      |    |
|           | 190    | 0.00                                    | 0.36                     | 0.40 | 0.44 | 0.47 | 0.51 | 0.55 | 0.59 | 190    |         |      |      |      |      |    |
|           | 195    | 0.00                                    | 0.34                     | 0.38 | 0.41 | 0.45 | 0.49 | 0.53 | 0.56 | 195    |         |      |      |      |      |    |
|           | 200    | 0.00                                    | 0.32                     | 0.36 | 0.39 | 0.43 | 0.46 | 0.50 | 0.54 | 200    |         |      |      |      |      |    |
|           | 205    | 0.00                                    | 0.31                     | 0.34 | 0.37 | 0.41 | 0.44 | 0.48 | 0.51 | 205    |         |      |      |      |      |    |
|           | 210    | 0.00                                    | 0.29                     | 0.32 | 0.36 | 0.39 | 0.42 | 0.45 | 0.49 | 210    |         |      |      |      |      |    |
|           | 215    | 0.00                                    | 0.28                     | 0.31 | 0.34 | 0.37 | 0.40 | 0.43 | 0.46 | 215    |         |      |      |      |      |    |
|           | 220    | 0.00                                    | 0.27                     | 0.30 | 0.32 | 0.35 | 0.38 | 0.41 | 0.44 | 220    |         |      |      |      |      |    |

(c) *Required simulations.*

- (1) To develop a comprehensive assessment of vehicle performance, simulations shall be performed for a variety of scenarios using MCAT. These simulations shall be performed on tangent or curved track, or both, depending on the level of cant deficiency and speed (track class) as summarized in table 3 of this appendix.

**TABLE 3 OF APPENDIX D TO PART 213 SUMMARY OF REQUIRED VEHICLE PERFORMANCE ASSESSMENT USING SIMULATIONS**

|                                         | <b>New vehicle types</b>                                                                                                                       | <b>Previously qualified vehicle types</b>                                                                                                         |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Curved track: cant deficiency ≤6 inches | Curving performance simulation: not required for track Classes 1 through 5; optional for track Class 6; required for track Classes 7 through 9 | Curving performance simulation: not required for track Classes 1 through 6; optional for track Classes 7 through 9 for cant deficiency >5 inches. |
| Curved track: cant deficiency >6 inches | Curving performance simulation required for all track classes                                                                                  | Curving performance simulation optional for all track classes.                                                                                    |
| Tangent track                           | Tangent performance simulation: not required for track Classes 1 through 5; optional for track Class 6; required for track Classes 7 through 9 | Tangent performance simulation not required for any track class.                                                                                  |

- (i) All simulations shall be performed using the design wheel profile and a nominal track gage of 56.5 inches, using tables 4, 5, 6, or 7 of this appendix, as appropriate. In addition, all simulations involving the assessment of curving performance shall be repeated using a nominal track gage of 57.0 inches, using tables 5, 6, or 7 of this appendix, as appropriate.
- (ii) If the wheel profile is different than American Public Transportation Administration (APTA) wheel profiles 320 or 340, then for tangent track segments all simulations shall be repeated using either APTA wheel profile 320 or 340, depending on the established conicity that is common for the operation, as specified in APTA SS-M-015-06, Standard for Wheel Flange Angle of Passenger Equipment (2007). This APTA standard is incorporated by reference into this appendix with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in this appendix, FRA must publish notice of change in the FEDERAL REGISTER and the material must be made available to the public. All approved material is available for inspection at the Federal Railroad Administration, Docket Clerk, 1200 New Jersey Avenue SE., Washington, DC 20590 (telephone 202-493-6030), and is available from the American Public Transportation Association, 1666 K Street NW., Suite 1100, Washington, DC 20006 (telephone 202-496-4800; [www.apta.com](http://www.apta.com)). It is also available for inspection at the National Archives and Records Administration (NARA). For information on the availability of this

material at NARA, call 202-741-6030 or go to [http://www.archives.gov/federal\\_register/code\\_of\\_federal\\_regulations/ibr\\_locations.html](http://www.archives.gov/federal_register/code_of_federal_regulations/ibr_locations.html). An alternative worn wheel profile may be used in lieu of either APTA wheel profile, if approved by FRA.

(iii) All simulations shall be performed using a wheel/rail coefficient of friction of 0.5.

(2) **Vehicle performance on tangent track Classes 6 through 9.** For maximum vehicle speeds corresponding to track Class 6 and higher, the MCAT segments described in paragraphs (b)(1)(i) through (vii) of this appendix shall be used to assess vehicle performance on tangent track. For track Class 9, simulations must also include the combined perturbation segment described in paragraph (b)(1)(ix) of this appendix. A parametric matrix of MCAT simulations shall be performed using the following range of conditions:

(i) **Vehicle speed.** Simulations shall demonstrate that at up to 5 m.p.h. above the proposed maximum operating speed, the vehicle type shall not exceed the wheel/rail force and acceleration criteria defined in the Vehicle/Track Interaction Safety Limits table in § 213.333. Simulations shall also demonstrate acceptable vehicle dynamic response by incrementally increasing speed from 95 m.p.h. (115 m.p.h. if a previously qualified vehicle type on an untested route) to 5 m.p.h. above the proposed maximum operating speed (in 5 m.p.h. increments).

(ii) **Perturbation wavelength.** For each speed, a set of three separate MCAT simulations shall be performed. In each MCAT simulation for the perturbation segments described in paragraphs (b)(1)(ii) through (vii) and (b)(1)(ix) of this appendix, every perturbation shall have the same wavelength. The following three wavelengths,  $\lambda$ , shall be used: 31, 62, and 124 feet. The hunting perturbation segment described in paragraph (b)(1)(i) of this appendix has a fixed wavelength,  $\lambda$ , of 10 feet.

(iii) **Amplitude parameters.** Table 4 of this appendix provides the amplitude values for the MCAT segments described in paragraphs (b)(1)(i) through (vii) and (b)(1)(ix) of this appendix for each speed of the required parametric MCAT simulations. The last set of simulations shall be performed at 5 m.p.h. above the proposed maximum operating speed using the amplitude values in table 4 that correspond to the proposed maximum operating speed. For qualification of vehicle types at speeds greater than track Class 6 speeds, the following additional simulations shall be performed:

(A) For vehicle types being qualified for track Class 7 speeds, one additional set of simulations shall be performed at 115 m.p.h. using the track Class 6 amplitude values in table 4 (i.e., a 5 m.p.h. overspeed on Class 6 track).

(B) For vehicle types being qualified for track Class 8 speeds, two additional sets of simulations shall be performed. The first set at 115 m.p.h. using the track Class 6 amplitude values in table 4 (i.e., a 5 m.p.h. overspeed on Class 6 track), and a second set at 130 m.p.h. using the track Class 7 amplitude values in table 4 (i.e., a 5 m.p.h. overspeed on Class 7 track).

(C) For vehicle types being qualified for track Class 9 speeds, three additional sets of simulations shall be performed. The first set at 115 m.p.h. using the track Class 6 amplitude values in table 4 (i.e., a 5 m.p.h. overspeed on Class 6 track), a second set at 130 m.p.h. using the track Class 7 amplitude values in table 4 (i.e., a 5 m.p.h. overspeed on Class 7 track), and a third set at 165 m.p.h. using the track Class 8 amplitude values in table 4 (i.e., a 5 m.p.h. overspeed on Class 8 track).

**Table 4 of Appendix D to Part 213**  
**Track Class 6 through 9 Amplitude Parameters (in inches)**  
**for MCAT Simulations on Tangent Track**

|                                | Gage 56.5" |         |         |         |
|--------------------------------|------------|---------|---------|---------|
|                                | Class 6    | Class 7 | Class 8 | Class 9 |
| Max. Operating Speed (m.p.h.)  | 110        | 125     | 160     | 220     |
| Max. Simulation Speed (m.p.h.) | 115        | 130     | 165     | 225     |

| MCAT Segments         | Parameter          | Segment Description |
|-----------------------|--------------------|---------------------|
| Hunting               | $a_1$              | (b)(1)(i)           |
| Gage Narrowing        | $a_2$              | (b)(1)(ii)          |
| Gage Widening         | $a_3$              | (b)(1)(iii)         |
| Repeated Surface      | $a_9$              | (b)(1)(iv)          |
| Repeated Alinement    | $a_4$              | (b)(1)(v)           |
| Single Surface        | $a_{10}, a_{11}$   | (b)(1)(vi)          |
| Single Alinement      | $a_5, a_6$         | (b)(1)(vii)         |
| Short Warp            | $a_{12}$           | (b)(1)(ix)          |
| Combined Perturbation | $a_7, a_8, a_{13}$ |                     |

| Amplitude Parameters (inches)      |       |       |       |       |       |
|------------------------------------|-------|-------|-------|-------|-------|
| Wavelength $\lambda = 10\text{ft}$ | $a_1$ | 0.250 | 0.250 | 0.250 | 0.250 |

|                                    |          |  |  |  |  |
|------------------------------------|----------|--|--|--|--|
| Wavelength $\lambda = 20\text{ft}$ | $a_{12}$ |  |  |  |  |
|------------------------------------|----------|--|--|--|--|

|                                    |          |       |       |       |       |
|------------------------------------|----------|-------|-------|-------|-------|
| Wavelength $\lambda = 31\text{ft}$ | $a_2$    | 0.500 | 0.500 | 0.500 | 0.250 |
|                                    | $a_3$    | 0.750 | 0.500 | 0.500 | 0.500 |
|                                    | $a_4$    | 0.375 | 0.375 | 0.375 | 0.375 |
|                                    | $a_5$    | 0.500 | 0.500 | 0.500 | 0.500 |
|                                    | $a_6$    | 0.000 | 0.000 | 0.000 | 0.000 |
|                                    | $a_7$    |       |       |       | 0.333 |
|                                    | $a_8$    |       |       |       | 0.000 |
|                                    | $a_9$    | 0.750 | 0.750 | 0.500 | 0.375 |
|                                    | $a_{10}$ | 1.000 | 1.000 | 0.750 | 0.500 |
|                                    | $a_{11}$ | 0.000 | 0.000 | 0.000 | 0.000 |
|                                    | $a_{13}$ |       |       |       | 0.333 |

|  |       |       |       |       |       |
|--|-------|-------|-------|-------|-------|
|  | $a_2$ | 0.500 | 0.500 | 0.500 | 0.250 |
|  | $a_3$ | 0.750 | 0.500 | 0.500 | 0.500 |
|  | $a_4$ | 0.500 | 0.375 | 0.375 | 0.375 |
|  | $a_5$ | 0.750 | 0.750 | 0.750 | 0.500 |
|  | $a_6$ | 0.000 | 0.250 | 0.250 | 0.000 |
|  |       |       |       |       | 0.333 |
|  | $a_8$ |       |       |       | 0.000 |
|  | $a_9$ | 0.750 | 0.750 | 0.750 | 0.500 |



- (3) **Vehicle performance on curved track Classes 6 through 9.** For maximum vehicle speeds corresponding to track Class 6 and higher, the MCAT segments described in paragraphs (b)(1)(ii) through (viii) of this appendix shall be used to assess vehicle performance on curved track. For curves less than 1 degree, simulations must also include the hunting perturbation segment described in paragraph (b)(1)(i) of this appendix. For track Class 9 and for cant deficiencies greater than 5 inches, simulations must also include the combined perturbation segment described in paragraph (b)(1)(ix) of this appendix. A parametric matrix of MCAT simulations shall be performed using the following range of conditions:
- (i) **Vehicle speed.** Simulations shall demonstrate that at up to 5 m.p.h. above the proposed maximum operating speed, the vehicle type shall not exceed the wheel/rail force and acceleration criteria defined in the Vehicle/Track Interaction Safety Limits table in § 213.333. Simulations shall also demonstrate acceptable vehicle dynamic response by incrementally increasing speed from 95 m.p.h. (115 m.p.h. if a previously qualified vehicle type on an untested route) to 5 m.p.h. above the proposed maximum operating speed (in 5 m.p.h. increments).
  - (ii) **Perturbation wavelength.** For each speed, a set of three separate MCAT simulations shall be performed. In each MCAT simulation for the perturbation segments described in paragraphs (b)(1)(ii) through (vii) and paragraph (b)(1)(ix) of this appendix, every perturbation shall have the same wavelength. The following three wavelengths,  $\lambda$ , shall be used: 31, 62, and 124 feet. The hunting perturbation segment described in paragraph (b)(1)(i) of this appendix has a fixed wavelength,  $\lambda$ , of 10 feet, and the short warp perturbation segment described in paragraph (b)(1)(viii) of this appendix has a fixed wavelength,  $\lambda$ , of 20 feet.
  - (iii) **Track curvature.** For each speed, a range of curvatures shall be used to produce cant deficiency conditions ranging from greater than 3 inches up to the maximum intended for qualification (in 1 inch increments). The value of curvature,  $D$ , shall be determined using the equation defined in paragraph (b)(3) of this appendix. Each curve shall include representations of the MCAT segments described in paragraphs (b)(1)(i) through (ix) of this appendix, as appropriate, and have a fixed superelevation of 6 inches.
  - (iv) **Amplitude parameters.** Table 5 of this appendix provides the amplitude values for each speed of the required parametric MCAT simulations for cant deficiencies greater than 3 inches and not more than 5 inches. Table 6 of this appendix provides the amplitude values for each speed of the required parametric MCAT simulations for cant deficiencies greater than 5 inches. The last set of simulations at the maximum cant deficiency shall be performed at 5 m.p.h. above the proposed maximum operating speed using the amplitude values in table 5 or 6 of this appendix, as appropriate, that correspond to the proposed maximum operating speed and cant deficiency. For these simulations, the value of curvature,  $D$ , shall correspond to the proposed maximum operating speed and cant deficiency. For qualification of vehicle types at speeds greater than track Class 6 speeds, the following additional simulations shall be performed:
    - (A) For vehicle types being qualified for track Class 7 speeds, one additional set of simulations shall be performed at 115 m.p.h. using the track Class 6 amplitude values in table 5 or 6 of this appendix, as appropriate (i.e., a 5 m.p.h. overspeed on Class 6 track) and a value of curvature,  $D$ , that corresponds to 110 m.p.h. and the proposed maximum cant deficiency.

- (B) For vehicle types being qualified for track Class 8 speeds, two additional set of simulations shall be performed. The first set of simulations shall be performed at 115 m.p.h. using the track Class 6 amplitude values in table 5 or 6 of this appendix, as appropriate (i.e., a 5 m.p.h. overspeed on Class 6 track) and a value of curvature, D, that corresponds to 110 m.p.h. and the proposed maximum cant deficiency. The second set of simulations shall be performed at 130 m.p.h. using the track Class 7 amplitude values in table 5 or 6, as appropriate (i.e., a 5 m.p.h. overspeed on Class 7 track) and a value of curvature, D, that corresponds to 125 m.p.h. and the proposed maximum cant deficiency.
- (C) For vehicle types being qualified for track Class 9 speeds, three additional sets of simulations shall be performed. The first set of simulations shall be performed at 115 m.p.h. using the track Class 6 amplitude values in table 5 or 6 of this appendix, as appropriate (i.e., a 5 m.p.h. overspeed on Class 6 track) and a value of curvature, D, that corresponds to 110 m.p.h. and the proposed maximum cant deficiency. The second set of simulations shall be performed at 130 m.p.h. using the track Class 7 amplitude values in table 5 or 6, as appropriate (i.e., a 5 m.p.h. overspeed on Class 7 track) and a value of curvature, D, that corresponds to 125 m.p.h. and the proposed maximum cant deficiency. The third set of simulations shall be performed at 165 m.p.h. using the track Class 8 amplitude values in table 5 or 6, as appropriate (i.e., a 5 m.p.h. overspeed on Class 8 track) and a value of curvature, D, that corresponds to 160 m.p.h. and the proposed maximum cant deficiency.

**Table 5 of Appendix D to Part 213**

**Track Classes 6 through 9 Amplitude Parameters (in inches)  
for MCAT Simulations on Curved Track with Cant Deficiency > 3 and ≤ 5 Inches**

|                                |  | Gage 56.5" |         |         |         | Gage 57.0" |         |         |         |
|--------------------------------|--|------------|---------|---------|---------|------------|---------|---------|---------|
|                                |  | Class 6    | Class 7 | Class 8 | Class 9 | Class 6    | Class 7 | Class 8 | Class 9 |
| Max. Operating Speed (m.p.h.)  |  | 110        | 125     | 160     | 220     | 110        | 125     | 160     | 220     |
| Max. Simulation Speed (m.p.h.) |  | 115        | 130     | 165     | 225     | 115        | 130     | 165     | 225     |

| MCAT Segments         | Parameter                                         | Segment Description    |  |  |  |
|-----------------------|---------------------------------------------------|------------------------|--|--|--|
| Hunting               | a <sub>1</sub>                                    | (b)(1)(i) <sup>1</sup> |  |  |  |
| Gage Narrowing        | a <sub>2</sub>                                    | (b)(1)(ii)             |  |  |  |
| Gage Widening         | a <sub>3</sub>                                    | (b)(1)(iii)            |  |  |  |
| Repeated Surface      | a <sub>9</sub>                                    | (b)(1)(iv)             |  |  |  |
| Repeated Alinement    | a <sub>4</sub>                                    | (b)(1)(v)              |  |  |  |
| Single Surface        | a <sub>10</sub> , a <sub>11</sub>                 | (b)(1)(vi)             |  |  |  |
| Single Alinement      | a <sub>5</sub> , a <sub>6</sub>                   | (b)(1)(vii)            |  |  |  |
| Short Warp            | a <sub>12</sub>                                   | (b)(1)(viii)           |  |  |  |
| Combined Perturbation | a <sub>7</sub> , a <sub>8</sub> , a <sub>13</sub> | (b)(1)(ix)             |  |  |  |

|                      |                 | Amplitude Parameters (inches) |                    |                    |                    |
|----------------------|-----------------|-------------------------------|--------------------|--------------------|--------------------|
| Wavelength λ = 10ft  | a <sub>1</sub>  | 0.250 <sup>1</sup>            | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> |
| Wavelength λ = 20ft  | a <sub>12</sub> | 0.625                         | 0.563              | 0.500              | 0.375              |
| Wavelength λ = 31ft  | a <sub>2</sub>  | 0.500                         | 0.500              | 0.500              | 0.250              |
|                      | a <sub>3</sub>  | 0.750                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>4</sub>  | 0.375                         | 0.375              | 0.375              | 0.375              |
|                      | a <sub>5</sub>  | 0.500                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>6</sub>  | 0.000                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>7</sub>  |                               |                    |                    | 0.333              |
|                      | a <sub>8</sub>  |                               |                    |                    | 0.000              |
|                      | a <sub>9</sub>  | 0.750                         | 0.750              | 0.500              | 0.375              |
|                      | a <sub>10</sub> | 1.000                         | 1.000              | 0.750              | 0.500              |
|                      | a <sub>11</sub> | 0.000                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>13</sub> |                               |                    |                    | 0.333              |
| Wavelength λ = 62ft  | a <sub>2</sub>  | 0.500                         | 0.500              | 0.500              | 0.250              |
|                      | a <sub>3</sub>  | 0.750                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>4</sub>  | 0.500                         | 0.375              | 0.375              | 0.375              |
|                      | a <sub>5</sub>  | 0.625                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>6</sub>  | 0.000                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>7</sub>  |                               |                    |                    | 0.333              |
|                      | a <sub>8</sub>  |                               |                    |                    | 0.000              |
|                      | a <sub>9</sub>  | 0.750                         | 0.750              | 0.750              | 0.500              |
|                      | a <sub>10</sub> | 1.000                         | 1.000              | 1.000              | 0.750              |
|                      | a <sub>11</sub> | 0.000                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>13</sub> |                               |                    |                    | 0.500              |
| Wavelength λ = 124ft | a <sub>2</sub>  | 0.500                         | 0.500              | 0.500              | 0.250              |
|                      | a <sub>3</sub>  | 0.750                         | 0.750              | 0.750              | 0.750              |
|                      | a <sub>4</sub>  | 1.000                         | 0.875              | 0.500              | 0.500              |
|                      | a <sub>5</sub>  | 1.500                         | 1.250              | 0.750              | 0.750              |
|                      | a <sub>6</sub>  | 0.750                         | 0.500              | 0.000              | 0.000              |
|                      | a <sub>7</sub>  |                               |                    |                    | 0.500              |
|                      | a <sub>8</sub>  |                               |                    |                    | 0.000              |
|                      | a <sub>9</sub>  | 1.250                         | 1.000              | 0.875              | 0.625              |
|                      | a <sub>10</sub> | 1.750                         | 1.500              | 1.250              | 1.000              |
|                      | a <sub>11</sub> | 0.250                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>13</sub> |                               |                    |                    | 0.667              |

|                      |                 | Amplitude Parameters (inches) |                    |                    |                    |
|----------------------|-----------------|-------------------------------|--------------------|--------------------|--------------------|
| Wavelength λ = 10ft  | a <sub>1</sub>  | 0.250 <sup>1</sup>            | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> |
| Wavelength λ = 20ft  | a <sub>12</sub> | 0.625                         | 0.563              | 0.500              | 0.375              |
| Wavelength λ = 31ft  | a <sub>2</sub>  | 0.500                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>3</sub>  | 0.250                         | 0.250              | 0.250              | 0.500              |
|                      | a <sub>4</sub>  | 0.375                         | 0.375              | 0.375              | 0.375              |
|                      | a <sub>5</sub>  | 0.500                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>6</sub>  | 0.250                         | 0.250              | 0.250              | 0.250              |
|                      | a <sub>7</sub>  |                               |                    |                    | 0.333              |
|                      | a <sub>8</sub>  |                               |                    |                    | 0.083              |
|                      | a <sub>9</sub>  | 0.750                         | 0.750              | 0.500              | 0.375              |
|                      | a <sub>10</sub> | 1.000                         | 1.000              | 0.750              | 0.500              |
|                      | a <sub>11</sub> | 0.000                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>13</sub> |                               |                    |                    | 0.333              |
| Wavelength λ = 62ft  | a <sub>2</sub>  | 0.500                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>3</sub>  | 0.250                         | 0.250              | 0.250              | 0.250              |
|                      | a <sub>4</sub>  | 0.500                         | 0.375              | 0.375              | 0.375              |
|                      | a <sub>5</sub>  | 0.625                         | 0.500              | 0.500              | 0.500              |
|                      | a <sub>6</sub>  | 0.375                         | 0.250              | 0.250              | 0.250              |
|                      | a <sub>7</sub>  |                               |                    |                    | 0.333              |
|                      | a <sub>8</sub>  |                               |                    |                    | 0.083              |
|                      | a <sub>9</sub>  | 0.750                         | 0.750              | 0.750              | 0.500              |
|                      | a <sub>10</sub> | 1.000                         | 1.000              | 1.000              | 0.750              |
|                      | a <sub>11</sub> | 0.000                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>13</sub> |                               |                    |                    | 0.500              |
| Wavelength λ = 124ft | a <sub>2</sub>  | 1.000                         | 1.000              | 1.000              | 0.750              |
|                      | a <sub>3</sub>  | 0.250                         | 0.250              | 0.250              | 0.250              |
|                      | a <sub>4</sub>  | 1.000                         | 0.875              | 0.500              | 0.500              |
|                      | a <sub>5</sub>  | 1.500                         | 1.250              | 0.750              | 0.750              |
|                      | a <sub>6</sub>  | 1.250                         | 1.000              | 0.500              | 0.500              |
|                      | a <sub>7</sub>  |                               |                    |                    | 0.500              |
|                      | a <sub>8</sub>  |                               |                    |                    | 0.250              |
|                      | a <sub>9</sub>  | 1.250                         | 1.000              | 0.875              | 0.625              |
|                      | a <sub>10</sub> | 1.750                         | 1.500              | 1.250              | 1.000              |
|                      | a <sub>11</sub> | 0.250                         | 0.000              | 0.000              | 0.000              |
|                      | a <sub>13</sub> |                               |                    |                    | 0.667              |

<sup>1</sup> For curves <1 degree

**Table 6 of Appendix D to Part 213**  
**Track Class 6 through 9 Amplitude Parameters (in inches)**  
**for MCAT Simulations on Curved Track with Cant Deficiency > 5 Inches**

|                                |  | Gage 56.5" |         |         |         | Gage 57.0" |         |         |         |
|--------------------------------|--|------------|---------|---------|---------|------------|---------|---------|---------|
|                                |  | Class 6    | Class 7 | Class 8 | Class 9 | Class 6    | Class 7 | Class 8 | Class 9 |
| Max. Operating Speed (m.p.h.)  |  | 110        | 125     | 160     | 220     | 110        | 125     | 160     | 220     |
| Max. Simulation Speed (m.p.h.) |  | 115        | 130     | 165     | 225     | 115        | 130     | 165     | 225     |

| MCAT Segments         | Parameter          | Segment Description    |  |  |  |
|-----------------------|--------------------|------------------------|--|--|--|
| Hunting               | $a_1$              | (b)(1)(i) <sup>1</sup> |  |  |  |
| Gage Narrowing        | $a_2$              | (b)(1)(ii)             |  |  |  |
| Gage Widening         | $a_3$              | (b)(1)(iii)            |  |  |  |
| Repeated Surface      | $a_9$              | (b)(1)(iv)             |  |  |  |
| Repeated Alinement    | $a_4$              | (b)(1)(v)              |  |  |  |
| Single Surface        | $a_{10}, a_{11}$   | (b)(1)(vi)             |  |  |  |
| Single Alinement      | $a_5, a_6$         | (b)(1)(vii)            |  |  |  |
| Short Warp            | $a_{12}$           | (b)(1)(viii)           |  |  |  |
| Combined Perturbation | $a_7, a_8, a_{13}$ | (b)(1)(ix)             |  |  |  |

|                                     |          | Amplitude Parameters (inches) |                    |                    |                    |
|-------------------------------------|----------|-------------------------------|--------------------|--------------------|--------------------|
| Wavelength $\lambda = 10\text{ft}$  | $a_1$    | 0.250 <sup>1</sup>            | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> |
| Wavelength $\lambda = 20\text{ft}$  | $a_{12}$ | 0.625                         | 0.500              | 0.500 <sup>2</sup> | 0.375              |
| Wavelength $\lambda = 31\text{ft}$  | $a_2$    | 0.500                         | 0.500              | 0.500              | 0.250              |
|                                     | $a_3$    | 0.750                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_4$    | 0.375                         | 0.375              | 0.375              | 0.375              |
|                                     | $a_5$    | 0.500                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_6$    | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_7$    | 0.333                         | 0.333              | 0.333              | 0.333              |
|                                     | $a_8$    | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_9$    | 0.750                         | 0.750              | 0.500              | 0.375              |
|                                     | $a_{10}$ | 1.000                         | 1.000              | 0.750              | 0.500              |
|                                     | $a_{11}$ | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_{13}$ | 0.667                         | 0.667              | 0.500              | 0.333              |
| Wavelength $\lambda = 62\text{ft}$  | $a_2$    | 0.500                         | 0.500              | 0.500              | 0.250              |
|                                     | $a_3$    | 0.750                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_4$    | 0.500                         | 0.375              | 0.375              | 0.375              |
|                                     | $a_5$    | 0.625                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_6$    | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_7$    | 0.417                         | 0.333              | 0.333              | 0.333              |
|                                     | $a_8$    | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_9$    | 0.750                         | 0.750              | 0.750              | 0.500              |
|                                     | $a_{10}$ | 1.000                         | 1.000              | 1.000              | 0.750              |
|                                     | $a_{11}$ | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_{13}$ | 0.667                         | 0.667              | 0.667              | 0.500              |
| Wavelength $\lambda = 124\text{ft}$ | $a_2$    | 0.500                         | 0.500              | 0.500              | 0.250              |
|                                     | $a_3$    | 0.750                         | 0.750              | 0.750              | 0.750              |
|                                     | $a_4$    | 1.000                         | 0.875              | 0.500              | 0.500              |
|                                     | $a_5$    | 1.250                         | 1.000              | 0.750              | 0.750              |
|                                     | $a_6$    | 0.500                         | 0.250              | 0.000              | 0.000              |
|                                     | $a_7$    | 0.833                         | 0.667              | 0.500              | 0.500              |
|                                     | $a_8$    | 0.083                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_9$    | 1.250                         | 1.000              | 0.875              | 0.625              |
|                                     | $a_{10}$ | 1.500                         | 1.250              | 1.250              | 1.000              |
|                                     | $a_{11}$ | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_{13}$ | 1.000                         | 0.833              | 0.833              | 0.667              |

|                                     |          | Amplitude Parameters (inches) |                    |                    |                    |
|-------------------------------------|----------|-------------------------------|--------------------|--------------------|--------------------|
| Wavelength $\lambda = 10\text{ft}$  | $a_1$    | 0.250 <sup>1</sup>            | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> | 0.250 <sup>1</sup> |
| Wavelength $\lambda = 20\text{ft}$  | $a_{12}$ | 0.625                         | 0.500              | 0.500 <sup>2</sup> | 0.375              |
| Wavelength $\lambda = 31\text{ft}$  | $a_2$    | 0.500                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_3$    | 0.250                         | 0.250              | 0.250              | 0.500              |
|                                     | $a_4$    | 0.375                         | 0.375              | 0.375              | 0.375              |
|                                     | $a_5$    | 0.500                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_6$    | 0.250                         | 0.250              | 0.250              | 0.250              |
|                                     | $a_7$    | 0.333                         | 0.333              | 0.333              | 0.333              |
|                                     | $a_8$    | 0.083                         | 0.083              | 0.083              | 0.083              |
|                                     | $a_9$    | 0.750                         | 0.750              | 0.500              | 0.375              |
|                                     | $a_{10}$ | 1.000                         | 1.000              | 0.750              | 0.500              |
|                                     | $a_{11}$ | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_{13}$ | 0.667                         | 0.667              | 0.500              | 0.333              |
| Wavelength $\lambda = 62\text{ft}$  | $a_2$    | 0.500                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_3$    | 0.250                         | 0.250              | 0.250              | 0.250              |
|                                     | $a_4$    | 0.500                         | 0.375              | 0.375              | 0.375              |
|                                     | $a_5$    | 0.625                         | 0.500              | 0.500              | 0.500              |
|                                     | $a_6$    | 0.375                         | 0.250              | 0.250              | 0.250              |
|                                     | $a_7$    | 0.417                         | 0.333              | 0.333              | 0.333              |
|                                     | $a_8$    | 0.167                         | 0.083              | 0.083              | 0.083              |
|                                     | $a_9$    | 0.750                         | 0.750              | 0.750              | 0.500              |
|                                     | $a_{10}$ | 1.000                         | 1.000              | 1.000              | 0.750              |
|                                     | $a_{11}$ | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_{13}$ | 0.667                         | 0.667              | 0.667              | 0.500              |
| Wavelength $\lambda = 124\text{ft}$ | $a_2$    | 1.000                         | 1.000              | 1.000              | 0.750              |
|                                     | $a_3$    | 0.250                         | 0.250              | 0.250              | 0.250              |
|                                     | $a_4$    | 1.000                         | 0.875              | 0.500              | 0.500              |
|                                     | $a_5$    | 1.250                         | 1.000              | 0.750              | 0.750              |
|                                     | $a_6$    | 1.000                         | 0.750              | 0.500              | 0.500              |
|                                     | $a_7$    | 0.833                         | 0.667              | 0.500              | 0.500              |
|                                     | $a_8$    | 0.583                         | 0.417              | 0.250              | 0.250              |
|                                     | $a_9$    | 1.250                         | 1.000              | 0.875              | 0.625              |
|                                     | $a_{10}$ | 1.500                         | 1.250              | 1.250              | 1.000              |
|                                     | $a_{11}$ | 0.000                         | 0.000              | 0.000              | 0.000              |
|                                     | $a_{13}$ | 1.000                         | 0.833              | 0.833              | 0.667              |

- (4) **Vehicle performance on curved track Classes 1 through 5 at high cant deficiency.** For maximum vehicle speeds corresponding to track Classes 1 through 5, the MCAT segments described in paragraphs (b)(1)(ii) through (ix) of this appendix shall be used to assess vehicle performance on curved track if the proposed maximum cant deficiency is greater than 6 inches. A parametric matrix of MCAT simulations shall be performed using the following range of conditions:
- (i) **Vehicle speed.** Simulations shall demonstrate that at up to 5 m.p.h. above the proposed maximum operating speed, the vehicle shall not exceed the wheel/rail force and acceleration criteria defined in the Vehicle/Track Interaction Safety Limits table in § 213.333. Simulations shall also demonstrate acceptable vehicle dynamic response at 5 m.p.h. above the proposed maximum operating speed.
  - (ii) **Perturbation wavelength.** For each speed, a set of two separate MCAT simulations shall be performed. In each MCAT simulation for the perturbation segments described in paragraphs (b)(1)(ii) through (vii) and paragraph (b)(1)(ix) of this appendix, every perturbation shall have the same wavelength. The following two wavelengths,  $\lambda$ , shall be used: 31 and 62 feet. The short warp perturbation segment described in paragraph (b)(1)(viii) of this appendix has a fixed wavelength,  $\lambda$ , of 20 feet.
  - (iii) **Track curvature.** For a speed corresponding to 5 m.p.h. above the proposed maximum operating speed, a range of curvatures shall be used to produce cant deficiency conditions ranging from 6 inches up to the maximum intended for qualification (in 1 inch increments). The value of curvature,  $D$ , shall be determined using the equation in paragraph (b)(3) of this appendix. Each curve shall contain the MCAT segments described in paragraphs (b)(1)(ii) through (ix) of this appendix and have a fixed superelevation of 6 inches.
  - (iv) **Amplitude parameters.** Table 7 of this appendix provides the amplitude values for the MCAT segments described in paragraphs (b)(1)(ii) through (ix) of this appendix for each speed of the required parametric MCAT simulations.

**Table 7 of Appendix D to Part 213**  
**Track Class 1 through 5 Amplitude Parameters (in inches)**  
**for MCAT Simulations on Curved Track with Cant Deficiency > 6 Inches**

|                                |  | Gage 56.5" |         |         |         |         | Gage 57.0" |         |         |         |         |
|--------------------------------|--|------------|---------|---------|---------|---------|------------|---------|---------|---------|---------|
|                                |  | Class 1    | Class 2 | Class 3 | Class 4 | Class 5 | Class 1    | Class 2 | Class 3 | Class 4 | Class 5 |
| Max. Operating Speed (m.p.h.)  |  | 15         | 30      | 60      | 80      | 90      | 15         | 30      | 60      | 80      | 90      |
| Max. Simulation Speed (m.p.h.) |  | 20         | 35      | 65      | 85      | 95      | 20         | 35      | 65      | 85      | 95      |

| MCAT Segments         | Parameter                                         | Segment Description |  |  |  |  |
|-----------------------|---------------------------------------------------|---------------------|--|--|--|--|
| Hunting               | a <sub>1</sub>                                    | (b)(1)(i)           |  |  |  |  |
| Gage Narrowing        | a <sub>2</sub>                                    | (b)(1)(ii)          |  |  |  |  |
| Gage Widening         | a <sub>3</sub>                                    | (b)(1)(iii)         |  |  |  |  |
| Repeated Surface      | a <sub>4</sub>                                    | (b)(1)(iv)          |  |  |  |  |
| Repeated Alinement    | a <sub>5</sub>                                    | (b)(1)(v)           |  |  |  |  |
| Single Surface        | a <sub>10</sub> , a <sub>11</sub>                 | (b)(1)(vi)          |  |  |  |  |
| Single Alinement      | a <sub>9</sub> , a <sub>6</sub>                   | (b)(1)(vii)         |  |  |  |  |
| Short Warp            | a <sub>12</sub>                                   | (b)(1)(viii)        |  |  |  |  |
| Combined Perturbation | a <sub>7</sub> , a <sub>8</sub> , a <sub>13</sub> | (b)(1)(ix)          |  |  |  |  |

|                     |                 | Amplitude Parameters (inches) |       |       |       |       | Amplitude Parameters (inches) |       |       |       |       |
|---------------------|-----------------|-------------------------------|-------|-------|-------|-------|-------------------------------|-------|-------|-------|-------|
| Wavelength λ = 10ft | a <sub>1</sub>  |                               |       |       |       |       |                               |       |       |       |       |
| Wavelength λ = 20ft | a <sub>12</sub> | 1.000                         | 1.000 | 0.875 | 0.875 | 0.750 | 1.000                         | 1.000 | 0.875 | 0.875 | 0.750 |
| Wavelength λ = 31ft | a <sub>2</sub>  | 0.500                         | 0.500 | 0.500 | 0.500 | 0.500 | 1.250                         | 1.250 | 1.250 | 0.500 | 0.500 |
|                     | a <sub>3</sub>  | 1.250                         | 1.250 | 1.250 | 0.500 | 0.500 | 0.750                         | 0.750 | 0.750 | 0.500 | 0.500 |
|                     | a <sub>4</sub>  | 0.750                         | 0.750 | 0.750 | 0.750 | 0.500 | 0.750                         | 0.750 | 0.750 | 0.750 | 0.500 |
|                     | a <sub>5</sub>  | 0.750                         | 0.750 | 0.750 | 0.750 | 0.500 | 0.750                         | 0.750 | 0.750 | 0.750 | 0.500 |
|                     | a <sub>6</sub>  | 0.000                         | 0.000 | 0.000 | 0.250 | 0.000 | 0.000                         | 0.000 | 0.000 | 0.250 | 0.000 |
|                     | a <sub>7</sub>  | 0.500                         | 0.500 | 0.500 | 0.500 | 0.333 | 0.500                         | 0.500 | 0.500 | 0.500 | 0.333 |
|                     | a <sub>8</sub>  | 0.000                         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000                         | 0.000 | 0.000 | 0.000 | 0.000 |
|                     | a <sub>9</sub>  | 1.000                         | 1.000 | 1.000 | 1.000 | 1.000 | 1.000                         | 1.000 | 1.000 | 1.000 | 1.000 |
|                     | a <sub>10</sub> | 1.000                         | 1.000 | 1.000 | 1.000 | 1.000 | 1.000                         | 1.000 | 1.000 | 1.000 | 1.000 |
|                     | a <sub>11</sub> | 0.000                         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000                         | 0.000 | 0.000 | 0.000 | 0.000 |
|                     | a <sub>13</sub> | 0.667                         | 0.667 | 0.667 | 0.667 | 0.667 | 0.667                         | 0.667 | 0.667 | 0.667 | 0.667 |
| Wavelength λ = 62ft | a <sub>2</sub>  | 0.500                         | 0.500 | 0.500 | 0.500 | 0.500 | 1.250                         | 1.250 | 1.250 | 0.500 | 0.500 |
|                     | a <sub>3</sub>  | 1.250                         | 1.250 | 1.250 | 0.500 | 0.500 | 0.750                         | 0.750 | 0.750 | 0.500 | 0.500 |
|                     | a <sub>4</sub>  | 1.250                         | 1.250 | 1.250 | 0.875 | 0.625 | 1.250                         | 1.250 | 1.250 | 0.875 | 0.625 |
|                     | a <sub>5</sub>  | 1.250                         | 1.250 | 1.250 | 0.875 | 0.625 | 1.250                         | 1.250 | 1.250 | 0.875 | 0.625 |
|                     | a <sub>6</sub>  | 0.000                         | 0.000 | 0.000 | 0.375 | 0.125 | 0.500                         | 0.500 | 0.500 | 0.375 | 0.125 |
|                     | a <sub>7</sub>  | 0.833                         | 0.833 | 0.833 | 0.583 | 0.417 | 0.833                         | 0.833 | 0.833 | 0.583 | 0.417 |
|                     | a <sub>8</sub>  | 0.000                         | 0.000 | 0.000 | 0.083 | 0.000 | 0.083                         | 0.083 | 0.083 | 0.083 | 0.000 |
|                     | a <sub>9</sub>  | 1.750                         | 1.750 | 1.750 | 1.250 | 1.000 | 1.750                         | 1.750 | 1.750 | 1.250 | 1.000 |
|                     | a <sub>10</sub> | 1.750                         | 1.750 | 1.750 | 1.250 | 1.000 | 1.750                         | 1.750 | 1.750 | 1.250 | 1.000 |
|                     | a <sub>11</sub> | 0.000                         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000                         | 0.000 | 0.000 | 0.000 | 0.000 |
|                     | a <sub>13</sub> | 1.167                         | 1.167 | 1.167 | 0.833 | 0.667 | 1.167                         | 1.167 | 1.167 | 0.833 | 0.667 |

[78 FR 16116, Mar. 13, 2013]